Supplement to Bayesian Bootstrap Spike-and-Slab LASSO

December 26, 2021

Contents

A Proofs

This section presents proofs of the main theoretical statements: Section A.1 shows the proof of Theorem 4.1. Section A.2 presents the proof of Corollary 4.1. Section A.3 shows an example for the remark under Theorem 4.1. Section A.4 proves Theorem 4.2 and Theorem 4.3. Section A.5 proves Corollary 4.1. Section A.6 introduces a motivation for Section 3.1.

A.1 Proof of Theorem 4.1

A.1.1 Definitions and Lemmas used in the proof of Theorem 4.1

Denote constants

$$c_+^{\lambda_0,\lambda_1} = (1 + \sqrt{1 - 4/(\lambda_0 - \lambda_1)^2})/2$$

and

$$\delta_{c+}^{\lambda_0,\lambda_1} = \frac{1}{\lambda_0 - \lambda_1} \log \left[\frac{1-\theta}{\theta} \frac{\lambda_0}{\lambda_1} \frac{c_+}{1 - c_+} \right].$$

With slight abuse of notation, we denote the marginal prior (after integrating out γ_i) by

$$\pi(\beta_i \mid \theta, \lambda_0, \lambda_1) = \theta \lambda_1 e^{-|\beta_i|\lambda_1}/2 + (1-\theta)\lambda_0 e^{-|\beta_i|\lambda_0}/2.$$

Similarly as in Ročková (2018), the conditional mixing weight between the spike and the slab will be denoted with

$$p^*(t; \lambda_0, \lambda_1) = \frac{\theta \lambda_1 e^{-|t|\lambda_1}}{\theta \lambda_1 e^{-|t|\lambda_1} + (1-\theta)\lambda_0 e^{-|t|\lambda_0}} \tag{1}$$

and the weighted penalty with

$$\lambda^*(t; \lambda_0, \lambda_1) = \lambda_1 p^*(t; \lambda_0, \lambda_1) + \lambda_0(1 - p^*(t; \lambda_0, \lambda_1)).$$

The quantities $p^*(t; \lambda_0, \lambda_1)$ and $\lambda^*(t; \lambda_0, \lambda_1)$ are analogues of $p_\theta^\star$ and $\lambda_\theta^\star$ defined in (5) (main manuscript). Here, we have emphasized the dependence on λ_1 and λ_0 and suppressed the dependence on θ since this parameter is treated as fixed in our theory. Finally, we denote

$$\rho(t \mid \theta, \lambda_0, \lambda_1) = -\lambda_1|t| + \log[p^*(0; \lambda_0, \lambda_1)/p^*(t; \lambda_0, \lambda_1)]$$

and (similarly as in Ročková (2018))

$$g(t; \lambda_0, \lambda_1) = [\lambda^*(t; \lambda_0, \lambda_1) - \lambda_1]^2 + 2 \log p^*(t; \lambda_0, \lambda_1).$$

The following Lemma (a version of Theorem 3.1 in Ročková (2018)) will be useful for characterizing the BB-SSL posterior properties.

Lemma A.1. *Assume Y_i's arise from the normal means model (14) with $\sigma = 1$ for $1 \leq i \leq n$. Under the SSL prior (2), the mode $\widehat{\boldsymbol{\beta}} = (\widehat{\beta}_1, \cdots, \widehat{\beta}_p)^T$ maximizing $L(\boldsymbol{\beta}, \boldsymbol{Y}) = -\frac{1}{2}\sum_{i=1}^n (Y_i - \beta_i)^2 + \sum_{i=1}^n \rho(\beta_i \mid \theta, \lambda_0, \lambda_1)$ satisfies*

$$\widehat{\beta}_i = \begin{cases} 0, & \text{if } |Y_i| \leq \Delta(\lambda_0, \lambda_1). \\ [|Y_i| - \lambda^*(\widehat{\beta}_i; \lambda_0, \lambda_1)]_+ sign(Y_i), & \text{otherwise.} \end{cases} \tag{2}$$

where

$$\Delta(\lambda_0, \lambda_1) = \inf_{t>0}[t/2 - \rho(t \mid \theta, \lambda_0, \lambda_1)/t].$$

Notice that $\Delta(\lambda_0, \lambda_1)$ is the normal-means analogue of Δ_j (defined below (4) in the main manuscript) where we have added (λ_0, λ_1) to emphasize its dependence on λ_0 and λ_1. Next, we will need an upper and lower bound for $\Delta(\lambda_0, \lambda_1)$. When $\lambda_0 - \lambda_1 > 2$ and $g(0; \lambda_0, \lambda_1) > 0$, $-\frac{1}{2}(Y_i - \beta)^2 + \rho(\beta \mid \theta, \lambda_0, \lambda_1)$ has two modes and Ročková (2018) shows that $\Delta(\lambda_0, \lambda_1)$ can also be written as

$$\Delta(\lambda_0, \lambda_1) = \sqrt{2\log[1/p^*(0; \lambda_0, \lambda_1)] + g(\widetilde{\beta}; \lambda_0, \lambda_1)} + \lambda_1,$$

where $\widetilde{\beta}$ is the local mode (not the global mode) of $-\frac{1}{2}(Y_i - \beta)^2 + \rho(\beta \mid \theta, \lambda_0, \lambda_1)$. It follows from Theorem 3.1 in Ročková (2018) and Lemma 1.2 in the Appendix of Ročková (2018) that

$$\Delta^L(\lambda_0, \lambda_1) < \Delta(\lambda_0, \lambda_1) \leq \Delta^U(\lambda_0, \lambda_1), \tag{3}$$

where

$$\Delta^L(\lambda_0, \lambda_1) = \sqrt{2\log[1/p^*(0; \lambda_0, \lambda_1)] - 2} + \lambda_1 \quad \text{and} \quad \Delta^U(\lambda_0, \lambda_1) = \sqrt{2\log[1/p^*(0; \lambda_0, \lambda_1)]} + \lambda_1.$$

Based on these facts, we then obtain the following simple Lemma.

Lemma A.2. *Under assumptions in Theorem 4.1, for any $w_i > 0$ which satisfies*

$$g(0; \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) > 0 \quad \text{and} \quad \lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2} > 2$$

we have

$$\Delta^L(\lambda_0, \lambda_1) + \lambda_1(w_i^{-1/2} - 1) \leq \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) \leq \Delta^U(\lambda_0, \lambda_1) + \lambda_1(w_i^{-1/2} - 1). \tag{4}$$

Proof. Since $g(0; \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) > 0$ and $\lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2} > 2$, from Lemma A.1 (the discussion below) and the fact that $p^*(0; \lambda_0, \lambda_1) = p^*(0; \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})$, we have

$$\Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) \leq \Delta^U(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) = \Delta^U(\lambda_0, \lambda_1) + \lambda_1(w_i^{-1/2} - 1).$$

The other inequality is obtained similarly. □

Lemma A.3. *For any fixed constant $C > 0$, $\mu \in \mathbb{R}$ and a random variable $Y \sim N(0, 1)$, we have*

$$\begin{aligned}&\mathbb{E}_Y\left[|Y-\mu|^2\mathbb{I}(|Y-\mu| \geq C)\right]\\ &\quad = 1+\mu^2 - \Phi(\mu+C) + \Phi(\mu-C) - (\mu-C)\phi(\mu-C) + (\mu+C)\phi(\mu+C),\end{aligned}$$

where Φ and ϕ are the standard normal distribution and density functions.

Proof. Denote a truncated normal random variable with $Z = [Y \mid Y \in [a, b]]$ and recall that

$$\mathbb{E}\left[Z\right] = \frac{\phi(a)-\phi(b)}{\Phi(b)-\Phi(a)} \quad \text{and} \quad \mathbb{E}\left[Z^2\right] = 1 + \frac{a\phi(a)-b\phi(b)}{\Phi(b)-\Phi(a)}.$$

Then we have

$$\begin{aligned}&\mathbb{E}_Y\left[|Y-\mu|^2\mathbb{I}(|Y-\mu| \geq C)\right]\\ &= \mathbb{E}_Y\left[|Y-\mu|^2\right] - \mathbb{E}_Y\left[|Y-\mu|^2\mathbb{I}(|Y-\mu| < C)\right]\\ &= 1+\mu^2 - \left[\Phi(\mu+C)-\Phi(\mu-C)\right]\\ &\quad \times \left[\mathbb{E}\left[Y^2 \mid Y \in [\mu-C, \mu+C]\right] - 2\mu\mathbb{E}\left[Y \mid Y \in [\mu-C, \mu+C]\right] + \mu^2\right]\\ &= 1+\mu^2 - \Phi(\mu+C) + \Phi(\mu-C) - (\mu-C)\phi(\mu-C) + (\mu+C)\phi(\mu+C).\end{aligned}$$

□

A.1.2 Proof of Theorem 4.1

First assume that $\sigma = 1$. Throughout this section, we will simply denote $\boldsymbol{Y} = \boldsymbol{Y}^{(n)}$. We want to obtain the expression for BB-SSL estimate $\widetilde{\boldsymbol{\beta}}$. Notice that the objective function (13) for BB-SSL can be written as

$$\begin{aligned}\widetilde{\boldsymbol{\beta}} &= \arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p}\left\{-\sum_{i=1}^n \frac{w_i}{2}(Y_i-\beta_i)^2 + \sum_{i=1}^n \log\pi(\beta_i-\mu_i \mid \theta)\right\}\\ &= \arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p}\left\{-\sum_{i=1}^n \frac{1}{2}(Y_i^*-\beta_i^*)^2 + \sum_{i=1}^n \log\pi_i^*(\beta_i^* \mid \theta)\right\},\end{aligned} \tag{5}$$

where $Y_i^* = w_i^{1/2}(Y_i-\mu_i)$, $\beta_i^* = w_i^{1/2}(\beta_i-\mu_i)$, and $\pi_i^*(\beta_i^* \mid \theta) = \pi(\beta_i^* \mid \theta, \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})$. So from (2), the mode estimator $\widehat{\beta_i^*}$ for β_i^* satisfies

$$\widehat{\beta_i^*} = \begin{cases}0, & \text{if } |Y_i^*| \leq \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}).\\ [|Y_i^*| - \lambda^*(\widehat{\beta_i^*}; \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})]_+ \mathrm{sign}(Y_i^*), & \text{otherwise.}\end{cases}$$

The BB-SSL estimate $\widetilde{\boldsymbol{\beta}} = (\widetilde{\beta}_1, \cdots, \widetilde{\beta}_p)^T$ can then be calculated via $\widetilde{\beta}_i = \widehat{\beta_i^*} w_i^{-1/2} + \mu_i$. Equation (3.2) in Ročková (2018) says that

$$\lambda^*(\beta_i; \lambda_0, \lambda_1) = -\frac{\partial\rho(\beta_i \mid \theta, \lambda_0, \lambda_1)}{\partial|\beta_i|}.$$

From the chain rule and the fact that

$$\rho\left(w_i^{1/2}(\widetilde{\beta}_i - \mu_i); \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}\right) = \rho(\widetilde{\beta}_i - \mu_i; \lambda_0, \lambda_1),$$

we know

$$\begin{aligned}
&\lambda^*\left(\widehat{\beta_i^*}; \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}\right) = \lambda^*\left(w_i^{1/2}\left(\widetilde{\beta}_i - \mu_i\right); \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}\right) \\
&= -\frac{\partial \rho\left(w_i^{1/2}(\widetilde{\beta}_i - \mu_i); \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}\right)}{\partial \left|w_i^{1/2}(\widetilde{\beta}_i - \mu_i)\right|} \\
&= -\frac{\partial \rho(\widetilde{\beta}_i - \mu_i; \lambda_0, \lambda_1)}{\partial \left|\widetilde{\beta}_i - \mu_i\right|} \frac{\partial \left|\widetilde{\beta}_i - \mu_i\right|}{\partial \left|w_i^{1/2}\left(\widetilde{\beta}_i - \mu_i\right)\right|} = w_i^{-1/2}\lambda^*\left(\widetilde{\beta}_i - \mu_i; \lambda_0, \lambda_1\right).
\end{aligned}$$

Thus,

$$\widetilde{\beta}_i = \begin{cases} \mu_i, \quad \text{if } |w_i^{1/2}(Y_i - \mu_i)| \leq \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}). \\ \mu_i + [|Y_i - \mu_i| - w_i^{-1}\lambda^*(\widetilde{\beta}_i - \mu_i; \lambda_0, \lambda_1)]_+ \text{sign}(Y_i - \mu_i), \quad \text{otherwise.} \end{cases} \tag{6}$$

From the Markov's inequality, we know that for any $M_n > 0$,

$$\mathbb{E}_{\boldsymbol{Y}} \mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^0\|_2^2 > M_n q \log\left(\frac{n}{q}\right) \mid \boldsymbol{Y}\right) \leq \frac{\mathbb{E}_{\boldsymbol{Y}} \mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}\left[\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^0\|_2^2 \mid \boldsymbol{Y}\right]}{M_n q \log \frac{n}{q}}.$$

Thus, in order to prove the desired statement, we only need to bound $\mathbb{E}_{\boldsymbol{Y}}\left[\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}\left[\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^0\|_2^2 \mid \boldsymbol{Y}\right]\right]$. Notice that for the separable SSL prior (i.e. with a fixed value of θ), the posterior is separable and we obtain

$$\mathbb{E}_{\boldsymbol{Y}}\left[\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}\left[\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^0\|_2^2 \mid \boldsymbol{Y}\right]\right] = \sum_{i=1}^n \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - \beta_i^0)^2 \mid Y_i\right]\right].$$

We will bound the risk $\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - \beta_i^0)^2 \mid Y_i\right]\right]$ separately for active and inactive coordinates.

Active coordinates For active coordinates, we have

$$\begin{aligned}
&\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - \beta_i^0)^2 \mid Y_i\right]\right] = \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - Y_i + Y_i - \beta_i^0)^2 \mid Y_i\right]\right] \\
&\leq \mathbb{E}_{Y_i}\left[2\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - Y_i)^2 \mid Y_i\right] + 2\mathbb{E}_{\mu_i,w_i}\left[(Y_i - \beta_i^0)^2 \mid Y_i\right]\right] \\
&\leq 2\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - Y_i)^2 \mid Y_i\right]\right] + 2 \\
&= 2\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - Y_i)^2 \mathbb{I}\left(w_i^{1/2}|Y_i - \mu_i| \leq \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right] \\
&\quad + 2\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - Y_i)^2 \mathbb{I}\left(w_i^{1/2}|Y_i - \mu_i| > \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right] + 2 \\
&= 2U_1 + 2U_2 + 2,
\end{aligned} \tag{7}$$

where

$$U_1 = \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - Y_i)^2\mathbb{I}\left(w_i^{1/2}|Y_i-\mu_i| \le \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right],$$
$$U_2 = \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - Y_i)^2\mathbb{I}\left(w_i^{1/2}|Y_i-\mu_i| > \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right].$$

Now we bound U_1 and U_2 separately. First,

$$\begin{aligned}
U_1 &= \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - Y_i)^2\mathbb{I}\left(w_i^{1/2}|Y_i-\mu_i| \le \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right] \\
&\overset{(a)}{\le} \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\mu_i - Y_i)^2\mathbb{I}\left(|Y_i-\mu_i| \le w_i^{-1/2}\Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right] \qquad (8) \\
&\le \mathbb{E}_{w_i}\left[\Delta^2(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\frac{1}{w_i}\right],
\end{aligned}$$

where (a) utilizes the observation $|\widetilde{\beta}_i - Y_i| \le |\mu_i - Y_i|$ from Equation (6). Deploying Lemma A.2 in equation (8), we get (when n is sufficiently large)

$$\begin{aligned}
U_1 &\le \mathbb{E}_{w_i}\left[\Delta^2(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\frac{1}{w_i}\mathbb{I}\left(g(0;\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) > 0 \text{ and } \lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2} > 2\right)\right] \\
&\quad + \mathbb{E}_{w_i}\left[\Delta^2(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\frac{1}{w_i}\mathbb{I}\left(g(0;\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) \le 0 \text{ or } \lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2} \le 2\right)\right] \\
&\le \mathbb{E}_{w_i}\left[\frac{1}{w_i}\left(\Delta^U(\lambda_0,\lambda_1) + \lambda_1 w_i^{-1/2} - \lambda_1\right)^2\right] \\
&\quad + \mathbb{E}_{w_i}\left[\Delta^2(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\frac{1}{w_i}\mathbb{I}\left(w_i \ge \frac{(\lambda_0-\lambda_1)^2(1-p^*(0;\lambda_0,\lambda_1))^2}{2\log(1/p^*(0;\lambda_0,\lambda_1))} \text{ or } w_i \ge \frac{(\lambda_0-\lambda_1)^2}{4}\right)\right] \\
&\overset{(a)}{\le} 2\left(\Delta^U(\lambda_0,\lambda_1) - \lambda_1\right)^2\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right) + 2\lambda_1^2\mathbb{E}_{w_i}\left(\frac{1}{w_i^2}\right) + \mathbb{E}_{w_i}\left(\frac{\lambda_0^2}{w_i}\frac{1}{w_i}\mathbb{I}\left(w_i \ge C(\lambda_0,\lambda_1)\right)\right) \\
&\le 2\left(\Delta^U(\lambda_0,\lambda_1) - \lambda_1\right)^2\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right) + 2\mathbb{E}_{w_i}\left(\frac{1}{w_i^2}\right) + \frac{\lambda_0^2}{C^2(\lambda_0,\lambda_1)}\mathbb{P}\left(w_i \ge C(\lambda_0,\lambda_1)\right) \\
&\overset{(b)}{\le} 2\left(\Delta^U(\lambda_0,\lambda_1) - \lambda_1\right)^2\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right) + 2\mathbb{E}_{w_i}\left(\frac{1}{w_i^2}\right) + \frac{\lambda_0^2}{C^3(\lambda_0,\lambda_1)} \\
&\le 2\left(\Delta^U(\lambda_0,\lambda_1) - \lambda_1\right)^2\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right) + 2\mathbb{E}_{w_i}\left(\frac{1}{w_i^2}\right) + 1,
\end{aligned} \qquad (9)$$

where (a) uses the fact that $(a+b)^2 \le 2a^2 + 2b^2$, $\Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) \le \lambda_0 w_i^{-1/2}$, and we denote $C(\lambda_0,\lambda_1) = \min\left\{\frac{(\lambda_0-\lambda_1)^2(1-p^*(0;\lambda_0,\lambda_1))^2}{2\log(1/p^*(0;\lambda_0,\lambda_1))}, \frac{(\lambda_0-\lambda_1)^2}{4}\right\}$, and (b) follows from the Markov's inequality.

For U_2, we have

$$
\begin{aligned}
U_2 =& \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i - Y_i)^2 \mathbb{I}\left(w_i^{1/2}(Y_i-\mu_i) > \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right] \\
\overset{(a)}{\leq}& \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(w_i^{-1}\lambda^*(\widetilde{\beta}_i-\mu_i)\right)^2 \mathbb{I}\left(w_i^{1/2}|Y_i-\mu_i| > \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right] \\
\leq& \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(w_i^{-1}\lambda^*(\widetilde{\beta}_i-\mu_i)\right)^2 \mathbb{I}\left(w_i^{1/2}|Y_i-\mu_i| > \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right)\right.\right. \\
& \qquad \left.\left. \mathbb{I}\left(\lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2} > 2 \text{ and } g(0;\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) > 0\right) \mid Y_i\right]\right] \\
&+ \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(w_i^{-1}\lambda^*(\widetilde{\beta}_i-\mu_i)\right)^2 \mathbb{I}\left(w_i^{1/2}|Y_i-\mu_i| > \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right)\right.\right. \\
& \qquad \left.\left. \mathbb{I}\left(\lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2} \leq 2\right) \mid Y_i\right]\right] \\
&+ \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(w_i^{-1}\lambda^*(\widetilde{\beta}_i-\mu_i)\right)^2 \mathbb{I}\left(w_i^{1/2}|Y_i-\mu_i| > \Delta(\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2})\right)\right.\right. \\
& \qquad \left.\left. \mathbb{I}\left(g(0;\lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}) \leq 0\right) \mid Y_i\right]\right] \\
=& U_3 + U_4 + U_5,
\end{aligned} \tag{10}
$$

where (a) utilizes the fact that $|\widetilde{\beta}_i - Y_i| \leq w_i^{-1}\lambda^*(\widetilde{\beta}_i - \mu_i)$ from Equation (6), and we abbreviate $\lambda^*(\widetilde{\beta}_i - \mu_i) = \lambda^*(\widetilde{\beta}_i - \mu_i; \lambda_0, \lambda_1)$. In order to bound (10), we bound U_3, U_4, U_5 separately.

- To bound U_3: When $\lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2} > 2$ and $g\left(0; \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}\right) > 0$, following the proof of Theorem 4.1 in Ročková (2018), we know that $|\widehat{\beta_i^*}| > \delta_{c+}^{\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i}}$ and thus $p^*\left(\widehat{\beta_i^*}; \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}\right) > c_+^{\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i}}$. This implies $p^*\left(\widetilde{\beta}_i - \mu_i; \lambda_0, \lambda_1\right) > c_+^{\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i}}$, so we have

$$
\begin{aligned}
&\lambda^*\left(\widetilde{\beta}_i - \mu_i; \lambda_0 w_i^{-1/2}, \lambda_1 w_i^{-1/2}\right) < c_+^{\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i}}\left(\lambda_1 w_i^{-1/2} - \lambda_0 w_i^{-1/2}\right) + \lambda_0 w_i^{-1/2} \\
=& \left(1 - c_+^{\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i}}\right)\left(\lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2}\right) + \lambda_1 w_i^{-1/2} \overset{(a)}{<} \frac{2w_i^{1/2}}{\lambda_0 - \lambda_1} + \lambda_1 w_i^{-1/2} < 1 + \lambda_1 w_i^{-1/2},
\end{aligned}
$$

  where (a) follows from the fact that

$$
c_+^{\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i}}\left(1 - c_+^{\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i}}\right) = \frac{1}{\left(\lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2}\right)^2} = \frac{w_i}{(\lambda_0 - \lambda_1)^2},
$$

  and

$$
c_+^{\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i}} > \frac{1}{2}.
$$

In view that $\lambda^*\left(\widetilde{\beta}_i-\mu_i;\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2}\right)=w_i^{-1/2}\lambda^*\left(\widetilde{\beta}_i-\mu_i;\lambda_0,\lambda_1\right)$, we have

$$U_3 \le \mathbb{E}_{w_i}\left(\frac{w_i^{1/2}+\lambda_1}{w_i}\right)^2 \le 2\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right)+2\lambda_1^2\mathbb{E}_{w_i}\left(\frac{1}{w_i^2}\right).$$

- To bound U_4: When $\lambda_0 w_i^{-1/2}-\lambda_1 w_i^{-1/2}\le 2$, we have

$$\lambda^*\left(\widetilde{\beta}_i-\mu_i;\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2}\right)\le \lambda_0 w_i^{-1/2}\le \lambda_1 w_i^{-1/2}+2.$$

  So in view that $\lambda^*\left(\widetilde{\beta}_i-\mu_i;\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2}\right)=w_i^{-1/2}\lambda^*\left(\widetilde{\beta}_i-\mu_i;\lambda_0,\lambda_1\right)$, we have

$$U_4 \le \mathbb{E}_{w_i}\left(\frac{\lambda_1+2w_i^{1/2}}{w_i}\right)^2 \le 2\lambda_1^2\mathbb{E}_{w_i}\left(\frac{1}{w_i^2}\right)+8\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right).$$

- To bound U_5: When $g(0;\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\le 0$, we have $w_i\ge \frac{(\lambda_0-\lambda_1)^2(1-p^*(0;\lambda_0,\lambda_1))^2}{2\log(1/p^*(0;\lambda_0,\lambda_1))}$, thus when n is sufficiently large,

$$U_5 \le \mathbb{E}_{w_i}\left(\frac{\lambda_0}{w_i}\mathbb{I}\left(w_i\ge \frac{(\lambda_0-\lambda_1)^2(1-p^*(0;\lambda_0,\lambda_1))^2}{2\log(1/p^*(0;\lambda_0,\lambda_1))}\right)\right)^2 \le \frac{4[\log(1/p^*(0;\lambda_0,\lambda_1))]^2\lambda_0^2}{(\lambda_0-\lambda_1)^4(1-p^*(0;\lambda_0,\lambda_1))^4}\le \frac{1}{\lambda_0}.$$

So plugging the above bounds into Equation (10), we obtain

$$U_2 \le U_3+U_4+U_5 \le 10\mathbb{E}_{w_i}\left(\frac{1}{w_i^2}\right)+4\lambda_1^2\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right)+\frac{1}{\lambda_0}. \tag{11}$$

Thus, from (7), (9), (11) and condition (2), we know that for active coordinates, when n is sufficiently large,

$$\begin{aligned}&\mathbb{E}_{Y_i}\mathbb{E}_{\mu_i,w_i}[(\widetilde{\beta}_i-\beta_i^0)^2\mid Y]\le 6+\left[4\left(\Delta^U(\lambda_0,\lambda_1)-\lambda_1\right)^2+8\lambda_1^2\right]\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right)+24\mathbb{E}_{w_i}\left(\frac{1}{w_i^2}\right)\\&\le C_3\left(\Delta^U(\lambda_0,\lambda_1)\right)^2.\end{aligned}$$

Inactive coordinates For inactive coordinates, we have

$$
\begin{aligned}
&\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}[(\widetilde{\beta}_i-\beta_i^0)^2 \mid Y_i]\right] = \mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}[(\widetilde{\beta}_i)^2 \mid Y_i]\right]\\
=&\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i-\mu_i+\mu_i)^2\mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i|\geq\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right]\\
&+\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i)^2\mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i|<\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right]\\
\overset{(a)}{\leq}&\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(|Y_i-\mu_i|+|\mu_i|)^2\,\mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i|\geq\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right) \mid Y_i\right]\right]+\mathbb{E}_{\mu_i}\mu_i^2\\
\leq&2\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(|Y_i-\mu_i|^2+|\mu_i|^2\right)\mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i|\geq\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right)\right.\right.\\
&\qquad\left.\left.\mathbb{I}\left(w_i\leq\eta+\gamma\right)\mathbb{I}\left(|\mu_i|\leq\frac{1}{\lambda_0}+\frac{1}{\sqrt{\lambda_0}}\right) \mid Y_i\right]\right]\\
&+2\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(|Y_i-\mu_i|^2+|\mu_i|^2\right)\mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i|\geq\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right)\right.\right.\\
&\qquad\left.\left.\mathbb{I}(w_i\leq\eta+\gamma)\mathbb{I}\left(|\mu_i|>\frac{1}{\lambda_0}+\frac{1}{\sqrt{\lambda_0}}\right) \mid Y_i\right]\right]\\
&+2\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(|Y_i-\mu_i|^2+|\mu_i|^2\right)\mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i|\geq\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right)\right.\right.\\
&\qquad\mathbb{I}(w_i>\eta+\gamma)\mid Y_i]]+\frac{2}{\lambda_0^2}\\
=&2U_1+2U_2+2U_3+\frac{2}{\lambda_0^2},
\end{aligned}
\tag{12}
$$

where (a) uses the fact that $|\widetilde{\beta}_i-\mu_i|\leq|Y_i-\mu_i|$ when $\sqrt{w_i}|Y_i-\mu_i|>\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})$ and $\widetilde{\beta}_i=\mu_i$ when $\sqrt{w_i}|Y_i-\mu_i|\leq\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2}))$ in view of Equation (6), and we denote

$$
\begin{aligned}
U_1=&\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(|Y_i-\mu_i|^2+|\mu_i|^2\right)\mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i|\geq\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right)\right.\right.\\
&\qquad\left.\left.\mathbb{I}(w_i\leq\eta+\gamma)\mathbb{I}\left(|\mu_i|\leq\frac{1}{\lambda_0}+\frac{1}{\sqrt{\lambda_0}}\right) \mid Y_i\right]\right],\\
U_2=&\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(|Y_i-\mu_i|^2+|\mu_i|^2\right)\mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i|\geq\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right)\right.\right.\\
&\qquad\left.\left.\mathbb{I}(w_i\leq\eta+\gamma)\mathbb{I}\left(|\mu_i|>\frac{1}{\lambda_0}+\frac{1}{\sqrt{\lambda_0}}\right) \mid Y_i\right]\right],\\
U_3=&\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(|Y_i-\mu_i|^2+|\mu_i|^2\right)\mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i|\geq\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right)\right.\right.\\
&\qquad\mathbb{I}(w_i>\eta+\gamma)\mid Y_i]].
\end{aligned}
$$

Now we bound U_1, U_2, U_3 separately.

For the first term U_1 in (12), when n is sufficiently large, we have

$$\begin{aligned}
2U_1 \overset{(a)}{\leq} & 2\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[|Y_i-\mu_i|^2\mathbb{I}\left(|Y_i-\mu_i| \geq \frac{\Delta^L(\lambda_0,\lambda_1)+\lambda_1(w_i^{-1/2}-1)}{\sqrt{\eta+\gamma}}\right)\mathbb{I}\left(|\mu_i| \leq \frac{1}{\lambda_0}+\frac{1}{\sqrt{\lambda_0}}\right) \mid Y_i\right]\right] \\
& + 2\mathbb{E}\mu_i^2 \\
\leq & 2\mathbb{E}_{\mu_i}\left[\mathbb{E}_{Y_i}\left[|Y_i-\mu_i|^2\mathbb{I}\left(|Y_i-\mu_i| \geq \frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\right) \mid \mu_i\right]\mathbb{I}\left(|\mu_i| \leq \frac{1}{\lambda_0}+\frac{1}{\sqrt{\lambda_0}}\right)\right] + \frac{4}{\lambda_0^2} \\
\overset{(b)}{=} & 2\mathbb{E}_{\mu_i}\left[\mathbb{I}\left(|\mu_i| \leq 1/\lambda_0+1/\sqrt{\lambda_0}\right)\left[\left(1-\Phi\left(\mu_i+\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\right)\right)+\mu_i^2\right.\right. \\
& \quad + \Phi\left(\mu_i-\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\right)-\left(\mu_i-\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\right)\phi\left(\mu_i-\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\right) \\
& \quad \left.\left.+\left(\mu_i+\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\right)\phi\left(\mu_i+\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\right)\right]\right]+\frac{4}{\lambda_0^2} \\
\leq & \frac{8}{\lambda_0^2}+4\left[1-\Phi\left(-1/\lambda_0-1/\sqrt{\lambda_0}+\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\right)\right] \\
& + 2\left[\left(\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}+1/\lambda_0+1/\sqrt{\lambda_0}\right)\phi\left(1/\lambda_0+1/\sqrt{\lambda_0}-\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\right)\right] \\
\overset{(c)}{\leq} & \frac{8}{\lambda_0^2}+4\left[\frac{1}{-1/\lambda_0-1/\sqrt{\lambda_0}+\left(\Delta^L(\lambda_0,\lambda_1)-\lambda_1\right)/\sqrt{\eta+\gamma}}\right. \\
& \qquad \left.\phi\left(-1/\lambda_0-1/\sqrt{\lambda_0}+\left(\Delta^L(\lambda_0,\lambda_1)-\lambda_1\right)/\sqrt{\eta+\gamma}\right)\right] \\
& + 4\left[\left(\left(\Delta^L(\lambda_0,\lambda_1)-\lambda_1\right)/\sqrt{\eta+\gamma}\right)\phi\left(1/\lambda_0+1/\sqrt{\lambda_0}-\left(\Delta^L(\lambda_0,\lambda_1)-\lambda_1\right)/\sqrt{\eta+\gamma}\right)\right] \\
\leq & 5\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\left[\phi\left(1/\lambda_0+1/\sqrt{\lambda_0}-\left(\Delta^L(\lambda_0,\lambda_1)-\lambda_1\right)/\sqrt{\eta+\gamma}\right)\right],
\end{aligned} \tag{13}$$

where (a) follows from the fact that when n is sufficiently large, $w_i \leq \eta+\gamma \Rightarrow g(0;\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2}) > 0, \lambda_0 w_i^{-1/2} - \lambda_1 w_i^{-1/2} > 2$, which ensures that Lemma A.2 holds, (b) uses Lemma A.3, (c) follows from the fact that $1-\Phi(x) \leq [\phi(x)/x]$ for all $x > 0$, and here $x = -1/\lambda_0 - 1/\sqrt{\lambda_0} + (\Delta^L(\lambda_0,\lambda_1)-\lambda_1)/\sqrt{\eta+\gamma} > 0$ always holds when n is sufficiently large.

For the second term in (12),

$$
\begin{aligned}
2U_2 = 2\mathbb{E}_{Y_i} \Big[\mathbb{E}_{\mu_i,w_i} \Big[\left(|Y_i-\mu_i|^2 + |\mu_i|^2 \right) \mathbb{I}\left(\sqrt{w_i}|Y_i-\mu_i| \geq \Delta(\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i}) \right) \\
\mathbb{I}(w_i \leq \eta+\gamma) \mathbb{I}\left(|\mu_i| > \frac{1}{\lambda_0} + \frac{1}{\sqrt{\lambda_0}} \right) \mid Y_i \Big] \Big] \\
\leq 2\mathbb{E}_{Y_i} \left[\mathbb{E}_{\mu_i,w_i} \left[\left(|Y_i-\mu_i|^2 + |\mu_i|^2 \right) \mathbb{I}\left(|\mu_i| > \frac{1}{\lambda_0} + \frac{1}{\sqrt{\lambda_0}} \right) \mid Y_i \right] \right] \\
\leq 2\mathbb{E}_{Y_i} \left[\mathbb{E}_{\mu_i,w_i} \left[(2Y_i^2 + 3\mu_i^2) \mathbb{I}\left(|\mu_i| > \frac{1}{\lambda_0} + \frac{1}{\sqrt{\lambda_0}} \right) \mid Y_i \right] \right] \\
= 4\mathbb{P}\left(|\mu_i| > \frac{1}{\lambda_0} + \frac{1}{\sqrt{\lambda_0}} \right) + 6\mathbb{E}_{\mu_i} \left[\mu_i^2 \mathbb{I}\left(|\mu_i| > \frac{1}{\lambda_0} + \frac{1}{\sqrt{\lambda_0}} \right) \right] \\
\overset{(a)}{=} 4e^{-1-\sqrt{\lambda_0}} + 6\left[\left(\frac{1}{\lambda_0} + \frac{1}{\sqrt{\lambda_0}} \right)^2 + \frac{2}{\lambda_0}\left(\frac{2}{\lambda_0} + \frac{1}{\sqrt{\lambda_0}} \right) \right] e^{-1-\sqrt{\lambda_0}} \\
< \frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}} \left[\phi\left(1/\lambda_0 + 1/\sqrt{\lambda_0} - \left(\Delta^L(\lambda_0,\lambda_1) - \lambda_1 \right)/\sqrt{\eta+\gamma} \right) \right],
\end{aligned}
$$

where (a) follows from integration by parts.

For the third term in (12), utilizing condition (3), when n is sufficiently large, we have

$$
\begin{aligned}
2U_3 &= 2\mathbb{E}_{Y_i} \left[\mathbb{E}_{\mu_i,w_i} \left[\left(|Y_i-\mu_i|^2 + |\mu_i|^2 \right) \mathbb{I}(\sqrt{w_i}|Y_i-\mu_i| \geq \Delta(\lambda_0/\sqrt{w_i}, \lambda_1/\sqrt{w_i})) \mathbb{I}(w_i > \eta+\gamma) \mid Y_i \right] \right] \\
&\leq 2\mathbb{E}_{Y_i} \left[\mathbb{E}_{\mu_i,w_i} \left[\left(2|Y_i|^2 + 3|\mu_i|^2 \right) \mathbb{I}(w_i > \eta+\gamma) \mid Y_i \right] \right] \\
&\leq \left(4 + \frac{12}{\lambda_0^2} \right) \mathbb{P}\left(w_i > \eta+\gamma \right) \leq \widetilde{C_4} \frac{q}{n} \sqrt{\log\left(\frac{n}{q} \right)} \\
&\leq C_4 \frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}} \left[\phi\left(1/\lambda_0 + 1/\sqrt{\lambda_0} - \left(\Delta^L(\lambda_0,\lambda_1) - \lambda_1 \right)/\sqrt{\eta+\gamma} \right) \right].
\end{aligned}
$$

In (12), combining the bound on U_1, U_2, U_3, the risk for inactive coordinates will be bounded

$$
\begin{aligned}
&\mathbb{E}_{Y_i} \left[\mathbb{E}_{\mu_i,w_i} [(\tilde{\beta}_i - \beta_i^0)^2 \mid Y_i] \right] = 2U_1 + 2U_2 + 2U_3 + (2/\lambda_0^2) \\
&\leq C_5 \frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}} \left[\phi\left(1/\lambda_0 + 1/\sqrt{\lambda_0} - \left(\Delta^L(\lambda_0,\lambda_1) - \lambda_1 \right)/\sqrt{\eta+\gamma} \right) \right].
\end{aligned}
$$

Combining the risk for active and inactive coordinates, we obtain

$$
\begin{aligned}
&\mathbb{E}_{\boldsymbol{Y}}\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}[\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}^0\|_2^2 \mid \boldsymbol{Y}] \\
\leq & qC_3\left[\Delta^U(\lambda_0,\lambda_1)\right]^2+(n-q)C_5\frac{\Delta^L(\lambda_0,\lambda_1)-\lambda_1}{\sqrt{\eta+\gamma}}\left[\phi\left(1/\lambda_0+1/\sqrt{\lambda_0}-\left(\Delta^L(\lambda_0,\lambda_1)-\lambda_1\right)/\sqrt{\eta+\gamma}\right)\right] \\
= & qC_3\left[\Delta^U(\lambda_0,\lambda_1)\right]^2+(n-q)C_6\frac{\Delta^L(\lambda_0,\lambda_1)}{\sqrt{\eta+\gamma}}\exp\left\{-\frac{\left(1/\lambda_0+1/\sqrt{\lambda_0}-\left(\Delta^L(\lambda_0,\lambda_1)-\lambda_1\right)/\sqrt{\eta+\gamma}\right)^2}{2}\right\} \\
\leq & qC_3\left[\Delta^U(\lambda_0,\lambda_1)\right]^2+(n-q)C_7\Delta^L(\lambda_0,\lambda_1)\exp\left\{-\frac{\left(\Delta^L(\lambda_0,\lambda_1)\right)^2}{2(\eta+\gamma)}\right\} \\
\leq & qC_3\left[\Delta^U(\lambda_0,\lambda_1)\right]^2+(n-q)C_7\Delta^L(\lambda_0,\lambda_1)\frac{q}{n} \\
\leq & qC_8\left[\Delta^U(\lambda_0,\lambda_1)\right]^2.
\end{aligned}
$$

Then from the Markov's inequality, for any $M_n \to \infty$,

$$
\mathbb{E}_{\boldsymbol{Y}}\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}^0\|_2^2 > M_n q\log\left(\frac{n}{q}\right) \mid \boldsymbol{Y}\right) \leq \mathbb{E}_{\boldsymbol{Y}}\frac{\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}\left[\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}^0\|_2^2 \mid \boldsymbol{Y}\right]}{M_n q\log\frac{n}{q}} \leq \frac{qC_8\left[\Delta^U(\lambda_0,\lambda_1)\right]^2}{M_n q\log\frac{n}{q}},
$$

where $\frac{qC_3\left[\Delta^U(\lambda_0,\lambda_1)\right]^2}{M_n q\log\frac{n}{q}} \to 0$. This means when $\sigma=1$, for any $M_n \to \infty$,

$$
\mathbb{E}_{\boldsymbol{Y}}\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}^0\|_2^2 > M_n q\log\left(\frac{n}{q}\right) \mid \boldsymbol{Y}\right) \to 0.
$$

For a general fixed value $\sigma > 0$, notice that we can always rescale the model so that it becomes

$$
Y_i/\sigma = \beta_i^0/\sigma + N(0,1).
$$

And thus all previous analysis holds for the rescaled model. Thus, for any $M_n \to \infty$,

$$
\begin{aligned}
&\mathbb{E}_{\boldsymbol{Y}}\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}^0\|_2^2 > M_n q\log\left(\frac{n}{q}\right) \mid \boldsymbol{Y}\right) = \mathbb{E}_{\boldsymbol{Y}}\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\|\widetilde{\boldsymbol{\beta}}/\sigma-\boldsymbol{\beta}^0/\sigma\|_2^2 > \sigma^{-2}M_n q\log\left(\frac{n}{q}\right) \mid \boldsymbol{Y}\right) \\
\leq & \mathbb{E}_{\boldsymbol{Y}}\frac{\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}\left[\|\widetilde{\boldsymbol{\beta}}/\sigma-\boldsymbol{\beta}^0/\sigma\|_2^2 \mid \boldsymbol{Y}\right]}{\sigma^{-2}M_n q\log\left(\frac{n}{q}\right)} \to 0.
\end{aligned}
$$

A.2 Proof of Corollary 4.1

The condition (1) of Theorem 4.1 is satisfied. With $\alpha \geq 2$, the condition (2) also holds because when $w_i \sim \frac{1}{\alpha}\text{Gamma}(\alpha,1)$, $\frac{1}{w_i} \sim \alpha \times \text{Inverse-Gamma}(\alpha,1)$ and when $\boldsymbol{w} \sim n\text{Dir}(\alpha,\cdots,\alpha)$, $w_i \sim n \times \text{Beta}(\alpha,(n-1)\alpha)$. Both of them satisfy condition (2). So we only need to check condition (3).

When $\boldsymbol{w} \sim n\text{Dir}(\alpha, \cdots, \alpha)$, the following equation holds for any $t \geq \frac{\alpha+1}{\alpha}$,

$$
\begin{aligned}
&\mathbb{P}_{w_i}(w_i > t) = \mathbb{P}\left(\text{Beta}(\alpha, n\alpha-\alpha) > \frac{t}{n}\right) = \frac{\int_{t/n}^{1} v^{\alpha-1}(1-v)^{n\alpha-\alpha-1}dv}{\text{Beta}(\alpha, n\alpha-\alpha)} \\
&\overset{z=\alpha nv}{=} \frac{1}{\text{Beta}(\alpha, n\alpha-\alpha)}\int_{\alpha t}^{\alpha n}\left(\frac{z}{\alpha n}\right)^{\alpha-1}\left(1-\frac{z}{\alpha n}\right)^{n\alpha-\alpha-1}d\left(\frac{z}{\alpha n}\right) \\
&= \frac{1}{\text{Beta}(\alpha, n\alpha-\alpha)(\alpha n)^{\alpha}}\int_{\alpha t}^{\alpha n} z^{\alpha-1}\left[\left(1-\frac{1}{\alpha n/z}\right)^{-\alpha n/z+1}\right]^{\frac{n\alpha-\alpha-1}{1-\alpha n/z}}dz \\
&\overset{(a)}{\leq} \frac{1}{\text{Beta}(\alpha, n\alpha-\alpha)(\alpha n)^{\alpha}}\int_{\alpha t}^{\alpha n} z^{\alpha-1}\exp\left\{\frac{n\alpha-\alpha-1}{z-\alpha n}z\right\}dz \\
&\leq \frac{1}{\text{Beta}(\alpha, n\alpha-\alpha)(\alpha n)^{\alpha}}\int_{\alpha t}^{\alpha n} z^{\alpha-1}\exp\left\{\frac{n\alpha-\alpha-1}{\alpha t-\alpha n}z\right\}dz \\
&\overset{(b)}{\leq} \frac{1}{\text{Beta}(\alpha, n\alpha-\alpha)(\alpha n)^{\alpha}}\int_{\alpha t}^{\alpha n} z^{\alpha-1}e^{-z}dz = \frac{\Gamma(n\alpha)}{\Gamma(n\alpha-\alpha)(\alpha n)^{\alpha}\Gamma(\alpha)}\int_{\alpha t}^{\alpha n} z^{\alpha-1}e^{-z}dz \\
&\leq \frac{\Gamma(n\alpha)}{\Gamma(n\alpha-\alpha)(\alpha n)^{\alpha}}\mathbb{P}(\text{Gamma}(\alpha,1) > \alpha t) \leq \mathbb{P}(\text{Gamma}(\alpha,1) > \alpha t) \\
&\overset{(c)}{\leq} \frac{\mathbb{E}_{v\sim\text{Gamma}(\alpha,1)}e^{xv}}{e^{\alpha tx}} = \frac{(1-x)^{-\alpha}}{e^{\alpha tx}} = e^{\alpha(-\log(1-x)-tx)},
\end{aligned}
$$

where (a) uses the fact $\left(1-\frac{1}{x}\right)^{-x+1} \leq e$ for any $x > 0$. Inequality (b) uses $t \geq \frac{\alpha+1}{\alpha}$. Inequality (c) uses Chernoff bound and we need $x \in (0,1)$ for the above inequality to hold. Notice that when $w_i \sim \frac{1}{\alpha}\text{Gamma}(\alpha, 1)$, we can directly get

$$\mathbb{P}_{w_i}(w_i > t) = \mathbb{P}(\text{Gamma}(\alpha,1) > \alpha t) \leq e^{\alpha(-\log(1-x)-tx)}.$$

Setting $x = 1-\frac{1}{t}$ and $\alpha = \frac{\log[\frac{(1-\theta)\lambda_0}{\theta\lambda_1}]}{(t-1-\log t)(\eta+\gamma)}$, we have

$$\mathbb{P}_{w_i}(w_i > t) \leq e^{\alpha(-\log(1-x)-tx)} = e^{\alpha(\log(t)+1-t)} \asymp \frac{q}{n}. \tag{14}$$

Setting $t = \eta+\gamma$, we notice that $t = \eta+\gamma \geq \frac{\alpha+1}{\alpha}$ when n is sufficiently large and thus condition (3) always holds when n is sufficiently large. We can get Lemma 4.1 by applying Theorem 4.1.

Now let us consider the case when α depends on σ^2. From the proof of Theorem 4.1, we know that for any fixed $\sigma > 0$,

$$
\begin{aligned}
&\mathbb{E}_{\boldsymbol{Y}}\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}^0\|_2^2 > M_n q\log\left(\frac{n}{q}\right) \mid \boldsymbol{Y}\right) = \mathbb{E}_{\boldsymbol{Y}}\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\|\widetilde{\boldsymbol{\beta}}/\sigma-\boldsymbol{\beta}^0/\sigma\|_2^2 > \sigma^{-2}M_n q\log\left(\frac{n}{q}\right) \mid \boldsymbol{Y}\right) \\
&\leq \mathbb{E}_{\boldsymbol{Y}}\frac{\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}\left[\|\widetilde{\boldsymbol{\beta}}/\sigma-\boldsymbol{\beta}^0/\sigma\|_2^2 \mid \boldsymbol{Y}\right]}{\sigma^{-2}M_n q\log\left(\frac{n}{q}\right)} \overset{(a)}{\lesssim} \frac{\sigma^2}{M_n q\log\frac{n}{q}}q\log\left(\frac{n}{q}\right)\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right),
\end{aligned}
$$

where (a) follows from the fact that the dominating term in the upper bound for risk $\mathbb{E}_{\boldsymbol{Y}}\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\|\widetilde{\boldsymbol{\beta}}/\sigma-\boldsymbol{\beta}^0/\sigma\|_2^2>M_n q\log\left(\frac{n}{q}\right)\mid\boldsymbol{Y}\right)$ comes from $4q[\Delta^U(\lambda_0,\lambda_1)]^2\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right)$. When $w_i\sim\frac{1}{\alpha}\text{Gamma}(\alpha,1)$,we have $\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right)=1+\frac{1}{\alpha-1}$. Thus

$$\mathbb{E}_{\boldsymbol{Y}}\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}^0\|_2^2>M_n q\log\left(\frac{n}{q}\right)\mid\boldsymbol{Y}\right)\lesssim\frac{1}{M_n}\left(\sigma^2+\frac{\sigma^2}{\alpha-1}\right).$$

An ideal choice of α should thus satisfy $\alpha-1\propto\sigma^2$. We thereby suggest choosing $\alpha\gtrsim\sigma^2\log\left[\frac{(1-\theta)\lambda_0}{\theta\lambda_1}\right]$ when $\boldsymbol{w}\sim n\times\text{Dir}(\alpha,\cdots,\alpha)$, observing that $w_i\sim n\times\text{Beta}(\alpha,(n-1)\alpha)$ and thus

$$\mathbb{E}_{w_i}\left(\frac{1}{w_i}\right)=\frac{1}{n}\frac{\alpha+(n-1)\alpha-1}{\alpha-1}=1+\frac{n-1}{n(\alpha-1)}.$$

A.3 Explanation of Remarks below Corollary 4.1

We utilize the notation introduced in Section A.1.1. Here we want to show that for $\boldsymbol{w}\sim n\times\text{Dir}(\alpha,\cdots,\alpha)$ where $0<\alpha<2$ is a fixed constant, the risk for BB-SSL arising from (14) can be arbitrarily large.

From the proof of Theorem 4.1, we know that for active coordinate i, when n is sufficiently large,

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{Y}}\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}[\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{Y}\|_2^2\mid\boldsymbol{Y}]\\
\geq&\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[(\widetilde{\beta}_i-Y_i)^2\mathbb{I}\left(\sqrt{w_i}(Y_i-\mu_i)>\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right)\mathbb{I}\left(|Y_i-\mu_i|\geq w_i^{-1}\lambda^*(\tilde{\beta}_i-\mu_i)\right)\mid Y_i\right]\right]\\
\stackrel{(a)}{=}&\mathbb{E}_{Y_i}\left[\mathbb{E}_{\mu_i,w_i}\left[\left(w_i^{-1}\lambda^*(\tilde{\beta}_i-\mu_i)\right)^2\mathbb{I}\left(\sqrt{w_i}(Y_i-\mu_i)>\Delta(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})\right)\right.\right.\\
&\qquad\left.\left.\mathbb{I}\left(|Y_i-\mu_i|\geq w_i^{-1}\lambda^*(\tilde{\beta}_i-\mu_i)\right)\mid Y_i\right]\right]\\
\geq&\lambda_1^2\mathbb{E}_{Y_i,\mu_i}\left[\mathbb{E}_{w_i}\left[\frac{1}{w_i^2}\mathbb{I}\left(w_i>\frac{\Delta^2(\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})}{(Y_i-\mu_i)^2}\right)\mathbb{I}\left(w_i\geq\frac{\lambda_0}{|Y_i-\mu_i|}\right)\right.\right.\\
&\qquad\left.\left.\mathbb{I}\left(g(0;\lambda_0 w_i^{-1/2},\lambda_1 w_i^{-1/2})>0,\ \lambda_0 w_i^{-1/2}-\lambda_1 w_i^{-1/2}>2\right)\mid Y_i,\mu_i\right]\right]\\
\stackrel{(b)}{\geq}&\lambda_1^2\mathbb{E}_{Y_i,\mu_i}\left[\mathbb{E}_{w_i}\left[\frac{1}{w_i^2}\mathbb{I}\left(w_i>\frac{[\Delta^U(\lambda_0,\lambda_1)+\lambda_1(w_i^{-1/2}-1)]^2}{(Y_i-\mu_i)^2}\right)\mathbb{I}\left(w_i\geq\frac{\lambda_0}{|Y_i-\mu_i|}\right)\mathbb{I}\left(w_i<B_n\right)\mid Y_i,\mu_i\right]\right]\\
\geq&\lambda_1^2\mathbb{E}_{Y_i,\mu_i}\left[\mathbb{E}_{w_i}\left[\frac{1}{w_i^2}\mathbb{I}\left(w_i>\frac{[\Delta^U(\lambda_0,\lambda_1)+\lambda_1(\sqrt{|Y_i-\mu_i|/\lambda_0}-1)]^2}{(Y_i-\mu_i)^2}\right)\right.\right.\\
&\qquad\left.\left.\mathbb{I}\left(w_i\geq\frac{\lambda_0}{|Y_i-\mu_i|}\right)\mathbb{I}\left(w_i<B_n\right)\mid Y_i,\mu_i\right]\right]\\
\geq&\lambda_1^2\mathbb{E}_{Y_i,\mu_i}\left[\mathbb{E}_{w_i}\left[\frac{1}{w_i^2}\mathbb{I}\left(w_i>C(Y_i,\mu_i)\right)\mathbb{I}\left(w_i<B_n\right)\mid Y_i,\mu_i\right]\right],
\end{aligned}\tag{15}$$

where (a) follows from Equation (6), (b) follows from Lemma A.2, and we denote

$$B_n = \min\left\{\frac{(\lambda_0-\lambda_1)^2(1-p^*(0;\lambda_0,\lambda_1))^2}{2\log[1/p^*(0;\lambda_0,\lambda_1)]}, \frac{(\lambda_0-\lambda_1)^2}{4}\right\},$$

$$C(Y_i,\mu_i) = \max\left\{\frac{[\Delta^U(\lambda_0,\lambda_1)+\lambda_1(\sqrt{|Y_i-\mu_i|/\lambda_0}-1)]^2}{(Y_i-\mu_i)^2}, \frac{\lambda_0}{|Y_i-\mu_i|}\right\}.$$

Notice that $n\text{Beta}(\alpha, n\alpha-\alpha) \xrightarrow{\text{d}} \frac{1}{\alpha}\text{Gamma}(\alpha,1)$ and as $n\to\infty$,

$$\mathbb{E}_{w_i}\left[\frac{1}{w_i^2}\mathbb{I}\left(B_n > w_i > C(Y_i,\mu_i)\right) \mid Y_i,\mu_i\right] \to \mathbb{E}_{v\sim\text{Gamma}(\alpha,1)}\left[\frac{\alpha^2}{v^2}\mathbb{I}\left(\alpha B_n > v > \alpha C(Y_i,\mu_i)\right) \mid Y_i,\mu_i\right],$$

$$\mathbb{E}_{v\sim\text{Gamma}(\alpha,1)}\left[\frac{\alpha^2}{v^2}\mathbb{I}\left(\alpha B_n > v > \alpha C(Y_i,\mu_i)\right) \mid Y_i,\mu_i\right] \le \frac{1}{C^2(Y_i,\mu_i)},$$

$$\frac{1}{C^2(Y_i,\mu_i)} \le \frac{|Y_i-\mu_i|^2}{\lambda_0^2},$$

where $\frac{|Y_i-\mu_i|^2}{\lambda_0^2}$ is integrable. Thus, direct application of the Dominated Convergence Theorem shows that as $n\to\infty$,

$$\begin{aligned}
&\mathbb{E}_{Y_i,\mu_i}\left[\mathbb{E}_{w_i}\left[\frac{1}{w_i^2}\mathbb{I}\left(w_i > C(Y_i,\mu_i)\right)\mathbb{I}\left(w_i < B_n\right) \mid Y_i,\mu_i\right]\right] \\
\to&\mathbb{E}_{Y_i,\mu_i}\left[\mathbb{E}_{v\sim\text{Gamma}(\alpha,1)}\left[\frac{\alpha^2}{v^2}\mathbb{I}\left(\alpha B_n > v > \alpha C(Y_i,\mu_i)\right) \mid Y_i,\mu_i\right]\right].
\end{aligned}$$

Plugging into Equation (15), we know that when n is sufficiently large,

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{Y}}\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}[\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{Y}\|_2^2 \mid \boldsymbol{Y}] \\
\ge&\frac{1}{2}\lambda_1^2\mathbb{E}_{Y_i,\mu_i}\left[\mathbb{E}_{v\sim\text{Gamma}(\alpha,1)}\left[\frac{\alpha^2}{v^2}\mathbb{I}\left(\alpha B_n > v > \alpha C(Y_i,\mu_i)\right) \mid Y_i,\mu_i\right]\right] \\
=&\frac{1}{2}\lambda_1^2\alpha^2\mathbb{E}_{Y_i,\mu_i}\left[\mathbb{E}_{z\sim\text{Inv-Gamma}(\alpha,1)}\left[z^2\mathbb{I}\left(\frac{1}{\alpha B_n} < z < \frac{1}{C(Y_i,\mu_i)\alpha}\right) \mid Y_i,\mu_i\right]\right].
\end{aligned}$$

For all constants M, m which satisfy $M > 2m > 0$, we have

$$\begin{aligned}
&\mathbb{E}_{z\sim\text{Inv-Gamma}(\alpha,1)}\left[z^2\mathbb{I}(m<z<M)\right] = \frac{1}{\Gamma(\alpha)}\int_m^M z^2\frac{1}{z^{\alpha+1}}e^{-1/z}dz = \frac{1}{\Gamma(\alpha)}\int_m^M z^{1-\alpha}e^{-1/z}dz \\
\overset{x=1/z}{=}&\frac{1}{\Gamma(\alpha)}\int_{1/M}^{1/m} x^{\alpha-3}e^{-x}dx > \frac{1}{\Gamma(\alpha)}\int_{1/M}^{2/M} x^{\alpha-3}e^{-x}dx = \frac{M^{2-\alpha}\exp\{-2/M\}}{\Gamma(\alpha)2^{3-\alpha}}.
\end{aligned}$$

We set

$$M = \frac{1}{C(Y_i,\mu_i)\alpha}, \quad m = \frac{1}{\alpha B_n}.$$

Then, suppose $\beta_i^0 > 2$, when n is sufficiently large, we have

$$
\begin{aligned}
&\mathbb{E}_{\boldsymbol{Y}}\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}[\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{Y}\|_2^2 \mid \boldsymbol{Y}] \\
\geq&\frac{1}{2}\lambda_1^2\alpha^2\mathbb{E}_{Y_i,\mu_i}\left[\mathbb{E}_{z\sim\text{Inv-Gamma}(\alpha,1)}\left[z^2\mathbb{I}(m<z<M)\mathbb{I}\left(M>2m\right) \mid Y_i,\mu_i\right]\right] \\
\geq&\frac{1}{2}\lambda_1^2\alpha^2\mathbb{E}_{Y_i,\mu_i}\left[\frac{M^{2-\alpha}\exp\left\{-2/M\right\}}{\Gamma(\alpha)2^{3-\alpha}}\mathbb{I}\left(M>2m\right)\right] \\
=&\frac{\lambda_1^2\alpha^\alpha}{2^{4-\alpha}\Gamma(\alpha)}\mathbb{E}_{Y_i,\mu_i}\left[\left(\frac{1}{C(Y_i,\mu_i)}\right)^{2-\alpha}\exp\left\{-2C(Y_i,\mu_i)\alpha\right\}\mathbb{I}\left(C(Y_i,\mu_i)<B_n/2\right)\right] \\
\geq&\frac{\lambda_1^2\alpha^\alpha}{2^{4-\alpha}\Gamma(\alpha)}\mathbb{E}_{Y_i,\mu_i}\left[\left(\frac{1}{C(Y_i,\mu_i)}\right)^{2-\alpha}\exp\left\{-2C(Y_i,\mu_i)\alpha\right\}\mathbb{I}\left(C(Y_i,\mu_i)<B_n/2\right)\mathbb{I}\left(|Y_i-\beta_i^0|\leq 1\right)\mathbb{I}(|\mu_i|<1)\right] \\
\geq&\frac{\lambda_1^2\alpha^\alpha}{2^{4-\alpha}\Gamma(\alpha)}\mathbb{E}_{Y_i,\mu_i}\left[\left(\frac{1}{C_n}\right)^{2-\alpha}\exp\left\{-2C_n\alpha\right\}\mathbb{I}\left(C_n<B_n/2\right)\mathbb{I}\left(|Y_i-\beta_i^0|\leq 1\right)\mathbb{I}(|\mu_i|<1)\right] \\
=&\frac{\lambda_1^2\alpha^\alpha}{2^{4-\alpha}\Gamma(\alpha)}\left(1/C_n\right)^{2-\alpha}\exp\left\{-2C_n\alpha\right\}\mathbb{I}\left(C_n<B_n/2\right)\mathbb{P}\left(|N(0,1)|\leq 1\right)\mathbb{P}(|\mu_i|<1) \\
=&C\frac{\lambda_1^2\alpha^\alpha}{2^{4-\alpha}\Gamma(\alpha)}\left(1/C_n\right)^{2-\alpha}\exp\left\{-2C_n\alpha\right\}\mathbb{I}\left(C_n<B_n/2\right),
\end{aligned}
$$

where

$$
C_n = \max\left\{\frac{[\Delta^U(\lambda_0,\lambda_1)+\lambda_1(\sqrt{(\beta_i^0+2)/\lambda_0}-1)]^2}{(\beta_i^0-2)^2}, \frac{\lambda_0}{|\beta_i^0-2|}\right\}.
$$

Notice that $C_n < B_n/2$ is always satisfied when n is sufficiently large, so we know that when n is sufficiently large,

$$
\mathbb{E}_{\boldsymbol{Y}}\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}[\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{Y}\|_2^2 \mid \boldsymbol{Y}] \geq C\frac{\lambda_1^2\alpha^\alpha}{2^{4-\alpha}\Gamma(\alpha)}\left(1/C_n\right)^{2-\alpha}\exp\left\{-2C_n\alpha\right\},
$$

where the lower bound depends on β_i^0 through C_n. For any fixed, sufficiently large n, when β_i^0 becomes larger and larger, C_n becomes smaller and smaller, and as $\beta_i^0 \to \infty$, $C_n \to 0$, and the lower bound goes to infinity, which implies that $\mathbb{E}_{\boldsymbol{Y}}\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}[\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{Y}\|_2^2 \mid \boldsymbol{Y}] \to \infty$ as $\beta_i^0 \to \infty$.

Thus, we conclude that the risk $\mathbb{E}_{\boldsymbol{Y}}\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}[\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{Y}\|_2^2 \mid \boldsymbol{Y}]$ depends on the truth $\boldsymbol{\beta^0}$ and for any fixed, sufficiently large n, when one of the coordinates β_i^0 becomes arbitrarily large, the risk can also be arbitrarily large.

A.4 Proof of Theorem 4.2 and Theorem 4.3

A.4.1 Definitions and Lemmas used in Theorem 4.2 and Theorem 4.3

We rewrite model (1) in a matrix form $\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ and denote

$$
pen(\boldsymbol{\beta} \mid \theta) = \log\left[\frac{\pi(\boldsymbol{\beta} \mid \theta)}{\pi(\mathbf{0}_p \mid \theta)}\right],
$$

where $\mathbf{0}_p \in \mathbb{R}^p$ is a vector of all 0's. We write $\boldsymbol{W} = \operatorname{diag}(\sqrt{w_1}, \cdots, \sqrt{w_n}) \in \mathbb{R}^{n\times n}$ and denote with

$$Q(\boldsymbol{\beta}) = -\frac{1}{2\sigma^2}\|\boldsymbol{W}\boldsymbol{Y} - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\mu} - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta)$$

and with

$$\widehat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta}} Q(\boldsymbol{\beta}). \tag{16}$$

Notice that the BB-SSL solution $\widetilde{\boldsymbol{\beta}}$ satisfies $\widetilde{\boldsymbol{\beta}} = \widehat{\boldsymbol{\beta}} + \boldsymbol{\mu}$. The matrix norm $\|\cdot\|_a$ is defined as $\|\boldsymbol{X}\|_a = \sup_{\boldsymbol{\beta}} \frac{\|\boldsymbol{X}\boldsymbol{\beta}\|_a}{\|\boldsymbol{\beta}\|_a}$ where $\|\cdot\|_a$ is the vector a-norm. Write $\boldsymbol{\Theta} = \widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0$. We use $\|\cdot\|_1$ to denote the vector 1-norm. We define

$$\Delta = \inf_{t>0}[nt/2 - \sigma^2\rho(t \mid \theta)/t], \tag{17}$$

with $\rho(t \mid \theta)$ defined as

$$\rho(t \mid \theta) = -\lambda_1|t| + \log[p^*(0)/p^*(t)],$$

where $p^*(t)$ is equal to the $p^*(t; \lambda_0, \lambda_1)$ defined in Equation (1). We fix λ_0, λ_1 in $p^*(t; \lambda_0, \lambda_1)$ and thus simply write $p^*(t)$. Notice that

$$pen(\boldsymbol{\beta} \mid \theta) = \sum_{j=1}^{p} \rho(\beta_j \mid \theta).$$

The proofs below use similar ideas and techniques as Ročková and George (2018). We first outline auxiliary Lemmas and then prove them later in this section.

Definition A.1. *Let $\tilde{\eta} \in (0, 1]$. We say that $\boldsymbol{X}$ satisfies the $\tilde{\eta}$-null consistency ($\tilde{\eta}$-NC) condition with a penalty function $pen(\boldsymbol{\beta} \mid \theta)$ if*

$$\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2}\|\boldsymbol{\epsilon}/\tilde{\eta} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta) \right\} = \mathbf{0}_p.$$

Lemma A.4. *Under Condition (5) in Theorem 4.2, we have*

$$\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0} \left(\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2}\|\boldsymbol{\epsilon}/\tilde{\eta} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta) \right\} = \mathbf{0}_p \right) = 1,$$

i.e. $\boldsymbol{X}$ satisfies the $\tilde{\eta}$-NC condition with probability approaching 1.

Lemma A.5. *Under Conditions (1)-(5) in Theorem 4.2, given that $\|\boldsymbol{\epsilon}\|_\infty \lesssim \sqrt{\log n}$ and $\boldsymbol{X}$ satisfies $\tilde{\eta}$-NC condition, we have*

$$\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}} \left(\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2}\|\boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta) \right\} = \mathbf{0}_p \,\Big|\, \boldsymbol{\epsilon} \right) = 1,$$

where $\eta^* = \max\left\{\widetilde{\eta} + C_n \frac{\|\boldsymbol{X}\|}{\lambda_1}, \frac{\widetilde{\eta}}{m}\right\}$ and C_n is any sequence that satisfies $C_n \to \infty$.

Lemma A.6. *If, for $\eta^* \in (0, 1]$,*

$$\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2} \|\boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta) \right\} = \mathbf{0}_p, \tag{18}$$

then $\widehat{\boldsymbol{\beta}}$ defined in (16) lies inside a cone

$$C(\eta^*; \boldsymbol{\beta}) = \{\boldsymbol{\Theta} \in \mathbb{R}^p : (\eta^* + 1) pen(\boldsymbol{\Theta}_S \mid \theta) \leq (1 - \eta^*) pen(\boldsymbol{\Theta}_{S^C} \mid \theta)\}$$

with high probability, where S is the active set of β_j^0's, $S^C = \{1, 2, \cdots, p\} \backslash S$ and

$$pen(\boldsymbol{\Theta}_S \mid \theta) = \sum_{j \in S} \rho(\hat{\beta}_j - \beta_j^0 \mid \theta) \quad \text{and} \quad pen(\boldsymbol{\Theta}_{S^C} \mid \theta) = \sum_{j \in S^C} \rho(\hat{\beta}_j - \beta_j^0 \mid \theta),$$

with $\hat{\beta}_j$ the j-th dimension of $\hat{\boldsymbol{\beta}}$.

Lemma A.7. *If (18) holds and $\max_{1 \leq i \leq n} w_i \leq M$, then*

$$\|\boldsymbol{X}^T \boldsymbol{W}^2 (\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})\|_\infty \leq \sqrt{M} \Delta^U \eta^*,$$

where $\Delta^U = \sqrt{2n\sigma^2 \log[1/p^*(0)]} + \sigma^2 \lambda_1$.

Definition A.2. *The minimal restricted eigenvalue is defined as*

$$c(\eta^*; \boldsymbol{\beta}) = \inf_{\boldsymbol{\Theta} \in \mathbb{R}^p} \left\{ \frac{\|\boldsymbol{X}\boldsymbol{\Theta}\|_2}{\|\boldsymbol{X}\|_2 \|\boldsymbol{\Theta}\|_2} : \boldsymbol{\Theta} \in C(\eta^*; \boldsymbol{\beta}) \right\}.$$

Definition A.3. *The compatibility number $\phi(C)$ of vectors in cone $C \subset \mathbb{R}^p$ is defined as*

$$\phi(C) = \inf_{\boldsymbol{\Theta} \in \mathbb{R}^p} \left\{ \frac{\|\boldsymbol{X}\boldsymbol{\Theta}\|_2 \|\boldsymbol{\Theta}\|_0^{1/2}}{\|\boldsymbol{X}\|_2 \|\boldsymbol{\Theta}\|_1} : \boldsymbol{\Theta} \in C(\eta^*; \boldsymbol{\beta}) \right\}.$$

A.4.2 Proof of Lemma A.4

This Lemma is a direct consequence of Proposition 3 of Zhang and Zhang (2012). This Proposition says the following. In a regression model (1), suppose $\delta \in (0, 1]$ and $\xi_0 > 0$, and

$$-\frac{\sigma^2}{n} \rho(t \mid \theta) \geq \min\left\{ \frac{\Delta^2}{2n^2}, \frac{\Delta |t|}{n} \right\} \quad \text{with} \quad \frac{\Delta}{n} \geq (1 + \xi_0) \frac{\sigma}{\widetilde{\eta}} n^{-1/2} \left(1 + \sqrt{2 \log(2p/\delta)} \right). \tag{19}$$

Then, the $\widetilde{\eta}$-NC condition is satisfied with probability at least $2 - e^{\delta/2} - \exp\left\{ -n \left(1 - 1/\sqrt{2}\right)^2 \right\}$, provided that

$$\max\left\{\lambda_{\max}^{1/2}\left(\frac{\boldsymbol{X}_B^\top \boldsymbol{P}_A \boldsymbol{X}_B}{n}\right) : B\cap A=\emptyset, |A| = \text{rank}(\boldsymbol{P}_A) = |B| = k,\right.$$
$$\left. k(1+\xi_0)^2\left(1+\sqrt{2\log(2p/\delta)}\right)^2 \le 2n\right\} \le \xi_0. \tag{20}$$

From the Condition (5), we know that (20) holds with $\delta = 2p^{-1/4}$. Thus, we only need to show (19) holds. Denote

$$f_1(t) = \frac{1}{t}\log\frac{p^*(t)}{p^*(0)} - \frac{1}{\sigma}\sqrt{2n\log[1/p^*(0)]}, \quad t>0,$$
$$f_2(t) = \lambda_1 t + \log p^*(t) - \frac{\sigma^2\lambda_1^2}{2n} - \lambda_1\sqrt{(2\sigma^2/n)\log[1/p^*(0)]}.$$

Notice that $\Delta \le \sqrt{2n\sigma^2\log[1/p^*(0)]} + \sigma^2\lambda_1$ from Proposition 5 in Moran et al. (2019) and (19) holds trivially when $t=0$. Thus, in order to show (19), we only need to show that

$$\max\{f_1(t), f_2(t)\} \ge 0, \quad \forall t>0,$$
$$\Delta \ge (1+\xi_0)\frac{\sigma}{\widetilde{\eta}}n^{1/2}\left(1+\sqrt{(5/2)\log p}\right).$$

Notice that

$$\lim_{t\to 0+}\frac{\partial f_1(t)}{\partial t} = \lim_{t\to 0+}\left[-\frac{1}{t^2}\log\frac{p^*(t)}{p^*(0)} + \frac{1}{t}\frac{\partial \log p^*(t)}{\partial t}\right]$$
$$\overset{(a)}{=} \lim_{t\to 0+}\left[-\frac{1}{2t}\frac{\partial\log p^*(t)}{\partial t} + \frac{1}{t}\frac{\partial\log p^*(t)}{\partial t}\right] = \lim_{t\to 0+}\frac{1}{2t}\frac{\partial \log p^*(t)}{\partial t} > 0$$

where (a) follows from L'Hopital's rule. It implies that there exists a positive constant $t_0>0$ such that, for all $t\in(0,t_0]$, $\frac{\partial f_1(t)}{\partial t}>0$ and thus we have

$$f_1(t) > \lim_{t\to 0+} f_1(t) = \lim_{t\to 0+}\left[\frac{1}{t}\log\frac{p^*(t)}{p^*(0)} - \frac{1}{\sigma}\sqrt{2n\log[1/p^*(0)]}\right]$$
$$\overset{(a)}{=} \lim_{t\to 0+}\frac{\partial\log p^*(t)}{\partial t} - \frac{1}{\sigma}\sqrt{2n\log[1/p^*(0)]}$$
$$= \frac{\lambda_0-\lambda_1}{\theta\lambda_1/[(1-\theta)\lambda_0]+1} - \frac{1}{\sigma}\sqrt{2n\log[1+(1-\theta)\lambda_0/(\theta\lambda_1)]}$$
$$\gtrsim p - \sqrt{n\log p}$$

where (a) follows from L'Hopital's rule. Thus, $f_1(t)>0$ for all $t\in(0,t_0]$, when n is sufficiently large. For $t\ge t_0$,

$$f_2(t) \ge f_2(t_0) = \lambda_1 t_0 + \log p^*(t_0) - \frac{\sigma^2\lambda_1^2}{2n} - \lambda_1\sqrt{2(\sigma^2/n)\log[1/p^*(0)]} > 0$$

is always satisfied when n is sufficiently large. Combining the above, we know that (when n is sufficiently large)

$$\max\{f_1(t), f_2(t)\} \geq 0, \quad \forall t > 0.$$

Recall that Proposition 5 of Moran et al. (2019) implies that

$$\lim_{n\to\infty} \frac{\Delta}{\sqrt{2n\sigma^2(\eta+\gamma)\log p}} = 1,$$

$$\lim_{n\to\infty} \frac{(1+\xi_0)\frac{\sigma}{\widetilde{\eta}}n^{1/2}\left(1+\sqrt{(5/2)\log p}\right)}{\sqrt{2n\sigma^2(\eta+\gamma)\log p}} = \frac{1+\xi_0}{\widetilde{\eta}}\sqrt{\frac{5}{4(\eta+\gamma)}}.$$

Thus, if $(1+\xi_0)/\widetilde{\eta} < \sqrt{4(\eta+\gamma)/5}$, when n is sufficiently large, we know that $\Delta \geq (1+\xi_0)\frac{\sigma}{\widetilde{\eta}}n^{1/2}\left(1+\sqrt{(5/2)\log p}\right)$. To sum up, if $(1+\xi_0)/\widetilde{\eta} < \sqrt{4(\eta+\gamma)/5}$ then (for n is sufficiently large)

$$\mathbb{P}_{\boldsymbol{\beta}_0}\left(\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p}\left\{-\frac{1}{2}\|\boldsymbol{\epsilon}/\tilde{\eta} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta}\mid\theta)\right\} = \mathbf{0}_p\right) \geq 2 - e^{1/p^{1/4} - e^{-n(1-1/\sqrt{2})^2}},$$

and thus

$$\lim_{n\to\infty}\mathbb{P}_{\boldsymbol{\beta}_0}\left(\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p}\left\{-\frac{1}{2}\|\boldsymbol{\epsilon}/\tilde{\eta} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta}\mid\theta)\right\} = \mathbf{0}_p\right) = 1.$$

A.4.3 Proof of Lemma A.5

On the event $\|\boldsymbol{\epsilon}\|_\infty \lesssim \sqrt{\log n}$, since $\mathbb{E}w_i = 1$, we have

$$\begin{aligned}
&\operatorname{Var}_{\boldsymbol{\mu},\boldsymbol{w}}\left[(\boldsymbol{\epsilon}-\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}^2\boldsymbol{X}\boldsymbol{\beta}\mid\boldsymbol{\epsilon}\right]=\operatorname{Var}_{\boldsymbol{\mu},\boldsymbol{w}}\left(\sum_i(\epsilon_i-\boldsymbol{x}_i^T\boldsymbol{\mu})w_i\boldsymbol{x}_i^T\boldsymbol{\beta}\mid\boldsymbol{\epsilon}\right)\\
=&\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}\left[\left(\sum_i(w_i-1)\epsilon_i\boldsymbol{x}_i^T\boldsymbol{\beta}-\sum_i w_i\boldsymbol{x}_i^T\boldsymbol{\mu}\boldsymbol{x}_i^T\boldsymbol{\beta}\right)^2\mid\boldsymbol{\epsilon}\right]\\
=&\mathbb{E}_{\boldsymbol{w}}\left[\left(\sum_i(w_i-1)\epsilon_i\boldsymbol{x}_i^T\boldsymbol{\beta}\right)^2\mid\boldsymbol{\epsilon}\right]+\mathbb{E}_{\boldsymbol{\mu},\boldsymbol{w}}\left[\sum_i w_i\boldsymbol{x}_i^T\boldsymbol{\mu}\boldsymbol{x}_i^T\boldsymbol{\beta}\right]^2\\
=&\sum_i\operatorname{Var}(w_i)(\epsilon_i\boldsymbol{x}_i^T\boldsymbol{\beta})^2+\sum_{i\neq j}\operatorname{Cov}(w_i,w_j)\epsilon_i\boldsymbol{x}_i^T\boldsymbol{\beta}\epsilon_j\boldsymbol{x}_j^T\boldsymbol{\beta}+\sum_{i,j}\boldsymbol{x}_i^T\boldsymbol{\beta}\boldsymbol{x}_j^T\boldsymbol{\beta}\boldsymbol{x}_i^T\mathbb{E}(\boldsymbol{\mu}\boldsymbol{\mu}^T)\boldsymbol{x}_j\mathbb{E}\left(w_iw_j\right)\\
\overset{(a)}{\leq}&\frac{C_1}{\log n}\sum_i(\epsilon_i\boldsymbol{x}_i^T\boldsymbol{\beta})^2+\frac{C_2}{n\log n}\sum_{i\neq j}\frac{1}{2}\left[(\epsilon_i\boldsymbol{x}_i^T\boldsymbol{\beta})^2+(\epsilon_j\boldsymbol{x}_j^T\boldsymbol{\beta})^2\right]+\mathbb{E}\left(w_iw_j\right)\frac{2}{\lambda_0^2}\sum_{i,j}\boldsymbol{x}_i^T\boldsymbol{\beta}\boldsymbol{x}_j^T\boldsymbol{\beta}(\boldsymbol{x}_i^T\boldsymbol{x}_j)\\
\leq&\frac{C_1}{\log n}\sum_i(\epsilon_i\boldsymbol{x}_i^T\boldsymbol{\beta})^2+\frac{C_2}{n\log n}(n-1)\sum_i(\epsilon_i\boldsymbol{x}_i^T\boldsymbol{\beta})^2+C\mathbb{E}\left(w_iw_j\right)\frac{2\max_{i\neq j}|\boldsymbol{x}_i^T\boldsymbol{x}_j|}{\lambda_0^2}(\sum_i|\boldsymbol{x}_i^T\boldsymbol{\beta}|)^2\\
\overset{(b)}{\leq}&\frac{\widetilde{C}_1}{\log n}\|\boldsymbol{\epsilon}\|_\infty^2\|\boldsymbol{X}\boldsymbol{\beta}\|_2^2+C_3\frac{2n\max_{i\neq j}|\boldsymbol{x}_i^T\boldsymbol{x}_j|}{\lambda_0^2}\sum_i(\boldsymbol{x}_i^T\boldsymbol{\beta})^2\\
\overset{(d)}{\leq}&\widetilde{C}_3\|\boldsymbol{X}\boldsymbol{\beta}\|_2^2,
\end{aligned}$$

where (a) uses the assumption (4) in Theorem 4.2, the fact $ab\leq\frac{1}{2}(a^2+b^2)$ and $\mu\sim$ Spike. The inequality (b) follows from $\|\boldsymbol{\epsilon}^T\boldsymbol{X}\boldsymbol{\beta}\|_2^2\leq\|\boldsymbol{\epsilon}\|_\infty^2\|\boldsymbol{X}\boldsymbol{\beta}\|_2^2$ and the Cauchy-Schwarz inequality $(\sum_i|\boldsymbol{x}_i^T\boldsymbol{\beta}|)^2\leq n\sum_i(\boldsymbol{x}_i^T\boldsymbol{\beta})^2$. The inequality (d) follows from the fact that $\lambda_0\asymp p^\gamma$ where $\gamma\geq 1$, $\|\boldsymbol{\epsilon}\|_\infty\lesssim\sqrt{\log n}$ and $\max_{i\neq j}|\boldsymbol{x}_i^T\boldsymbol{x}_j|\lesssim\lambda_0^2/n$. Thus, from the Markov's inequality, on the event that $\|\boldsymbol{\epsilon}\|_\infty\lesssim\sqrt{\log n}$, we have for any $t>0$,

$$\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(|(\boldsymbol{\epsilon}-\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}^2\boldsymbol{X}\boldsymbol{\beta}-\boldsymbol{\epsilon}^T\boldsymbol{X}\boldsymbol{\beta}|>t\mid\boldsymbol{\epsilon}\right)\leq\frac{\operatorname{Var}_{\boldsymbol{\mu},\boldsymbol{w}}\left[(\boldsymbol{\epsilon}-\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}^2\boldsymbol{X}\boldsymbol{\beta}\mid\boldsymbol{\epsilon}\right]}{t^2}\leq\frac{\widetilde{C}_3\|\boldsymbol{X}\boldsymbol{\beta}\|_2^2}{t^2}.$$

Set $t=C_n\|\boldsymbol{X}\boldsymbol{\beta}\|_2$ where $C_n\to\infty$, we have

$$\lim_{n\to\infty}\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(|(\boldsymbol{\epsilon}-\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}^2\boldsymbol{X}\boldsymbol{\beta}-\boldsymbol{\epsilon}^T\boldsymbol{X}\boldsymbol{\beta}|>C_n\|\boldsymbol{X}\boldsymbol{\beta}\|_2\mid\boldsymbol{\epsilon}\right)=0. \tag{21}$$

When $|(\boldsymbol{\epsilon}-\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}^2\boldsymbol{X}\boldsymbol{\beta}-\boldsymbol{\epsilon}^T\boldsymbol{X}\boldsymbol{\beta}|\leq C_n\|\boldsymbol{X}\boldsymbol{\beta}\|_2$, we have

$$(\boldsymbol{\epsilon}-\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}^2\boldsymbol{X}\boldsymbol{\beta}\leq|\boldsymbol{\epsilon}^T\boldsymbol{X}\boldsymbol{\beta}|+C_n\|\boldsymbol{X}\boldsymbol{\beta}\|_2. \tag{22}$$

Notice that

$$\|\boldsymbol{X}\boldsymbol{\beta}\|_2+\frac{\|\boldsymbol{X}\|_2}{\lambda_1}pen(\boldsymbol{\beta}\mid\theta)\overset{(e)}{\leq}\|\boldsymbol{X}\|_2\times\|\boldsymbol{\beta}\|_2+\frac{\|\boldsymbol{X}\|_2}{\lambda_1}(-\lambda_1\|\boldsymbol{\beta}\|_1)\leq 0,$$

where (e) follows from $pen(\boldsymbol{\beta} \,|\, \theta) = -\lambda_1 \|\boldsymbol{\beta}\|_1 + \sum_j \log \frac{p^*(0)}{p^*(\beta_j)} \leq -\lambda_1 \|\boldsymbol{\beta}\|_1$. Plugging this into (22), we have

$$(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} \leq |\boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta}| - C_n \frac{\|\boldsymbol{X}\|_2}{\lambda_1} pen(\boldsymbol{\beta} \,|\, \theta). \tag{23}$$

Since the $\tilde{\eta}$-NC condition holds, we have

$$-\frac{\tilde{\eta}}{2\sigma^2} \|\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta} + \tilde{\eta} \times pen(\boldsymbol{\beta} \,|\, \theta) \leq 0, \; \forall \boldsymbol{\beta}. \tag{24}$$

Thus, on condition that $m < \min_i w_i \leq \max_i w_i < M$, if we choose $\eta^* = \max\left\{\tilde{\eta} + C_n \frac{\|\boldsymbol{X}\|}{\lambda_1}, \frac{\tilde{\eta}}{m}\right\}$, we have $\forall \boldsymbol{\beta}$,

$$\begin{aligned}
&-\frac{\eta^*}{2\sigma^2} \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + (\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} + \eta^* pen(\boldsymbol{\beta} \,|\, \theta) \\
\overset{(f)}{\leq} &-\frac{\eta^* m}{2\sigma^2} \|\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + |\boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta}| + \left(\eta^* - C_n \frac{\|\boldsymbol{X}\|}{\lambda_1}\right) pen(\boldsymbol{\beta} \,|\, \theta) \\
\overset{(g)}{\leq} &-\frac{\tilde{\eta}}{2\sigma^2} \|\boldsymbol{X}\boldsymbol{\zeta}\|_2^2 + \boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\zeta} + \tilde{\eta} \times pen(\boldsymbol{\zeta} \,|\, \theta) \leq 0,
\end{aligned} \tag{25}$$

where (f) follows from Equation (23), (g) follows from the definition of η^* and the fact that $pen(\boldsymbol{\beta} \,|\, \theta) \leq 0$ for any $\boldsymbol{\beta}$. We set $\boldsymbol{\zeta} = \boldsymbol{\beta}$ if $\boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta} \geq 0$ and $\boldsymbol{\zeta} = -\boldsymbol{\beta}$ if $\boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta} < 0$. The last inequality directly follows from the $\tilde{\eta}$-NC condition (24).

Previous analysis implies that under conditions (1), (4) and (5) in Theorem 4.2 (and assuming $m < \min_i w_i \leq \max_i w_i < M$, $\|\boldsymbol{\epsilon}\|_\infty \lesssim \sqrt{\log n}$ and that the $\tilde{\eta}$-NC condition holds), whenever $|(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta}| \leq C_n \|\boldsymbol{X}\boldsymbol{\beta}\|_2$ holds, (25) holds. Thus, when $\|\boldsymbol{\epsilon}\|_\infty \lesssim \sqrt{\log n}$ and $\boldsymbol{X}$ satisfies $\tilde{\eta}$-NC condition,

$$\begin{aligned}
&\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(|(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta}| \leq C_n \|\boldsymbol{X}\boldsymbol{\beta}\|_2 \mid \boldsymbol{\epsilon}, m < \min_i w_i \leq \max_i w_i < M\right) \\
\leq &\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(-\frac{\eta^*}{2\sigma^2} \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + (\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} + \eta^* pen(\boldsymbol{\beta} \,|\, \theta) \leq 0 \,|\, \boldsymbol{\epsilon}, m < \min_i w_i \leq \max_i w_i < M\right).
\end{aligned}$$

Notice that the conditions (2) and (3) in Theorem 4.2 say

$$\mathbb{P}\left(m < \min_i w_i \leq \max_i w_i < M\right) \to 1.$$

Thus, as $n \to \infty$,

$$\begin{aligned}
&\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(|(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta}| \leq C_n \|\boldsymbol{X}\boldsymbol{\beta}\|_2 \mid \boldsymbol{\epsilon}, m < \min_i w_i \leq \max_i w_i < M\right) \\
\to &\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(|(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta}| \leq C_n \|\boldsymbol{X}\boldsymbol{\beta}\|_2 \mid \boldsymbol{\epsilon}\right),
\end{aligned}$$

$$\begin{aligned}
&\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(-\frac{\eta^*}{2\sigma^1} \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + (\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} + \eta^* pen(\boldsymbol{\beta} \,|\, \theta) \leq 0 \,|\, \boldsymbol{\epsilon}, m < \min_i w_i \leq \max_i w_i < M\right) \\
\to &\mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}}\left(-\frac{\eta^*}{2\sigma^2} \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + (\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} + \eta^* pen(\boldsymbol{\beta} \,|\, \theta) \leq 0 \,|\, \boldsymbol{\epsilon}\right).
\end{aligned}$$

So given that $\|\boldsymbol{\epsilon}\|_\infty \lesssim \sqrt{\log n}$ and that $\boldsymbol{X}$ satisfies $\tilde{\eta}$-NC condition, we have

$$\begin{aligned}
&\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}} \left(|(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{\epsilon}^T \boldsymbol{X}\boldsymbol{\beta}| \leq C_n \|\boldsymbol{X}\boldsymbol{\beta}\|_2 \mid \boldsymbol{\epsilon} \right) \\
&\leq \lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}} \left(-\frac{\eta^*}{2\sigma^2} \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + (\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} + \eta^* pen(\boldsymbol{\beta} \mid \theta) \leq 0 \mid \boldsymbol{\epsilon} \right).
\end{aligned}$$

Combined with Equation (21), we know (given that $\|\boldsymbol{\epsilon}\|_\infty \lesssim \sqrt{\log n}$ and $\boldsymbol{X}$ satisfies $\tilde{\eta}$-NC condition), the following holds

$$\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}} \left(-\frac{\eta^*}{2\sigma^2} \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + (\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}^2 \boldsymbol{X}\boldsymbol{\beta} + \eta^* pen(\boldsymbol{\beta} \mid \theta) \leq 0 \mid \boldsymbol{\epsilon} \right) = 1,$$

which is equivalent to the conclusion in Lemma A.5.

A.4.4 Proof of Lemma A.6

Starting from basic inequality $Q(\widehat{\boldsymbol{\beta}}) \geq Q(\boldsymbol{\beta}_0)$, we get

$$\|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta}\|_2^2 - 2(\boldsymbol{W}\boldsymbol{\epsilon} - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\mu})^T \boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta} + 2\sigma^2 pen(\boldsymbol{\beta}_0 \mid \theta) - 2\sigma^2 pen(\widehat{\boldsymbol{\beta}} \mid \theta) \leq 0. \tag{26}$$

From

$$\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2} \|\boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta) \right\} = \mathbf{0}_p,$$

we have, for all $\boldsymbol{\Theta} \in \mathbb{R}^p$,

$$-2\boldsymbol{\Theta}^T \boldsymbol{X}^T \boldsymbol{W}^2 (\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu}) \geq -\eta^* \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta}\|_2^2 + 2\eta^* \sigma^2 pen(\boldsymbol{\Theta} \mid \theta). \tag{27}$$

Notice that $pen(\cdot \mid \theta)$ is super-additive ($pen(\boldsymbol{a} \mid \theta) + pen(\boldsymbol{b} \mid \theta) \leq pen(\boldsymbol{a}+\boldsymbol{b} \mid \theta)$ for all $\boldsymbol{a}, \boldsymbol{b}$). Plugging (27) into (26) (denoting $\boldsymbol{\beta} = \boldsymbol{\beta}_0$), we get

$$\begin{aligned}
&(1-\eta^*)\|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta}\|_2^2 \leq -2\eta^*\sigma^2 pen(\boldsymbol{\Theta} \mid \theta) - 2\sigma^2 pen(\boldsymbol{\beta} \mid \theta) + 2\sigma^2 pen(\boldsymbol{\Theta} + \boldsymbol{\beta} \mid \theta) \\
&= -2\eta^*\sigma^2 pen(\boldsymbol{\Theta}_S \mid \theta) - 2\eta^*\sigma^2 pen(\boldsymbol{\Theta}_{S^C} \mid \theta) - 2\sigma^2 pen(\boldsymbol{\beta}_S \mid \theta) + 2\sigma^2 pen(\boldsymbol{\Theta}_S + \boldsymbol{\beta}_S \mid \theta) + 2\sigma^2 pen(\boldsymbol{\Theta}_{S^C} \mid \theta) \\
&\overset{(a)}{\leq} -2\eta^*\sigma^2 pen(\boldsymbol{\Theta}_S \mid \theta) - 2\eta^*\sigma^2 pen(\boldsymbol{\Theta}_{S^C} \mid \theta) - 2\sigma^2 pen(\boldsymbol{\Theta}_S \mid \Theta) + 2\sigma^2 pen(\boldsymbol{\Theta}_{S^C} \mid \theta) \\
&= -2(\eta^*+1)\sigma^2 pen(\boldsymbol{\Theta}_S \mid \theta) - 2(\eta^*-1)\sigma^2 pen(\boldsymbol{\Theta}_{S^C} \mid \theta),
\end{aligned}$$

where (a) utilizes the fact that

$$pen(\boldsymbol{\Theta}_S + \boldsymbol{\beta}_S \mid \theta) + pen(\boldsymbol{\Theta}_S \mid \Theta) = pen(\boldsymbol{\Theta}_S + \boldsymbol{\beta}_S \mid \theta) + pen(-\boldsymbol{\Theta}_S \mid \Theta) \leq pen(\boldsymbol{\beta}_S \mid \theta).$$

Since $(1-\eta^*)\|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\theta}\|_2^2 \geq 0$, we get the desired conclusion.

A.4.5 Proof of Lemma A.7

The proof follows from the proof of Lemma 1 in the Appendix of Zhang and Zhang (2012). For any $j \in 1, 2, \cdots, p$, denote with $\mathbf{1}_j \in \mathbb{R}^p$ the vector where the j-th element is 1 and all the other elements are 0. Denote

$$G(\boldsymbol{\beta}) = -\frac{1}{2}\|\boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta).$$

Notice that $pen(\boldsymbol{\beta} \mid \theta) = \sum_{j=1}^p \rho(\beta_j \mid \theta)$. For any $t > 0$, $G(\mathbf{0}_p) \geq G(t\mathbf{1}_j)$ leads to

$$t\left(\boldsymbol{W}\boldsymbol{X}_j\right)^T \boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* \leq \frac{t^2}{2}\|\boldsymbol{W}\boldsymbol{X}_j\|_2^2 - \sigma^2 pen\left(t\mathbf{1}_j \mid \theta\right) \overset{(a)}{\leq} \frac{t^2 Mn}{2} - \sigma^2\rho\left(t \mid \theta\right),$$

where (a) follows from the fact that $\max_i w_i \leq M$. Thus, for any $t > 0$,

$$\left(\boldsymbol{W}\boldsymbol{X}_j\right)^T \boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* \leq M\left(\frac{1}{2}nt - \frac{\sigma^2}{M}\frac{\rho(t \mid \theta)}{t}\right).$$

Again, from the definition of Δ (17) and Proposition 5 in Moran et al. (2019), we have

$$\begin{aligned}
&\|\boldsymbol{X}^T\boldsymbol{W}^2\left(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu}\right)\|_\infty \leq \eta^* M \inf_{t>0}\left(\frac{1}{2}nt - \frac{\sigma^2}{M}\rho(t \mid \theta)/t\right) \\
\leq& \eta^* M\left[\sqrt{(2n\sigma^2/M)\log[1/p^*(0)]} + \sigma^2\lambda_1/M\right] \\
\leq& \eta^*\sqrt{M}\left[\sqrt{2n\sigma^2\log[1/p^*(0)]} + \sigma^2\lambda_1\right] = \eta^*\sqrt{M}\Delta^U.
\end{aligned}$$

A.4.6 Proof of Theorem 4.2

The proof follows from the proof of Theorem 7 in Ročková and George (2018). First, we assume that (i) $\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p}\{-\frac{1}{2\sigma^2}\|\boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta)\} = \mathbf{0}_p$; (ii) $\max_i w_i \leq M$, (iii) $\min_i w_i \geq m$, (iv) $\|\boldsymbol{\epsilon}\|_\infty \lesssim \sqrt{\log n}$ and (v) $\boldsymbol{\Theta} \in C(\eta^*; \boldsymbol{\beta})$ holds. Then, from Lemma A.7, we know that

$$\|\boldsymbol{X}^T\boldsymbol{W}^2(\boldsymbol{\epsilon} - \boldsymbol{W}\boldsymbol{\mu})\|_\infty \leq \sqrt{M}\eta^*\Delta^U,$$

where Δ^U is defined in Lemma A.7. Following Ročková and George (2018), we denote $c_+^w = 0.5\left(1 + \sqrt{1 - \frac{4\|\boldsymbol{W}\boldsymbol{X}_j\|_2^2}{\sigma^2(\lambda_0-\lambda_1)^2}}\right)$. Notice that $\delta_{c_+^w} = \frac{1}{\lambda_0-\lambda_1}\log\left(\frac{1-\theta}{\theta}\frac{\lambda_0}{\lambda_1}\frac{c_+^w}{1-c_+^w}\right)$ is the inflection point of $Q(\boldsymbol{\beta})$ in the j-th direction while keeping the other coordinates fixed. Since $\|\boldsymbol{W}\boldsymbol{X}_j\|_2^2 \leq Mn \leq \sigma^2(\lambda_0 - \lambda_1)^2$ when n is sufficiently large, c_+^w is well-defined when n is sufficiently large. Denote $\hat{q} = \|\hat{\boldsymbol{\beta}}\|_0$ and $q = \|\boldsymbol{\beta}_0\|_0$. From the basic inequality $0 \geq Q(\boldsymbol{\beta}_0) - Q(\widehat{\boldsymbol{\beta}})$, we get

$$
\begin{aligned}
0 &\geq \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta}\|_2^2 - 2(\boldsymbol{W}\boldsymbol{\epsilon} - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta} + 2\sigma^2 \log \frac{\pi(\boldsymbol{\beta}_0 \mid \theta)}{\pi(\widehat{\boldsymbol{\beta}} \mid \theta)} \\
&\geq \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta}\|_2^2 - 2(\boldsymbol{W}\boldsymbol{\epsilon} - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta} + 2\sigma^2 \left[-\lambda_1\|\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}}\|_1 + \sum_{j=1}^p \log \frac{p^*(\hat{\beta}_j)}{p^*(0)} + \sum_{j=1}^p \log \frac{p^*(0)}{p^*(\beta_j^0)}\right] \\
&\overset{(a)}{\geq} \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta}\|_2^2 - 2(\boldsymbol{W}\boldsymbol{\epsilon} - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta} + 2\sigma^2 \left[-\lambda_1\|\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}}\|_1 + \hat{q}b^w + (\hat{q} - q)\log \frac{1}{p^*(0)}\right] \\
&\overset{(b)}{\geq} \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta}\|_2^2 - 2\|(\boldsymbol{W}\boldsymbol{\epsilon} - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}\boldsymbol{X}\|_\infty \times \|\boldsymbol{\Theta}\|_1 - 2\sigma^2\lambda_1\|\boldsymbol{\beta}_0 - \widehat{\boldsymbol{\beta}}\|_1 \\
&\qquad + 2\sigma^2\hat{q}b^w + 2(\hat{q} - q)\sigma^2 \log[1/p^*(0)],
\end{aligned}
\tag{28}
$$

where (a) follows from the fact that $p^*(\hat{\beta}_j) > c_+^w$ when $\hat{\beta}_j \neq 0$ and we denote $0 > b^w = \log c_+^w > \log(1/2)$, (b) uses Holder inequality. So, from Equation (28) and Lemma A.6, we have

$$
\begin{aligned}
0 &\geq \|\boldsymbol{W}\boldsymbol{X}\boldsymbol{\Theta}\|_2^2 - 2(\sqrt{M}\eta^*\Delta^U + \sigma^2\lambda_1)\|\boldsymbol{\Theta}\|_1 + 2\sigma^2\hat{q}b^w + 2(\hat{q} - q)\sigma^2 \log[1/p^*(0)] \\
&\overset{(a)}{\geq} mc^2(\eta^*;\boldsymbol{\beta})\|\boldsymbol{\Theta}\|_2^2\|\boldsymbol{X}\|_2^2 - 2(\sqrt{M}\eta^*\Delta^U + \sigma^2\lambda_1)\|\boldsymbol{\Theta}\|_1 + 2\sigma^2\hat{q}b^w + 2\sigma^2(\hat{q} - q)\log[1/p^*(0)] \\
&\geq mc^2(\eta^*;\boldsymbol{\beta})\|\boldsymbol{\Theta}\|_2^2\|\boldsymbol{X}\|_2^2 - 2(\sqrt{M}\eta^*\Delta^U + \sigma^2\lambda_1)\|\boldsymbol{\Theta}\|_2\|\boldsymbol{\Theta}\|_0^{1/2} + 2\sigma^2\hat{q}b^w + 2(\hat{q} - q)\sigma^2 \log[1/p^*(0)],
\end{aligned}
$$

where (a) follows from the definition of $c(\eta^*;\boldsymbol{\beta})$. This is equivalent to

$$
\begin{aligned}
&\left(\sqrt{m}c(\eta^*;\boldsymbol{\beta})\|\boldsymbol{\Theta}\|_2\|\boldsymbol{X}\|_2 - \frac{\sqrt{M}\eta^*\Delta^U + \sigma^2\lambda_1}{\sqrt{m}c(\eta^*;\boldsymbol{\beta})\|\boldsymbol{X}\|_2}\|\boldsymbol{\Theta}\|_0^{1/2}\right)^2 - \frac{(\sqrt{M}\eta^*\Delta^U + \sigma^2\lambda_1)^2}{mc^2(\eta^*;\boldsymbol{\beta})\|\boldsymbol{X}\|_2^2}\|\boldsymbol{\Theta}\|_0 \\
&\qquad\qquad + 2\sigma^2\hat{q}b^w + 2(\hat{q} - q)\sigma^2 \log \frac{1}{p^*(0)} \leq 0,
\end{aligned}
$$

and consequently,

$$
(\hat{q} - q)\log \frac{1}{p^*(0)} + \hat{q}b^w \leq \frac{(\sqrt{M}\eta^*\Delta^U + \sigma^2\lambda_1)^2}{2m\sigma^2c^2(\eta^*;\boldsymbol{\beta})\|\boldsymbol{X}\|_2^2}\|\boldsymbol{\Theta}\|_0 \overset{(a)}{\leq} \frac{(\sqrt{M}\eta^*\Delta^U + \sigma^2\lambda_1)^2}{2m\sigma^2c^2(\eta^*;\boldsymbol{\beta})n}(\hat{q} + q),
$$

where (a) follows from $\|\boldsymbol{X}\|_2^2 \geq n$. Thus,

$$
\hat{q} \leq q\frac{A + B}{B + b^w - A} = q\left(1 + \frac{2A - b^w}{B + b^w - A}\right) \leq q\left(1 + \frac{2r}{1 - r}\right),
$$

where

$$
A = \frac{(\sqrt{M}\eta^*\Delta^U + \sigma^2\lambda_1)^2}{2m\sigma^2c^2(\eta^*;\boldsymbol{\beta})n},\ B = \log[1/p^*(0)],\ b^w = \log c_+^w \in (\log(1/2), 0),\ r = \frac{A}{B}.
$$

For simplicity assume that $\frac{1-\theta}{\theta} = C_1 p^{\eta}, \lambda_0 = C_2 p^{\gamma}$ with $C_1 C_2 = 4$. Then $B = \log(1 + \frac{1-\theta}{\theta}\frac{\lambda_0}{\lambda_1}) > (\eta + \gamma - 1)\log p$, so

$$
\begin{aligned}
r = \frac{A}{B} &= \left(\frac{\sqrt{M}\eta^* \Delta^U}{\sigma c(\eta^*; \boldsymbol{\beta})\sqrt{2nmB}} + \frac{\sigma \lambda_1}{c(\eta^*; \boldsymbol{\beta})\sqrt{2mnB}} \right)^2 \\
&< \left(\frac{\eta^*}{c}\sqrt{\frac{M}{m}} + \frac{\sigma\lambda_1\sqrt{M}}{c\sqrt{2mn(\eta+\gamma-1)\log p}} \right)^2 = D.
\end{aligned}
$$

Thus, the desired conclusion holds when conditions (i)-(v) hold. Now, we only need to verify that $\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta},\boldsymbol{\mu},\boldsymbol{w}}$ (condition (i), (ii), (iii), (iv), (v) all holds) $= 1$. Notice that (i) implies (v) from Lemma A.6 and

$$
\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0} (\text{condition (iv) holds}) \geq \lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0} \left(\|\boldsymbol{\epsilon}\|_\infty \leq \sqrt{2\log n} \right) = 1.
$$

Combined with Lemma A.4 and Lemma A.5, we have

$$
\begin{aligned}
&\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0,\boldsymbol{\mu},\boldsymbol{w}} \left(\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \{ -\frac{1}{2\sigma^2} \|\boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta) \} = \mathbf{0}_p \right) \\
= &\lim_{n\to\infty} \mathbb{E}_{\boldsymbol{\beta}_0} \mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}} \left(\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \{ -\frac{1}{2\sigma^2} \|\boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta) \} = \mathbf{0}_p \mid \boldsymbol{\epsilon} \right) = 1.
\end{aligned}
$$

This means $\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0,\boldsymbol{\mu},\boldsymbol{w}}$ (condition (i) holds) $= 1$. In addition, assumptions (2) and (3) in Theorem 4.2 say that

$$
\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0,\boldsymbol{\mu},\boldsymbol{w}} (\text{condition (ii) holds}) = 1 \quad \text{and} \quad \lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0,\boldsymbol{\mu},\boldsymbol{w}} (\text{condition (iii) holds}) = 1.
$$

Thereby, from the union bound, $\lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0,\boldsymbol{\mu},\boldsymbol{w}}$ (condition (i), (ii), (iii), (iv), (v) all holds) $=$ 1. Since

$$
\begin{aligned}
\lim_{n\to\infty} \mathbb{E}_{\boldsymbol{\beta}_0} \mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}} \left(\hat{q} \leq q(1 + \frac{2D}{1-D}) \mid \boldsymbol{Y}^{(n)} \right) &= \lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0,\boldsymbol{\mu},\boldsymbol{w}} \left(\hat{q} \leq q(1 + \frac{2D}{1-D}) \right) \\
&\geq \lim_{n\to\infty} \mathbb{P}_{\boldsymbol{\beta}_0,\boldsymbol{\mu},\boldsymbol{w}} (\text{conditions (i) - (v) all hold}).
\end{aligned}
$$

We have

$$
\lim_{n\to\infty} \mathbb{E}_{\boldsymbol{\beta}_0} \mathbb{P}_{\boldsymbol{\mu},\boldsymbol{w}} \left(\hat{q} \leq q(1 + \frac{2D}{1-D}) \mid \boldsymbol{Y}^{(n)} \right) = 1.
$$

A.4.7 Proof of Theorem 4.3

The proof follows from the proof of Theorem 8 in Ročková and George (2018). Notice that

$$
\|\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2 \leq \|\widetilde{\boldsymbol{\beta}} - \widehat{\boldsymbol{\beta}}\|_2 + \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2.
$$

First, we prove the high probability bound for $\|\mathbf{\Theta}\|_2 = \|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\|_2$. Since $Q(\widehat{\boldsymbol{\beta}}) \geq Q(\boldsymbol{\beta}_0)$, we know if (i) $\arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p}\{-\frac{1}{2\sigma^2}\|\boldsymbol{W}(\boldsymbol{\epsilon} - \boldsymbol{X}\boldsymbol{\mu})/\eta^* - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta}\|_2^2 + pen(\boldsymbol{\beta} \mid \theta)\} = \mathbf{0}_p$; (ii) $\max w_i \leq M$, then

$$
\begin{aligned}
0 &\geq \|\boldsymbol{W}\boldsymbol{X}\mathbf{\Theta}\|_2^2 - 2(\boldsymbol{W}\boldsymbol{\epsilon} - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}\boldsymbol{X}\mathbf{\Theta} + 2\log\frac{\pi(\boldsymbol{\beta}_0 \mid \theta)}{\pi(\widehat{\boldsymbol{\beta}} \mid \theta)} \\
&\overset{(a)}{\geq} \|\boldsymbol{W}\boldsymbol{X}\mathbf{\Theta}\|_2^2 - 2\|(\boldsymbol{W}\boldsymbol{\epsilon} - \boldsymbol{W}\boldsymbol{X}\boldsymbol{\mu})^T\boldsymbol{W}\boldsymbol{X}\|_\infty\|\mathbf{\Theta}\|_1 - 2\lambda_1\|\mathbf{\Theta}\|_1 + 2q\log p^*(0) \\
&\overset{(b)}{\geq} \|\boldsymbol{W}\boldsymbol{X}\mathbf{\Theta}\|_2^2 - 2\left(\sqrt{M}\eta^*\Delta^U + \lambda_1\right)\|\mathbf{\Theta}\|_1 + 2q\log p^*(0),
\end{aligned}
\tag{29}
$$

where (a) follows from $\log\frac{\pi(\boldsymbol{\beta}_0 \mid \theta)}{\pi(\boldsymbol{\beta} \mid \theta)} \geq -\lambda_1\|\mathbf{\Theta}\|_1 + q\log p^*(0)$ and Holder Inequality, (b) follows from Lemma A.7. From Theorem 4.2, $\|\mathbf{\Theta}\|_0 \leq (1+K)q$, and using $4ab \leq a^2 + 4b^2$, we have

$$
\begin{aligned}
2(\sqrt{M}\eta^*\Delta^U + \lambda_1)\|\mathbf{\Theta}\|_1 &\leq 3(\sqrt{M}\eta^*\Delta^U + \lambda_1)\frac{\|\boldsymbol{X}\mathbf{\Theta}\|_2\sqrt{(K+1)q}}{\|\boldsymbol{X}\|_2\phi} - (\sqrt{M}\eta^*\Delta^U + \lambda_1)\|\mathbf{\Theta}\|_1 \\
&\leq \frac{m\|\boldsymbol{X}\mathbf{\Theta}\|_2^2}{2} + \frac{5(K+1)q(\sqrt{M}\eta^*\Delta^U + \lambda_1)^2}{m\|\boldsymbol{X}\|_2^2\phi^2} - (\sqrt{M}\eta^*\Delta^U + \lambda_1)\|\mathbf{\Theta}\|_1.
\end{aligned}
$$

Plugging into (29), we know (i), (ii), plus (iii) $\min w_i \geq m$ and (iv) $\|\boldsymbol{\epsilon}\|_\infty \lesssim \sqrt{\log n}$ implies the following:

$$
\begin{aligned}
0 &\geq \|\boldsymbol{W}\boldsymbol{X}\mathbf{\Theta}\|_2^2 - \frac{m\|\boldsymbol{X}\mathbf{\Theta}\|_2^2}{2} - \frac{5(K+1)q(\sqrt{M}\eta^*\Delta^U + \lambda_1)^2}{m\|\boldsymbol{X}\|_2^2\phi^2} + (\sqrt{M}\eta^*\Delta^U + \lambda_1)\|\mathbf{\Theta}\|_1 + 2q\log p^*(0) \\
&\geq \frac{m}{2}\|\boldsymbol{X}\mathbf{\Theta}\|_2^2 - \frac{5(K+1)q(\sqrt{M}\eta^*\Delta^U + \lambda_1)^2}{m\|\boldsymbol{X}\|_2^2\phi^2} + (\sqrt{M}\eta^*\Delta^U + \lambda_1)\|\mathbf{\Theta}\|_1 + 2q\log p^*(0).
\end{aligned}
$$

Thus, whenever (i)-(iv) holds, in view that $\|\boldsymbol{X}\|_2^2 \geq n$, we have

$$
\begin{aligned}
\frac{m}{2}\|\boldsymbol{X}\mathbf{\Theta}\|_2^2 + (\sqrt{M}\eta^*\Delta^U + \lambda_1)\|\mathbf{\Theta}\|_1 &\leq \frac{5(K+1)q(C_3\sqrt{M}\eta^*\sqrt{n\log p})^2}{mn\phi^2} + 2qC_4\log p \\
&< \frac{C_5^2 M(\eta^*)^2}{m\phi^2}q(1+K)\log p.
\end{aligned}
$$

Thus, whenever (i)-(iv) holds,

$$
\|\boldsymbol{X}\mathbf{\Theta}\|_2 \leq \frac{C_5\eta^*\sqrt{M}}{\sqrt{m}\,\phi}\sqrt{q(1+K)\log p}.
$$

It follows from definition of c that

$$
\lim_{n\to\infty}\mathbb{P}_{\boldsymbol{w},\boldsymbol{\mu},\boldsymbol{\beta}_0}\left(\|\mathbf{\Theta}\|_2 \leq \frac{C_5\eta^*\sqrt{M}}{\sqrt{m}\,\phi\, c}\sqrt{q(1+K)\frac{\log p}{n}}\right) \geq \lim_{n\to\infty}\mathbb{P}\,(\text{condition (i)-(iv) holds}) = 1.
$$

Notice that the difference between $\widetilde{\boldsymbol{\beta}}$ and $\widehat{\boldsymbol{\beta}}$ only depends on $\boldsymbol{\mu}$ and satisfies

$$\mathbb{P}_{\boldsymbol{\mu}}\left(\|\widetilde{\boldsymbol{\beta}}-\widehat{\boldsymbol{\beta}}\|_2^2 > t \mid \boldsymbol{Y}^{(n)}\right) \le \frac{\mathbb{E}\sum_{j=1}^p(\hat{\beta}_j-\tilde{\beta}_j)^2}{t} = \frac{1}{t}\sum_{j=1}^p\left(\frac{1}{\lambda_0^2}+\frac{2}{\lambda_0^2}\right) = \frac{1}{t}\frac{3p}{\lambda_0^2}.$$

Set $t = \frac{C_5^2(\eta^*)^2M}{m\,\phi^2\,c^2}q(1+K)\frac{\log p}{n}$, then $\frac{1}{t}\frac{3p}{\lambda_0^2} \to 0$ when $n, p \to \infty$. Thus, from triangle inequality,

$$\begin{aligned}
&\mathbb{E}_{\boldsymbol{\beta}_0}\mathbb{P}_{\boldsymbol{w},\boldsymbol{\mu}}\left(\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}_0\|_2 > \frac{C_5\eta^*\sqrt{M}}{\sqrt{m}\,\phi\,c}\sqrt{q(1+K)\frac{\log p}{n}} \mid \boldsymbol{Y}^{(n)}\right)\\
&\le \mathbb{E}_{\boldsymbol{\beta}_0}\mathbb{P}_{\boldsymbol{w},\boldsymbol{\mu}}\left(\|\widetilde{\boldsymbol{\beta}}-\widehat{\boldsymbol{\beta}}\|_2^2 > \frac{C_5\eta^*\sqrt{M}}{2\sqrt{m}\,\phi\,c}\sqrt{q(1+K)\frac{\log p}{n}} \mid \boldsymbol{Y}^{(n)}\right)\\
&+ \mathbb{E}_{\boldsymbol{\beta}_0}\mathbb{P}_{\boldsymbol{w},\boldsymbol{\mu}}\left(\|\boldsymbol{\Theta}\|_2 > \frac{C_5\eta^*\sqrt{M}}{2\sqrt{m}\,\phi\,c}\sqrt{q(1+K)\frac{\log p}{n}} \mid \boldsymbol{Y}^{(n)}\right)\\
&\le \frac{1}{t}\frac{3p}{\lambda_0^2} + \mathbb{P}_{\boldsymbol{\beta}_0,\boldsymbol{\mu},\boldsymbol{w}}\left(\|\boldsymbol{\Theta}\|_2 > \frac{C_5\eta^*\sqrt{M}}{2\sqrt{m}\,\phi\,c}\sqrt{q(1+K)\frac{\log p}{n}} \mid \boldsymbol{Y}^{(n)}\right),
\end{aligned}$$

where the right hand side goes to zero as n goes to infinity. Thus, we have

$$\lim_{n\to\infty}\mathbb{E}_{\boldsymbol{\beta}}\mathbb{P}_{\boldsymbol{w},\boldsymbol{\mu}}\left(\|\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}_0\|_2 > \frac{C_5\eta^*\sqrt{M}}{\sqrt{m}\,\phi\,c}\sqrt{q(1+K)\frac{\log p}{n}} \mid \boldsymbol{Y}^{(n)}\right) = 0.$$

A.5 Proof of Corollary 4.1

We notice that if $w_i \sim \frac{1}{\alpha}\text{Gamma}(\alpha,1)$, we have $\mathbb{E}w_i = 1$, $Var(w_i) = \frac{1}{\alpha} \lesssim \frac{1}{\log n}$, $Cov(w_i, w_j) = 0$. So we only need to prove the two high probability bounds for the order statistics for w_i. Let $\alpha = 2(\eta+\gamma)\log p$, from the union bound we have

$$\begin{aligned}
&\mathbb{P}\left(\min w_i < \frac{1}{e}\right) \le n\mathbb{P}\left(w_i < \frac{1}{e}\right) = n\mathbb{P}_{v\sim\text{Gamma}(\alpha,1)}(v < \alpha/e) = ne^{-\alpha/e}\sum_{i=\alpha}^{\infty}\frac{(\alpha/e)^i}{i!}\\
&\le ne^{-\alpha/e}\sum_{i=\alpha}^{\infty}\frac{(\alpha/e)^i}{\sqrt{2\pi}i^{i+1/2}e^{-i}} = n\frac{e^{-\alpha/e}}{\sqrt{2\pi}}\sum_{i=\alpha}^{\infty}\left(\frac{\alpha}{i}\right)^i \le n\frac{e^{-\alpha/e}}{\sqrt{2\pi}}\left[\sum_{i=\alpha+1}^{\infty}\left(\frac{\alpha}{i}\right)^i+1\right]\\
&\le n\frac{e^{-\alpha/e}}{\sqrt{2\pi}}\left[\sum_{i=\alpha+1}^{\infty}\left(\frac{\alpha}{\alpha+1}\right)^i+1\right] = n\frac{e^{-\alpha/e}}{\sqrt{2\pi}}\left[\left(\frac{\alpha}{\alpha+1}\right)^{\alpha+1}(\alpha+1)+1\right] \le C_0 n\frac{e^{-\alpha/e}}{\sqrt{2\pi}}\left[\frac{\alpha+1}{e}+1\right]\\
&\le C_1(\log p)\,n\,p^{-\frac{2(\eta+\gamma)}{e}}.
\end{aligned}$$

Since $2(\eta+\gamma)/e \ge 4/e > 1$, we know $\lim_{n\to\infty}\mathbb{P}(\min w_i < \frac{1}{e}) = 0$. From the proof of Corollary 4.1 and the union bound, for any $t \ge \frac{\alpha+1}{\alpha}$,

$$\mathbb{P}(\max w_i > t) = ne^{\alpha(log(t)+1-t} \le Cn\,(p)^{-\frac{\eta+\gamma}{t}}.$$

Set $t = \frac{2}{3}(\eta+\gamma) > \frac{4}{3} \geq \frac{\alpha+1}{\alpha}$ when n is sufficiently large, then $\lim_{n\to\infty} \mathbb{P}(\max w_i > t) = 0$. If $w_i \sim n\mathrm{Dir}(\alpha, \cdots, \alpha)$, we have $\mathbb{E}w_i = 1$, $Var(w_i) = n^2[\frac{1/n(1-1/n)}{n\alpha+1}] \lesssim \frac{1}{\log n}$, $Cov(w_i, w_j) = -n^2[\frac{1/n^2}{n\alpha+1}] \lesssim \frac{1}{n \log n}$. Using the fact that $w_i \xrightarrow{d} \frac{1}{\alpha}Gamma(\alpha, 1)$, we can prove

$$\lim_{n\to\infty} \mathbb{P}\left(\min w_i < \frac{1}{e}\right) = 0,$$

$$\lim_{n\to\infty} \mathbb{P}(\max w_i > 2(\eta+\gamma)/3) = 0.$$

Thus, conditions (1)-(4) hold for both $\boldsymbol{w} \sim n\mathrm{Dir}(\alpha, \cdots, \alpha)$ and $w_i \sim \frac{1}{\alpha}Gamma(\alpha, 1)$ where $\alpha \gtrsim \log p$.

A.6 Derivation of Observations in Section 3.1

In this section, we theoretically justify our statements from Section 3.1.

A.6.1 Notation

Define

$$c_0^{(-)} = (1-\theta)\lambda_0 e^{-y_i\lambda_0 + \lambda_0^2/2n}, \qquad c_0^{(+)} = (1-\theta)\lambda_0 e^{y_i\lambda_0 + \lambda_0^2/2n}.$$

Next, define

$$\phi_0^{(-)}(x) = \phi(x; y_i - \lambda_0/n, 1/n), \qquad \phi_0^{(+)}(x) = \phi(x; y_i + \lambda_0/n, 1/n),$$

where $\phi(x;\mu,\sigma^2)$ is the Gaussian density with mean μ and variance σ^2. The quantities $c_1^{(-)}$, $c_1^{(+)}$, $\phi_1^{(-)}(x)$, $\phi_1^{(+)}(x)$ are defined in a similar way in Section 3.1 in the main paper, with λ_0 replaced by λ_1. Throughout this section, we assume that the parameters λ_0 and λ_1 satisfy $(1-\theta)/\theta \asymp n^a$, $\lambda_0 \asymp n^d$ where $a, d \geq 2$ and $1/\sqrt{n} < \lambda_1 \leq c_0$ with some constant c_0.

A.6.2 Active Coordinates

Proposition 1. *Assume the true posterior defined in (8) with an active coordinate. Conditioning on the event $|y_i| > |\beta_i^0|/2$, we have*

$$w_0 = \pi(\gamma_i = 0 \,|\, y_i) \to 0.$$

Proof. Without loss of generality, we assume $y_i > \beta_i/2 > 0$.

$$
\begin{aligned}
w_1 = \pi(\gamma_i = 1 \,|\, y_i) &= \frac{\pi(y_i \,|\, \gamma_i = 1)\pi(\gamma_i = 1)}{\pi(y_i \,|\, \gamma_i = 1)\pi(\gamma_i = 1) + \pi(y_i \,|\, \gamma_i = 0)\pi(\gamma_i = 0)} \\
&= \frac{\int_{\beta_i} \pi(y_i \,|\, \beta_i)\pi(\beta_i \,|\, \gamma_i = 1)\pi(\gamma_i = 1) d\beta_i}{\int_{\beta_i} \pi(y_i \,|\, \beta_i)\pi(\beta_i \,|\, \gamma_i = 1)\pi(\gamma_i = 1) d\beta_i + \int_{\beta_i} \pi(y_i \,|\, \beta_i)\pi(\beta_i \,|\, \gamma_i = 0)\pi(\gamma_i = 0) d\beta_i} \\
&= \frac{\int_0^\infty c_1^{(-)}\phi_1^{(-)}(\beta_i)d\beta_i + \int_{-\infty}^0 c_1^{(+)}\phi_1^{(+)}(\beta_i)d\beta_i}{\int_0^\infty c_1^{(-)}\phi_1^{(-)}(\beta_i)d\beta_i + \int_0^\infty c_0^{(-)}\phi_0^{(-)}(\beta_i)d\beta_i + \int_{-\infty}^0 c_1^{(+)}\phi_1^{(+)}(\beta_i)d\beta_i + \int_{-\infty}^0 c_0^{(+)}\phi_0^{(+)}(\beta_i)d\beta_i}
\end{aligned} \tag{30}
$$

We consider the four terms in the denominator separately. It is helpful to divide each of them by $\theta\lambda_1$. Regarding the first term, we have

$$
\frac{1}{\theta\lambda_1}\int_0^\infty c_1^{(-)}\phi_1^{(-)}(\beta_i)d\beta_i = e^{-y_i\lambda_1 + \lambda_1^2/2n}\left(1 - \Phi\left(-\sqrt{n}(y_i - \frac{\lambda_1}{n})\right)\right) \to e^{-y_i\lambda_1}. \tag{31}
$$

Regarding the second term in the denominator, from the Mills ratio we have

$$
\begin{aligned}
\frac{1}{\theta\lambda_1}\int_0^\infty c_0^{(-)}\phi_0^{(-)}(\beta_i)d\beta_i &= \frac{1}{\theta\lambda_1}c_0^{(-)}\Phi\left(-\sqrt{n}\left(y_i - \frac{\lambda_0}{n}\right)\right) \\
&\le C\frac{(1-\theta)\lambda_0}{\theta\lambda_1}e^{-y_i\lambda_0 + \lambda_0^2/2n}\frac{\phi\left(\sqrt{n}(y_i - \frac{\lambda_0}{n})\right)}{\sqrt{n}(y_i - \frac{\lambda_0}{n})} = C\frac{(1-\theta)\lambda_0}{\sqrt{n}(y_i - \frac{\lambda_0}{n})\theta\lambda_1}\frac{1}{\sqrt{2\pi}}e^{-\frac{n}{2}y_i^2}
\end{aligned}
$$

where $\frac{(1-\theta)\lambda_0}{\sqrt{n}(y_i - \frac{\lambda_0}{n})\theta\lambda_1}\frac{1}{\sqrt{2\pi}}e^{-\frac{n}{2}y_i^2} \to 0$. Thus,

$$
\frac{1}{\theta\lambda_1}\int_0^\infty c_0^{(-)}\phi_0^{(-)}(\beta_i)d\beta_i \to 0. \tag{32}
$$

For the third term in the denominator, we write

$$
\frac{1}{\theta\lambda_1}\int_{-\infty}^0 c_1^{(+)}\phi_1^{(+)}(\beta_i)d\beta_i = e^{y_i\lambda_1 + \lambda_1^2/2n}\Phi\left(-\sqrt{n}(y_i + \frac{\lambda_1}{n})\right) \to 0. \tag{33}
$$

For the fourth term, we then have

$$
\begin{aligned}
\frac{1}{\theta\lambda_1}\int_{-\infty}^0 c_0^{(+)}\phi_0^{(+)}(\beta_i)d\beta_i &= \frac{1}{\theta\lambda_1}\int_{-\infty}^0 (1-\theta)\lambda_0\sqrt{\frac{n}{2\pi}}e^{-\frac{n}{2}(\beta_i - y_i)^2 + \beta_i\lambda_0}d\beta_i \\
&\le \frac{1}{\theta\lambda_1}\int_{-\infty}^0 (1-\theta)\lambda_0\sqrt{\frac{n}{2\pi}}e^{-\frac{n}{2}y_i^2 + \beta_i\lambda_0}d\beta_i = \frac{1}{\theta\lambda_1}(1-\theta)\sqrt{\frac{n}{2\pi}}e^{-\frac{n}{2}y_i^2}
\end{aligned}
$$

where $\frac{1}{\theta\lambda_1}(1-\theta)\sqrt{\frac{n}{2\pi}}e^{-\frac{n}{2}y_i^2} \to 0$. Thus,

$$
\frac{1}{\theta\lambda_1}\int_{-\infty}^0 c_0^{(+)}\phi_0^{(+)}(\beta_i)d\beta_i \to 0. \tag{34}
$$

From (31), (32), (33) and (34), we know that $w_1 \to \frac{e^{-y_i\lambda_1}}{e^{-y_i\lambda_1}} = 1$. Thus, $w_0 = 1 - w_1 \to 0$. $\square$

Proposition 2. *For the true posterior defined in* (8), *we have*

$$\pi\left(\sqrt{n}(\beta_i - y_i) \mid y_i, \gamma_i = 1\right) \to \phi(\sqrt{n}(\beta_i - y_i); 0, 1).$$

Proof. Setting $u_i = \sqrt{n}(\beta_i - y_i)$, we have

$$\pi\left(u_i \mid y_i, \gamma_i = 1\right) = \frac{1}{\sqrt{n}} \frac{\mathbb{I}(u_i \geq -\sqrt{n}y_i) c_1^{(-)} \phi_1^{(-)}(u_i/\sqrt{n} + y_i) + \mathbb{I}(u_i < -\sqrt{n}y_i) c_1^{(+)} \phi_1^{(+)}(u_i/\sqrt{n} + y_i)}{\int_0^\infty c_1^{(-)} \phi_1^{(-)}(\beta_i) d\beta_i + \int_{-\infty}^0 c_1^{(+)} \phi_1^{(+)}(\beta_i) d\beta_i}. \tag{35}$$

Notice that both

$$\frac{1}{\sqrt{n}} \phi_1^{(-)}(u_i/\sqrt{n} + y_i) \to \phi(u_i; 0, 1) \quad \text{and} \quad \frac{1}{\sqrt{n}} \phi_1^{(+)}(u_i/\sqrt{n} + y_i) \to \phi(u_i; 0, 1).$$

For any u_i, only one of $\mathbb{I}(u_i < -\sqrt{n}y_i)$ and $\mathbb{I}(u_i \geq -\sqrt{n}y_i)$ holds, and the denominator in (35) does not depend on u_i, so $\pi\left(u_i \mid y_i, \gamma_i = 1\right) \to \phi(u_i; 0, 1)$. ☐

Proposition 3. *Consider the fixed WBB estimator $\widehat{\beta}_i$. Conditioning on w_i and y_i, when $\widehat{\beta}_i \neq 0$ and $|y_i| > \frac{|\beta_i^0|}{2}$, we have*

$$n\left(\widehat{\beta}_i - y_i\right) \to -\frac{1}{w_i}\lambda_1.$$

Proof. Fixed WBB estimate $\widehat{\beta}_i$ satisfies equation (11) in the main text. Without loss of generality, we assume $y_i > \frac{\beta_i^0}{2} > 0$. Conditioning on y_i, w_i, when $\widehat{\beta}_i \neq 0$,

$$\begin{aligned} n\left(\widehat{\beta}_i - y_i\right) &= -\frac{1}{w_i}\lambda_1 - \frac{1}{w_i}(\lambda_0 - \lambda_1)(1 - p^*(\widehat{\beta}_i)) \\ &= -\frac{1}{w_i}\lambda_1 - \frac{1}{w_i} \frac{\frac{(1-\theta)\lambda_0}{\theta\lambda_1}(\lambda_0 - \lambda_1) e^{-|\widehat{\beta}_i|(\lambda_0 - \lambda_1)}}{1 + \frac{(1-\theta)\lambda_0}{\theta\lambda_1} e^{-|\widehat{\beta}_i|(\lambda_0 - \lambda_1)}} \to -\frac{1}{w_i}\lambda_1. \end{aligned}$$ ☐

Proposition 4. *Consider the fixed WBB estimator $\widehat{\beta}_i$. Conditioning on the event that $|y_i| > \frac{|\beta_i^0|}{2}$, we have*

$$\mathbb{P}_{w_i}\left(\widehat{\beta}_i = 0 \mid y_i\right) \to 0.$$

Proof. The fixed WBB estimator $\hat{\beta}_i$ satisfies Equation (11). When n is sufficiently large, from Proposition 5 in Moran et al. (2019), we have

$$\Delta_{w_i} \leq \sqrt{[2/(nw_i)] \log[1/p^*(0)]} + \lambda_1/(nw_i). \tag{36}$$

Thus, from the Markov's inequality, we find

$$
\begin{aligned}
&\mathbb{P}_{w_i}\left(\widehat{\beta}_i = 0 \mid y_i\right) \leq \mathbb{P}\left(|\widehat{\beta}_i - y_i|^{1/2} \geq |y_i|^{1/2} \mid y_i\right) \leq \frac{\mathbb{E}\left[|\widehat{\beta}_i - y_i|^{1/2} \mid y_i\right]}{|y_i|^{1/2}} \\
&\overset{(a)}{=} \frac{1}{|y_i|^{1/2}}\left\{\mathbb{E}\left[|y_i|^{1/2}\mathbb{I}\left(|y_i| \leq \Delta_{w_i}\right) \mid y_i\right] + \mathbb{E}\left[\left(\frac{1}{w_i n}\lambda^*(\widehat{\beta}_i)\right)^{1/2}\mathbb{I}\left(|y_i| > \Delta_{w_i}\right) \mid y_i\right]\right\} \\
&\overset{(b)}{\leq} \mathbb{P}\left(|\beta_i^0|/2 \leq \sqrt{[2/(nw_i)]\log[1/p^*(0)]} + \lambda_1/(nw_i)\right) + \frac{1}{|y_i|^{1/2}}\mathbb{E}\left(\frac{1}{nw_i}\lambda^*(\widehat{\beta}_i)\right)^{1/2} \\
&\leq \mathbb{P}\left(w_i^{1/2} < \frac{1}{|\beta_i^0|}\left[\sqrt{\frac{2}{n}\log[1/p^*(0)] + \frac{2\lambda_1}{n}|\beta_i^0|} + \sqrt{\frac{2}{n}\log[1/p^*(0)]}\right]\right) + \frac{1}{|y_i|^{1/2}}\mathbb{E}\left(\frac{1}{nw_i}\lambda^*(\widehat{\beta}_i)\right)^{1/2}.
\end{aligned} \tag{37}
$$

where (a) follows from Equation (11), (b) follows from the condition $|y_i| > |\beta_i^0|/2$ and (36). Notice that

$$
\mathbb{P}\left(w_i^{1/2} < \frac{1}{|\beta_i^0|}\left[\sqrt{\frac{2}{n}\log[1/p^*(0)] + \frac{2\lambda_1}{n}|\beta_i^0|} + \sqrt{\frac{2}{n}\log[1/p^*(0)]}\right]\right) \to 0.
$$

In order to bound $\mathbb{E}\left(\frac{1}{nw_i}\lambda^*(\widehat{\beta}_i)\right)^{1/2}$, notice that

$$
\widehat{\beta}_i = \arg\max_{\beta_i \in \mathbb{R}}\left\{-\frac{w_i n}{2}(y_i - \beta_i)^2 + \log\pi(\beta_i \mid \theta)\right\} = \arg\max_{\beta_i \in \mathbb{R}}\left\{-\frac{1}{2}(\sqrt{w_i n}y_i - \sqrt{w_i n}\beta_i)^2 + \log\pi(\beta_i \mid \theta)\right\}.
$$

Thus, following the same analysis as that for Equation (10), (11) in Section A.1.2, we know that

$$
\begin{aligned}
\mathbb{E}\left(\frac{1}{nw_i}\lambda^*(\widehat{\beta}_i)\right)^{1/2} &\leq \mathbb{E}\left(\frac{\sqrt{nw_i} + \lambda_1}{nw_i}\right)^{1/2} + \mathbb{E}\left(\frac{2\sqrt{nw_i} + \lambda_1}{nw_i}\right)^{1/2} + \frac{1}{\lambda_0} \\
\leq &(1 + \sqrt{2})\mathbb{E}\frac{1}{(nw_i)^{1/4}} + 2\mathbb{E}\left(\frac{\lambda_1}{nw_i}\right)^{1/2} + \frac{1}{\lambda_0},
\end{aligned}
$$

and thus, the right hand size of (37) satisfies

$$
\mathbb{P}\left(w_i^{1/2} < \frac{1}{|\beta_i^0|}\left[\sqrt{\frac{2}{n}\log[1/p^*(0)] + \frac{2\lambda_1}{n}|\beta_i^0|} + \sqrt{\frac{2}{n}\log[1/p^*(0)]}\right]\right) + \frac{1}{y_i^2}\mathbb{E}\left(\frac{1}{nw_i}\lambda^*(\widehat{\beta}_i)\right)^2 \to 0.
$$

Thus $\mathbb{P}_{w_i}\left(\widehat{\beta}_i = 0 \mid y_i\right) \to 0$ holds. □

A.6.3 Inactive Coordinates

Proposition 5. *For w_0 and w_1 defined in (8), when conditioning on $y_i \asymp \frac{1}{\sqrt{n}}$, we have $w_1 \to 0$ and $w_0 \to 1$.*

Proof. The expression for w_1 is given in (30). Again, we consider the four terms in the denominator separately and divide each one of them by $\theta\lambda_1$. For the first term, we write

$$\frac{1}{\theta\lambda_1}\int_0^\infty c_1^{(-)}\phi_1^{(-)}(\beta_i)d\beta_i = e^{-y_i\lambda_1+\lambda_1^2/2n}\left(1-\Phi\left(-\sqrt{n}(y_i-\frac{\lambda_1}{n})\right)\right) \to 1-\Phi\left(-\sqrt{n}y_i\right). \tag{38}$$

For the second term, for any fixed $\epsilon > 0$, we have

$$\begin{aligned}\frac{1}{\theta\lambda_1}\int_0^\infty c_0^{(-)}\phi_0^{(-)}(\beta_i)d\beta_i &= \frac{1}{\theta\lambda_1}\int_0^\infty (1-\theta)\lambda_0\sqrt{\frac{n}{2\pi}}e^{-\frac{n}{2}(\beta_i-y_i)^2-\beta_i\lambda_0}d\beta_i\\ &\geq \frac{1}{\theta\lambda_1}\int_0^{\epsilon y_i}(1-\theta)\lambda_0\sqrt{\frac{n}{2\pi}}e^{-\frac{n(\epsilon-1)^2}{2}y_i^2-\beta_i\lambda_0}d\beta_i\\ &= \frac{1}{\theta\lambda_1}(1-\theta)\sqrt{\frac{n}{2\pi}}e^{-\frac{n(\epsilon-1)^2}{2}y_i^2}\left(1-e^{-\epsilon y_i\lambda_0}\right)\end{aligned}$$

where $e^{-\epsilon y_i\lambda_0} \to 0$. Since the right-hand size term

$$\frac{1}{\theta\lambda_1}(1-\theta)\sqrt{\frac{n}{2\pi}}e^{-\frac{n(\epsilon-1)^2}{2}y_i^2}\left(1-e^{-\epsilon y_i\lambda_0}\right) \to \infty$$

because $y_i \asymp n^{-1/2}$ we know

$$\frac{1}{\theta\lambda_1}\int_0^\infty c_0^{(-)}\phi_0^{(-)}(\beta_i)d\beta_i \to \infty. \tag{39}$$

Regarding the third term in denominator, we obtain

$$\frac{1}{\theta\lambda_1}\int_{-\infty}^0 c_1^{(+)}\phi_1^{(+)}(\beta_i)d\beta_i = e^{y_i\lambda_1+\lambda_1^2/2n}\Phi\left(-\sqrt{n}(y_i+\frac{\lambda_1}{n})\right) \to \Phi\left(-\sqrt{n}y_i\right). \tag{40}$$

Finally, for the fourth term, for any fixed $\epsilon > 0$, we have

$$\begin{aligned}\frac{1}{\theta\lambda_1}\int_{-\infty}^0 c_0^{(+)}\phi_0^{(+)}(\beta_i)d\beta_i &= \frac{1}{\theta\lambda_1}\int_{-\infty}^0 (1-\theta)\lambda_0\sqrt{\frac{n}{2\pi}}e^{-\frac{n}{2}(\beta_i-y_i)^2+\beta_i\lambda_0}d\beta_i\\ &\geq \frac{1}{\theta\lambda_1}\int_{-\epsilon y_i}^0(1-\theta)\lambda_0\sqrt{\frac{n}{2\pi}}e^{-\frac{n(1+\epsilon)^2}{2}y_i^2+\beta_i\lambda_0}d\beta_i\\ &= \frac{1}{\theta\lambda_1}(1-\theta)\sqrt{\frac{n}{2\pi}}e^{-\frac{n(1+\epsilon)^2}{2}y_i^2}\left(1-e^{-\epsilon y_i\lambda_0}\right),\end{aligned}$$

where $e^{-\epsilon y_i\lambda_0} \to 0$. Since the right hand side term satisfies

$$\frac{1}{\theta\lambda_1}(1-\theta)\sqrt{\frac{n}{2\pi}}e^{-\frac{n(1+\epsilon)^2}{2}y_i^2}\left(1-e^{-\epsilon y_i\lambda_0}\right) \to \infty$$

we know

$$\frac{1}{\theta\lambda_1}\int_{-\infty}^{0} c_0^{(+)}\phi_0^{(+)}(\beta_i)d\beta_i \to \infty. \tag{41}$$

Combining (38), (39), (40) and (41), we know that in (30), the denominator $\to \infty$ and thus the numerator $\to 1$. Thus $w_1 \to 0$ and $w_0 = 1 - w_1 \to 1$. $\square$

We denote

$$d_{\mathrm{TV}}(P, Q) = \sup_{A\in\mathcal{F}} |P(A) - Q(A)|,$$

which is the total variation distance between two probability measures P and Q on a sigma-algebra $\mathcal{F}$ of subsets of the sample space. We have the following result:

Proposition 6. *When n is sufficiently large, conditioning on the event $|y_i| \asymp \frac{1}{\sqrt{n}}$, we have*

$$d_{TV}\left(\pi\left(\lambda_0\beta_i \,|\, y_i, \gamma_i = 0\right), \text{Laplace}(1)\right) \to 0,$$

where Laplace(1) is the Laplace distribution whose density is $f(t) = \frac{1}{2}e^{-|t|}$.

Proof. Notice that

$$\pi\left(\beta_i \,|\, y_i, \gamma_i = 0\right) \propto \pi\left(y_i \,|\, \beta_i, \gamma_i = 0\right)\pi\left(\beta_i \,|\, \gamma_i = 0\right) \propto e^{-\frac{n}{2}(\beta_i - y_i)^2}e^{-|\beta_i|\lambda_0}.$$

Thus, letting $\beta_i' = \lambda_0\beta_i$, since $\lambda_0 \asymp p^d$ with $d \geq 2$ we have for any fixed β_i'

$$\pi\left(\beta_i' \,|\, y_i, \gamma_i = 0\right) \propto e^{-\frac{n}{2}(\beta_i'/\lambda_0 - y_i)^2}e^{-|\beta_i'|} \to e^{-\frac{n}{2}(y_i)^2}e^{-|\beta_i'|}$$

which implies that

$$\pi\left(\beta_i' \,|\, y_i, \gamma_i = 0\right) \to e^{-|\beta_i'|}.$$

From Scheffé (1947), we know that β_i' converges in total variation to Laplace(1). $\square$

Proposition 7. *Consider the fixed WBB estimator $\widehat{\beta}_i$. Conditioning on $|y_i| \asymp n^{-1/2}$, we have*

$$\mathbb{P}_{w_i}\left(\widehat{\beta}_i = 0 \,|\, y_i\right) \to 1.$$

Proof. The definition of the fixed WBB sample $\widehat{\beta}_i$ is in (11). Notice that

$$\Delta_{w_i} = \inf_{t>0}\left[t/2 - \rho(t \mid \theta)/(nw_it)\right] = \frac{1}{nw_i}\inf_{t>0}\left[nw_it/2 - \rho(t \mid \theta)/t\right] \asymp \frac{\sqrt{2nw_i\log[1/p^*(0)]}}{nw_i}.$$

Then, when n is sufficiently large,

$$\begin{aligned}&\mathbb{P}_{w_i}\left(\widehat{\beta}_i = 0 \,|\, y_i\right) \geq \mathbb{P}_{w_i}\left(|y_i| \leq \Delta_{w_i} \,|\, y_i\right)\\ &\geq \mathbb{P}_{w_i}\left(|y_i| \leq \frac{1}{2}\frac{\sqrt{2nw_i\log[1/p^*(0)]}}{nw_i} \,|\, y_i\right) \geq \mathbb{P}_{w_i}\left(w_i \leq \sqrt{\frac{\log[1/p^*(0)]}{2n|y_i|^2}} \,|\, y_i\right),\end{aligned}$$

where the right-hand size

$$\mathbb{P}_{w_i}\left(w_i \leq \sqrt{\frac{\log[1/p^*(0)]}{2n|y_i|^2}} \,|\, y_i\right) \to 1$$

since $y_i \asymp n^{-1/2}$ and $\log[1/p^*(0)] \to \infty$. Thus,

$$\mathbb{P}_{w_i}\,(\widehat{\beta}_i = 0 \,|\, y_i) \to 1. \qquad \square$$

Remark A.1. *The random WBB is equivalent to the fixed WBB by setting weights to be $\boldsymbol{w}/w_p$ where w_p is the weight assigned to the prior term. Thus, using exactly the same arguments as fixed WBB, we can prove: $\mathbb{P}_{w_i}\left(\widehat{\beta}_i^{random} = 0 \,|\, y_i\right) \to 1$.*

B Details of Connection to NPL in Section 4.2

In Algorithm 3, we define the loss function l as

$$l(\boldsymbol{x}_i, y_i, \boldsymbol{\beta}) = -\frac{1}{2\sigma^2}(y_i - \boldsymbol{x}_i^T\boldsymbol{\beta})^2 + \frac{1}{n}\log\left[\int_\theta \prod_{j=1}^p \pi(\beta_j \,|\, \theta)d\pi(\theta)\right]. \tag{42}$$

Motivation for the Prior. For paired data $(\boldsymbol{x}_i, y_i)$, Fong et al. (2019) uses independent prior which assumes that y_i does not depend on $\boldsymbol{x}_i$:

$$\text{Prior 1: } \widetilde{\boldsymbol{x}}_k \sim \hat{F}_n(\boldsymbol{x}) = \frac{1}{n}\sum_{i=1}^n \delta(\boldsymbol{x}_i), \quad \widetilde{y}_k \,|\, \widetilde{\boldsymbol{x}}_k \sim N(0, \sigma^2).$$

However, this choice of F_π might be problematic when the sample size n is small. When n is small, a well-specified prior can help us better estimate $\boldsymbol{\beta}$ but this independent prior shrinks all coeffcients towards zero and will result in bias (see Figure 1).

One possible solution is to use $y = \boldsymbol{x}^T\widehat{\boldsymbol{\beta}} + \epsilon$ where $\widehat{\boldsymbol{\beta}}$ is the MAP of $\boldsymbol{\beta}$ under SSL penalty. This choice of $f_\pi(y|x)$ has an Empirical Bayes flavor (Martin and Walker (2014)). However, it includes information only from the posterior mode, ignoring shape information contained in the posterior variance. We could consider incorporating such information by adding noise $\boldsymbol{\mu}$ to $\widehat{\boldsymbol{\beta}}$. We would want $\boldsymbol{\mu}$ to be centered at the origin but not too far away from the origin. One choice that comes to mind is the spike distribution. So the prior F_π becomes

$$\widetilde{\boldsymbol{x}}_k \sim \hat{F}_n(\boldsymbol{x}) = \frac{1}{n}\sum_{i=1}^n \delta(\boldsymbol{x}_i), \quad \widetilde{y}_k \,|\, \widetilde{\boldsymbol{x}}_k = \widetilde{\boldsymbol{x}}_k^T(\widehat{\boldsymbol{\beta}} + \boldsymbol{\mu}) + \epsilon$$

where $\boldsymbol{\mu} \sim$ Spike and $\epsilon \sim N(0, \sigma^2)$. If $\widehat{\boldsymbol{\beta}}$ is close enough to the truth, we have $\widetilde{\boldsymbol{x}}_k^T\widehat{\boldsymbol{\beta}} \approx \widetilde{\boldsymbol{x}}_k^T\boldsymbol{\beta}_0$. Since $y_i = \boldsymbol{x}_i^T\boldsymbol{\beta}_0 + \epsilon_i$ and $\epsilon_i \overset{d}{=} \epsilon$, we can set $\widetilde{y}_k \,|\, \widetilde{\boldsymbol{x}}_k = y_i + \boldsymbol{x}_i^T\boldsymbol{\mu}$ where i satisfies $\widetilde{\boldsymbol{x}}_k = \boldsymbol{x}_i$. Then the above prior becomes

$$\text{Prior 2: } \widetilde{\boldsymbol{x}}_k \sim \hat{F}_n(\boldsymbol{x}) = \frac{1}{n}\sum_{i=1}^n \delta(\boldsymbol{x}_i), \quad \widetilde{y}_k \,|\, \widetilde{\boldsymbol{x}}_k = y_i + \boldsymbol{x}_i^T\boldsymbol{\mu} \text{ where } i \text{ satisfies } \widetilde{\boldsymbol{x}}_k = \boldsymbol{x}_i.$$

Derivation for Equation (17) When choosing $m = n$ in Algorithm 3, the NPL posterior of Fong et al. (2019) using Prior 2 becomes

$$
\begin{aligned}
\widetilde{\boldsymbol{\beta}}^t &= \arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2}\sum_{i=1}^n w_i (y_i - \boldsymbol{x}_i^T\boldsymbol{\beta})^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n \tilde{w}_i (y_i + \boldsymbol{x}_i'\boldsymbol{\mu} - \boldsymbol{x}_i'\boldsymbol{\beta})^2 + \frac{1}{n}\log\left[\int_\theta \prod_{j=1}^p \pi(\beta_j \mid \theta) d\pi(\theta)\right] \right\} \\
&= \arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2}\sum_{i=1}^n (w_i + \tilde{w}_i)\left(y_i + \frac{\tilde{w}_i}{w_i+\tilde{w}_i}\boldsymbol{x}_i^T\boldsymbol{\mu} - \boldsymbol{x}_i^T\boldsymbol{\beta}\right)^2 + \frac{1}{n}\log\left[\int_\theta \prod_{j=1}^p \pi(\beta_j \mid \theta) d\pi(\theta)\right] \right\} \\
&\approx \arg\max_{\boldsymbol{\beta}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2}\sum_{i=1}^n (w_i + \tilde{w}_i)\left(y_i + \frac{c}{c+n}\boldsymbol{x}_i^T\boldsymbol{\mu} - \boldsymbol{x}_i^T\boldsymbol{\beta}\right)^2 + \frac{1}{n}\log\left[\int_\theta \prod_{j=1}^p \pi(\beta_j \mid \theta) d\pi(\theta)\right] \right\} \\
&\overset{\tilde{\boldsymbol{\beta}}=\boldsymbol{\beta}-[c/(c+n)]\boldsymbol{\mu}}{=} \arg\max_{\tilde{\boldsymbol{\beta}}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2}\sum_{i=1}^n w_i^*(y_i - \boldsymbol{x}_i^T\tilde{\boldsymbol{\beta}})^2 + \log\left[\int_\theta \prod_{j=1}^p \pi\left(\tilde{\beta}_j + \frac{c}{c+n}\mu_j \mid \theta\right) d\pi(\theta)\right] \right\} \\
&\quad + \frac{c}{c+n}\boldsymbol{\mu},
\end{aligned}
$$

where $w_i^* = n(w_i + \tilde{w}_i)$. Since $\boldsymbol{\mu}$ and $-\boldsymbol{\mu}$ follow the same distribution, define $\boldsymbol{\mu}^* = -\boldsymbol{\mu}$ and

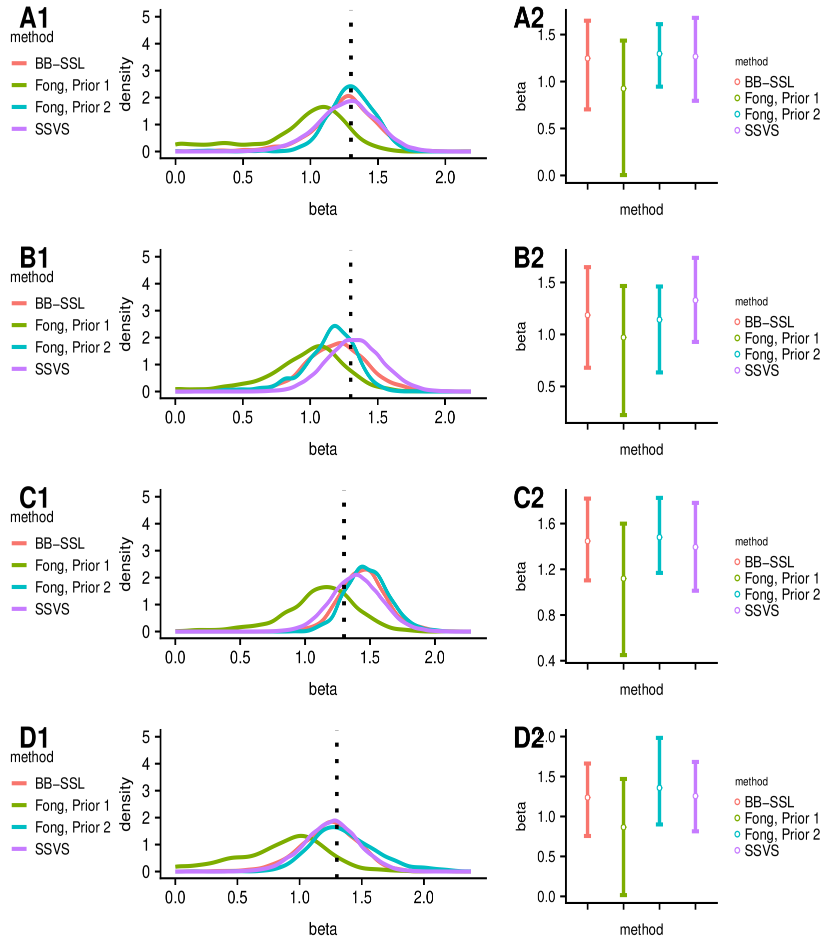
$$
\tilde{\boldsymbol{\beta}}^t \overset{\mathrm{D}}{=} \arg\max_{\tilde{\boldsymbol{\beta}}\in\mathbb{R}^p} \left\{ -\frac{1}{2\sigma^2}\sum_{i=1}^n w_i^*(y_i - \boldsymbol{x}_i^T\tilde{\boldsymbol{\beta}})^2 + \log\left[\int_\theta \prod_{j=1}^p \pi\left(\tilde{\beta}_j - \frac{c}{c+n}\mu_j^* \mid \theta\right) d\pi(\theta)\right] \right\} - \frac{c}{c+n}\boldsymbol{\mu}^*,
$$

where $(\boldsymbol{w}_{1:n}, \tilde{w}_{1:n}) \sim \mathrm{Dir}(1, \cdots, 1, c/n, c/n, \cdots, c/n)$ and thus $(w_1^*, w_2^*, \cdots, w_n^*) \sim n\mathrm{Dir}(1 + c/n, \cdots, 1 + c/n)$.

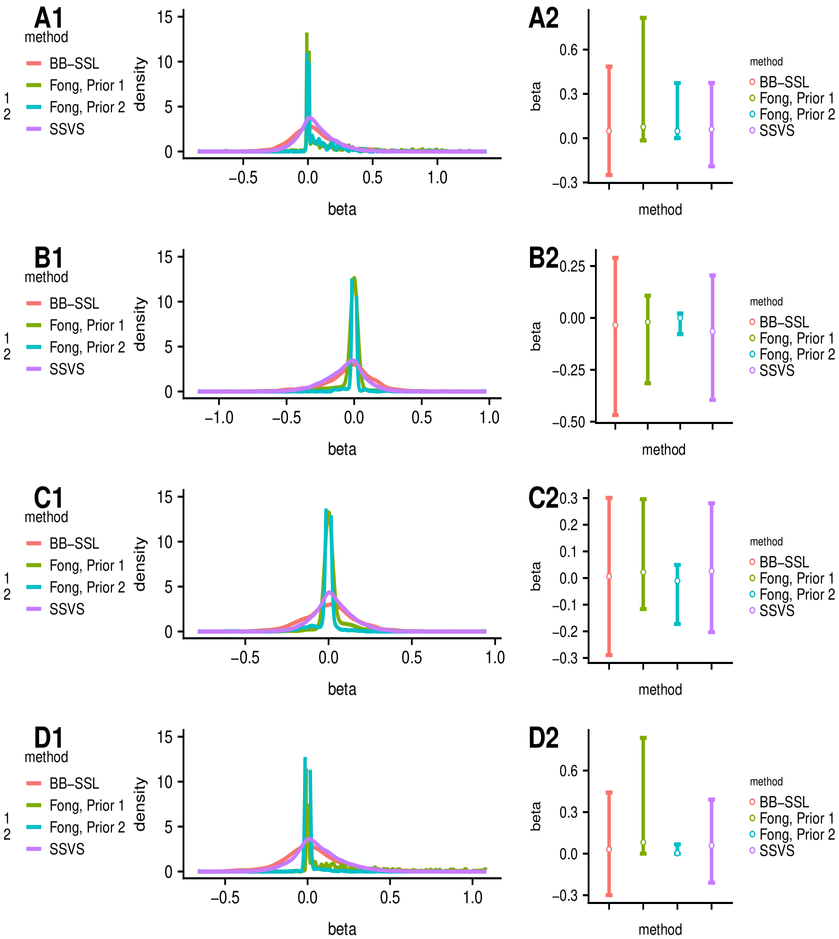
C Another Gibbs Sampler with Complexity $O(np)$

A referee suggested a one-site Gibbs sampler which can be implemented with our continuous Spike-and-Slab LASSO prior and which can potentially improve computational efficiency. The sampler is reminiscent of Geweke (1996) who proposed a variant for point mass spikes to avoid getting stuck after generating $\beta_i = 0$. However, although George and McCulloch (1997) mentioned that this implementation might also be useful under continuous spike-and-slab priors (where the variance ratio of slab to spike is large), we find this implementation to be rarely used for continuous priors. One-site Gibbs samplers are expected to converge slower and yield higher autocorrelation, which is one of the reasons why block-Gibbs samplers have been preferred. We nevertheless explore this Gibbs sampler further.

The algorithm to implement Geweke (1996)'s Gibbs sampler for the Spike-and-Slab LASSO prior is outlined in Algorithm 1 and we further refer to it as Gibbs II. We calculate the vector $\boldsymbol{r}_j$ through $\boldsymbol{r}_j = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) + \boldsymbol{X}_j\beta_j$ and $\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}$ is calculated once before running MCMC and updated after each update on β_j. When $p > n$, Algorithm 1 has complexity $O(np)$ compared to $O(p^3)$ for SSVS1 and $O(n^2p)$ for SSVS2. However, generating from the Truncated Normal distribution typically requires an accept-reject algorithm (Geweke,



(a) Active predictors, from top to bottom: $\beta_1, \beta_4, \beta_7, \beta_{10}$



(b) Inactive predictors, from top to bottom: $\beta_2, \beta_5, \beta_8, \beta_{11}$

Figure 1: Comparison of Fong et al. (2019)'s algorithm 2 with different choice of priors and loss function versus BB-SSL and SSVS under simulated dataset. We use $n = 50, p = 12$, $\boldsymbol{\beta} = (1.3, 0, 0, 1.3, 0, 0, 1.3, 0, 0, 1.3, 0, 0)^T$, predictors are grouped into 4 blockes with correlation coefficient $\rho = 0.6$, $\alpha = 2$, $m = 50$, $c = 10$, $\lambda_0 = 7, \lambda_1 = 0.15$. The dotted vertical lines in (a) mark the truth $\beta_i = 1.3, i = 1, 4, 7, 10$.

method	time per 100 iter (sec)	ESS per 100 iter	lag-one auto correlation on β_j's	time per 100 ESS (sec)
BB-SSL	0.7	100	-0.002	0.7
SSVS1	40	98	0.006	40.82
SSVS2	3.2	98	0.006	3.27
Gibbs II	3.1	83	0.100	3.73

Table 1: Performance comparison of Algorithm 1 (referred to as Gibbs II) against other methods. We set $n = 100, p = 1000$, variables are equi-correlated with $\rho = 0.9$, and the active coefficients are $\beta_{active} = (2, 4, -4, 6)'$.

1991) which creates some computational bottlenecks. We utilize the R package 'truncnorm' (Trautmann et al., 2015) which implements this accept-reject algorithm in C. The performance of Algorithm 1 against other methods is summarized in Table 1. We see that the per-iteration running time of Gibbs II is similar to SSVS2 (using the trick in Bhattacharya et al. (2016)), yet the serial correlation is higher than for both implementations of SSVS. Thus, contrary to the point-mass spike, Algorithm 1 (which is based on Geweke (1996)) trades computational efficiency for serial correlation. In general, the time used for generating the same number of effective samples is slightly longer than SSVS2, but improves significantly over SSVS1. From this observation, we believe Gibbs II is yet another useful method to speed up the standard SSVS1 under the Spike-and-Slab LASSO prior.

A less relevant, but perhaps interesting, observation is that since the Spike-and-Slab LASSO prior is a mixture of Laplace distributions, SSVS1 (which augments the data with τ_j's) can be seen as a generalization of the MCMC algorithm in Park and Casella (2008) while Gibbs II can be seen as a spike-and-slab version of the MCMC algorithm in Hans (2009). Our conclusion from this exercise is that, for the Spike-and-Slab LASSO prior, BB-SSL is actually doing better (both in terms of serial correlation and timing) compared with Gibbs II.

Algorithm 1 Gibbs II

Set: $\lambda_0 \gg \lambda_1$, $a, b > 0$, T (number of MCMC iterations), B (number of samples to discard as burn-in).

Initialize: $\boldsymbol{\beta}^0$ (e.g. LASSO solution after 10-fold cross validation) and $\boldsymbol{\tau}^0$.

for $t = 1, 2, \cdots, T$ **do**

for $j = 1, 2, \cdots, p$ **do**

(a) Sample $\gamma_j \sim \pi\left(\gamma_j|\theta, \boldsymbol{\beta}_{-j}, \boldsymbol{\gamma}_{-j}, \boldsymbol{y}\right) = \text{Bernoulli}\left(\frac{\pi_1}{\pi_1+\pi_0}\right)$, where

$$
\begin{aligned}
&\pi_1 = \theta c_1, \quad \pi_0 = (1-\theta)c_0,\\
&c_1 = \lambda_1 \left[e^{(\mu_+^1)^2/(2s_j)}\mathbb{P}\left(N(\mu_+^1, s_j) \geq 0\right) + e^{(\mu_-^1)^2/(2s_j)}\mathbb{P}\left(N(\mu_-^1, s_j) \leq 0\right)\right],\\
&c_0 = \lambda_0 \left[e^{(\mu_+^0)^2/(2s_j)}\mathbb{P}\left(N(\mu_+^0, s_j) \geq 0\right) + e^{(\mu_-^0)^2/(2s_j)}\mathbb{P}\left(N(\mu_-^0, s_j) \leq 0\right)\right],\\
&\mu_-^1 = \frac{\boldsymbol{r}_j^T\boldsymbol{X}_j + \sigma^2\lambda_1}{\|\boldsymbol{X}_j\|_2^2}, \quad \mu_+^1 = \frac{\boldsymbol{r}_j^T\boldsymbol{X}_j - \sigma^2\lambda_1}{\|\boldsymbol{X}_j\|_2^2}, \quad \boldsymbol{r}_j = \boldsymbol{y} - \sum_{k\neq j}\boldsymbol{X}_k\beta_k,\\
&\mu_-^0 = \frac{\boldsymbol{r}_j^T\boldsymbol{X}_j + \sigma^2\lambda_0}{\|\boldsymbol{X}_j\|_2^2}, \quad \mu_+^0 = \frac{\boldsymbol{r}_j^T\boldsymbol{X}_j - \sigma^2\lambda_0}{\|\boldsymbol{X}_j\|_2^2}, \quad s_j = \sigma^2\|\boldsymbol{X}_j\|_2^{-2}.
\end{aligned}
$$

(b) Sample $\beta_j \sim \pi(\beta_j|\gamma_j, \theta, \boldsymbol{\beta}_{-j}, \boldsymbol{\gamma}_{-j}, \boldsymbol{y}) = u_0\text{TruncNormal}^-\left(\mu_-^{\gamma_j}, s_j\right) + (1 - u_0)\text{TruncNormal}^+\left(\mu_+^{\gamma_j}, s_j\right)$ where

$$
\begin{aligned}
&u_0 = \frac{\mathbb{P}\left(N(\mu_-^{\gamma_j}, s_j) \leq 0\right)}{\mathbb{P}\left(N(\mu_-^{\gamma_j}, s_j) \leq 0\right) + \mathbb{P}\left(N(\mu_+^{\gamma_j}, s_j) \geq 0\right)\exp\left\{-\frac{2\boldsymbol{r}_j^T\boldsymbol{X}_j\lambda_{\gamma_j}}{\|\boldsymbol{X}_j\|_2^2}\right\}},\\
&\text{TruncNormal}^-(\mu, s) = \mathbb{P}\left(X \mid X \leq 0\right) \quad \text{where} \quad X \sim N(\mu, s),\\
&\text{TruncNormal}^+(\mu, s) = \mathbb{P}\left(X \mid X \geq 0\right) \quad \text{where} \quad X \sim N(\mu, s).
\end{aligned}
$$

end

(c) Sample $\theta \sim \text{Beta}(\sum_{j=1}^p \gamma_j + a, p - \sum_{j=1}^p \gamma_j + b)$.

end

Return: $\boldsymbol{\beta}^t, \boldsymbol{\gamma}^t, \theta^t$ where $t = B+1, B+2, \cdots, T$.

D Details on Computational Complexity Analysis

Below are details for the computational complexity analysis as shown in Table 1 of the main text.

BB-SSL, WBB1, WBB2. BB-SSL uses R package `SSLASSO` (Ročková and Moran (2017)) to implement the coordinate-descent algorithm in Ročková and George (2018). It iteratively updates $\boldsymbol{\beta}$ until convergence. At each iteration, we first update the active coordinates, then the candidate coordinates, and finally the inactive coordinates. The total number of iterations is limited to a pre-defined number. There are two ways to update each coordinate, one way is to keep track of a residual vector and for each coordinate we compute the inner product between the residual vector and $\boldsymbol{X}_j$ – this takes $O(n)$; another way is to pre-compute the Gram matrix and for each coordinate, we calculate the inner product between $\boldsymbol{X}^T\boldsymbol{X}_j$ and $\hat{\boldsymbol{\beta}}$ – this takes $O(p)$. For both ways, we update θ every c iterations where each update is $O(p)$. So for a single value of λ_0, SSLASSO is $O(\min(\text{maxiter} \times p(n + \frac{p}{c_1}), (n + \text{maxiter}) \times p^2))$. For a sequence of λ_0's, complexity is $O(L \times \min(\text{maxiter} \times p(n + \frac{p}{c_1}), (n + \text{maxiter}) \times p^2))$ where L is the length of λ_0's. Usually if the biggest λ_0 is large enough, we would expect that the larger λ_0's can reach convergence quickly using the estimated $\boldsymbol{\beta}$ from previous λ_0's, so usually it takes less than $O\left(L \times \min\left(\text{maxiter} \times p(n + \frac{p}{c_1}), (n + \text{maxiter}) \times p^2\right)\right)$.

SSVS1. The computational complexity for Algorithm 1 is $O(p^3)$ per iteration when $p > n$. Under this setting, we use the Woodbury matrix identity to simplify the matrix multiplication $(\boldsymbol{X}^T\boldsymbol{X} + \boldsymbol{D}_\tau^{-1})^{-1} = \boldsymbol{D}_\tau - \boldsymbol{D}_\tau\boldsymbol{X}^T(\boldsymbol{I}_n + \boldsymbol{X}\boldsymbol{D}_\tau\boldsymbol{X}^T)^{-1}\boldsymbol{X}\boldsymbol{D}_\tau$, whose complexity is $O(p^2n)$.

SSVS2. Using Bhattacharya et al. (2016)'s matrix inversion formula to generate the p-dimensional multivatiate Gaussian takes $O(n^2p + n^3) = O(n^2\max(n,p))$.

Skinny Gibbs. We modified Skinny Gibbs to sample from the posterior using SSL prior. Theoretically it is of complexity $O(np)$. However, sometimes when n is relatively small, we observe that the running time of Skinny Gibbs is slower than Bhattacharya's method. This is because Skinny Gibbs involves an $O(n)$ matrix product for each coordinate and the update for each coordinate is implemented via for-loop, whereas in Bhattacharya's method the $O(n^2p)$ operation is one matrix product which is very efficiently optimized in R. We will see that as n increases, this problem diminishes and Skinny Gibbs becomes faster than Bhattacharya's method.

WLB. Generating each WLB sample involves solving a least square problem whose complexity is $O(p^2n)$ when $p \leq n$. WLB is not applicable when $p > n$.

Setting	**Block-wise** $\rho = 0.6$, $\beta_{active} = (1.3, 1.3, 1.3, 1.3)'$									**Block-wise** $\rho = 0.9$, $\beta_{active} = (1.3, 1.3, 1.3, 1.3)'$								
	β_j's						γ_j's		Model	β_j's						γ_j's		Model
Metric	KL		JD of 90% CI		'Bias'		'Bias'		HD	KL		JD of 90% CI		'Bias'		'Bias'		HD
	+	−	+	−	+	−	+	−	all	+	−	+	−	+	−	+	−	all
Skinny Gibbs	0.24	0.25	0.29	0.34	**0.08**	0.07	**0.02**	0.0007	**0**	0.68	0.33	0.37	0.39	0.15	0.08	**0.04**	0.009	**0.2**
WBB1	0.22	1.66	0.28	0.62	0.09	0.04	**0.02**	0.0004	**0**	0.67	2.73	0.41	0.89	0.14	**0.08**	0.05	0.003	**0.2**
WBB2	0.20	1.69	0.28	0.62	**0.08**	0.04	**0.02**	0.0004	**0**	0.56	2.69	0.33	0.87	0.14	**0.07**	0.05	0.003	**0.2**
BB-SSL	**0.15**	**0.07**	**0.21**	**0.24**	0.09	**0.03**	**0.02**	**0.0003**	**0**	**0.14**	**0.09**	**0.21**	**0.24**	**0.13**	**0.07**	**0.04**	**0.002**	**0.2**

Setting	**Equi-correlation** $\rho = 0.6$, $\beta_{active} = (1.3, 1.3, 1.3, 1.3)'$									**Equi-correlation** $\rho = 0.9$, $\beta_{active} = (1, 1.5, -1.5, 2)'$								
	β_j's						γ_j's		Model	β_j's						γ_j's		Model
Metric	KL		JD of 90% CI		'Bias'		'Bias'		HD	KL		JD of 90% CI		'Bias'		'Bias'		HD
	+	−	+	−	+	−	+	−	all	+	−	+	−	+	−	+	−	all
Skinny Gibbs	0.11	0.20	0.20	0.31	**0.06**	0.06	0.05	0.002	**0.2**	0.66	0.29	0.27	0.40	0.30	0.06	0.01	0.016	1
WBB1	0.15	1.97	0.22	0.75	0.09	0.05	0.02	**0.001**	**0.2**	1.18	2.72	0.37	0.88	0.14	0.06	**0.01**	**0.001**	**0.2**
WBB2	0.14	1.98	0.23	0.75	0.09	0.05	0.02	**0.001**	**0.2**	1.20	**2.70**	0.37	0.87	0.14	**0.05**	**0.01**	**0.001**	**0.2**
BB-SSL	**0.10**	**0.07**	**0.19**	**0.24**	0.08	**0.03**	**0.02**	0.002	**0.2**	**0.23**	**0.07**	0.51	**0.20**	**0.12**	**0.05**	**0.01**	**0.001**	**0.2**

Table 2: Evaluation of approximation properties (relative to SSVS) in the low-dimensional setting with $n = 50$ and $p = 12$ based on 10 independent runs. We set $\lambda_0 = 7, \lambda_1 = 0.15$. The best performance is marked in bold font. KL is the Kullback-Leibler divergence, JD is the Jaccard distance of credible intervals (CI), HD is the Hamming distance of the median models. 'Bias' refers to the l_1 distance of estimated posterior means. We denote with $*$ all numbers smaller than 0.0001, with $+$ an average over active coordinates, and with $-$ an average over inactive coordinates.

E Additional Experimental Results

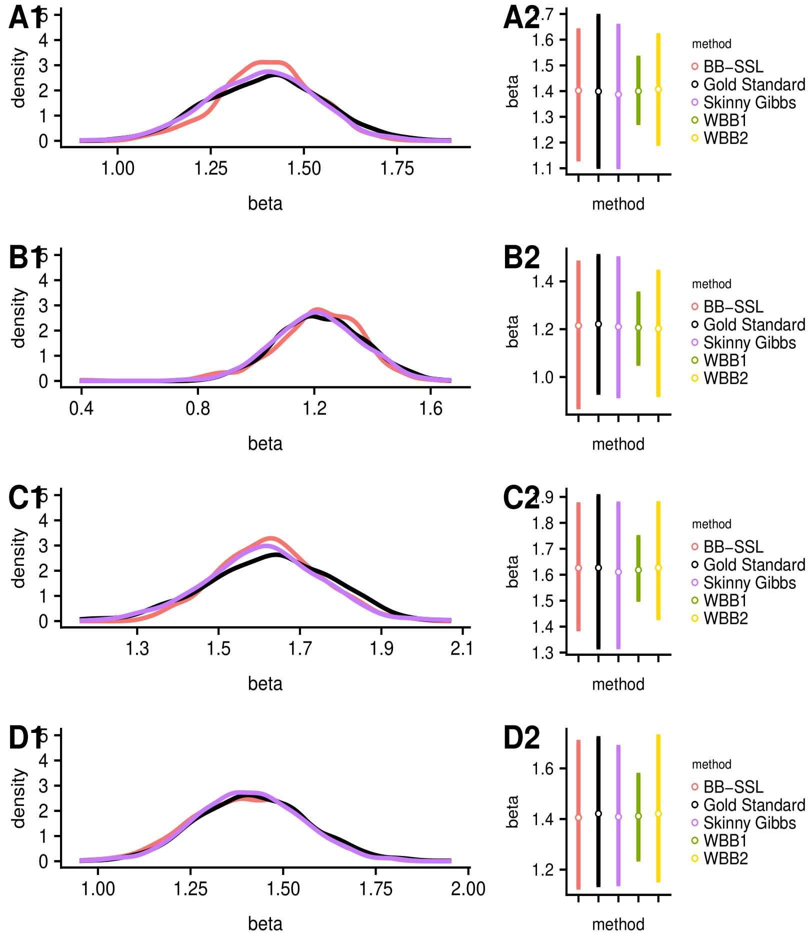
E.1 Low Dimensional Setting

When all covariates are mutually independent, we first investigate the marginal density of β_i's, which is shown in Figure 2. We find that all methods perform well for active β_i's. WBB1 and WBB2 are doing poorly for inactive β_i's. For the marginal mean of γ_i's, as shown in Figure 3, all methods perform well. All methods can detect over 95% of models. In Table 2, we quantify the performance of each method in the low-dimensional setting. In the settings we tried, BB-SSL consistently has the best performance.

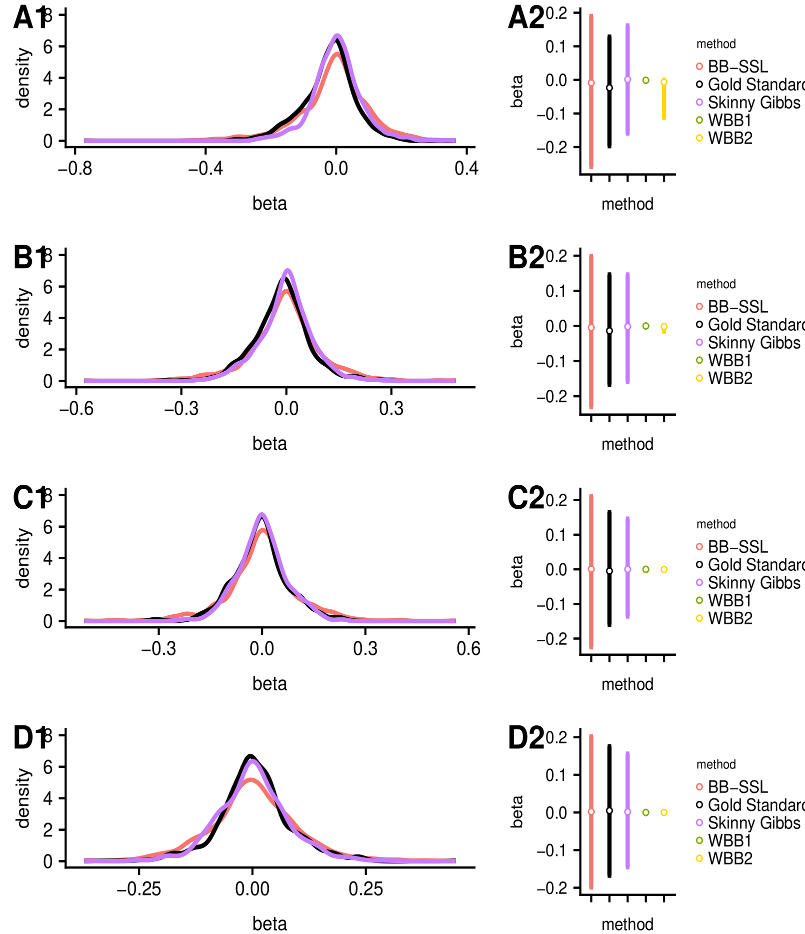
E.2 High Dimensional, Block-wise Correlation Structure

Figure 4 shows the posterior for β_i's when $\rho = 0$, Figure 5 is for $\rho = 0.6$ and Figure 6 is for $\rho = 0.9$ in the block-wise correlated setting. Skinny Gibbs slightly underestimates the variance for active coordinates. WBB1 and WBB2 does poorly for inactive coordinates. In general BB-SSL does well. Figure 7a shows the marginal inclusion probabilities (MIP) when $\rho = 0.6$ ($\rho = 0, 0.9$ are included in the main text) and we see that all methods perform well except that Skinny Gibbs sometimes overestimates MIP.

In the more extreme case where $\rho = 0.99$, SSVS suffers severely from local entrapment with the signal $\beta_{active} = (1, 2, -2, 3)'$, so we set a stronger signal with $\beta_{active} = (2, 4, -4, 6)'$. All the other settings are the same as before. We run two SSVS chains, one initialized at the truth and the other at origin. In Figure 8, we observe discrepancies in the posterior approximation for the two chains. This is why SSVS may not be a reliable gold standard to make comparisons with. We nevertheless observe that BB-SSL is close to the SSVS



(a) Active predictors, from top to bottom: $\beta_1, \beta_4, \beta_7, \beta_{10}$



(b) Inactive predictors, from top to bottom: $\beta_2, \beta_5, \beta_8, \beta_{11}$

Figure 2: Estimated posterior density (left panel) and credible intervals (right panel) of β_i's in the low-dimensional independent case. We have $n = 50, p = 12, \beta_{active} = (1.3, 1.3, 1.3, 1.3)', \lambda_0 = 7, \lambda_1 = 0.15, \rho = 0$. Each method has 5 000 sample points (after thinning for SSVS and Skinny Gibbs). BB-SSL is fitted using a single value $\lambda_0 = 7$. Since WBB1 and WBB2 produce a point mass at zero, we exclude them from density comparisons.

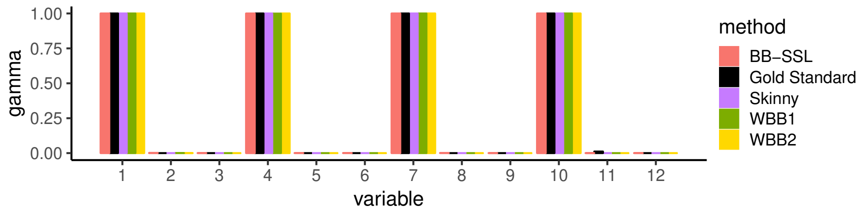


Figure 3: Mean of $\gamma_i, i = 1, 2, \cdots, 12$ in low-dimensional, independent case. $n = 50, p = 12, \beta_{active} = (1.3, 1.3, 1.3, 1.3)$ and predictors mutually independent. Each method is (thinned to) 1,000 sample points. $\lambda_0 = 13, \lambda_1 = 0.05$. `SSLASSO` is fitted using a single λ_0.

approximation obtained by initialization at the truth. Figure 7b shows the MIP and no method performs perfect in this extreme setting.

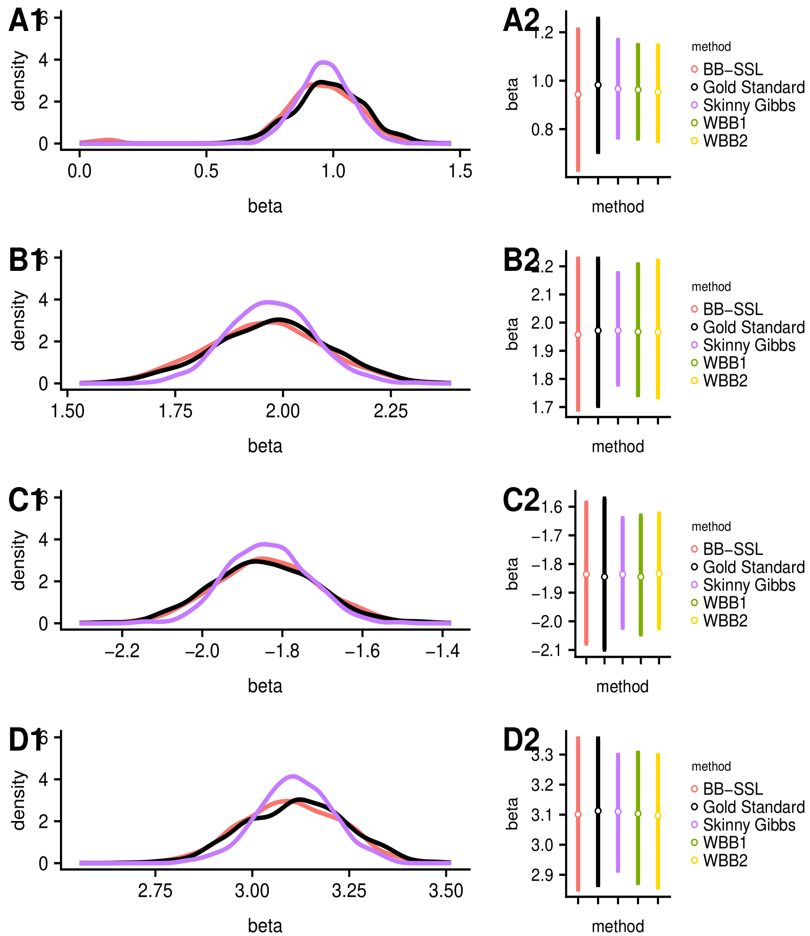
E.3 High Dimensional, Equi-correlation Structure

When all covariates are equi-correlated with $\rho = 0.9$, SSVS appears to suffer severely from a local trap when $\beta_0 = 2, \beta_2 = 3, \beta_3 = -3$ and $\beta_4 = 4$. Thus, we also try a setting with larger signals $\beta_0 = 2, \beta_2 = 4, \beta_3 = -4$ and $\beta_4 = 6$. Results for the smaller signal $\beta_{active} = (2, 3, -3, 4)'$ are summarized in Figure 9 and results for the larger signal $\beta_{active} = (2, 4, -4, 6)'$ are in Figure 10. We show results for both SSVS initializations (one at original and the other at the truth) and observe discrepancies in the posterior approximation. This is why SSVS may not be a reliable gold standard to make comparisons with. We nevertheless observe that BB-SSL is close to the SSVS approximation obtained by initialization at the truth.

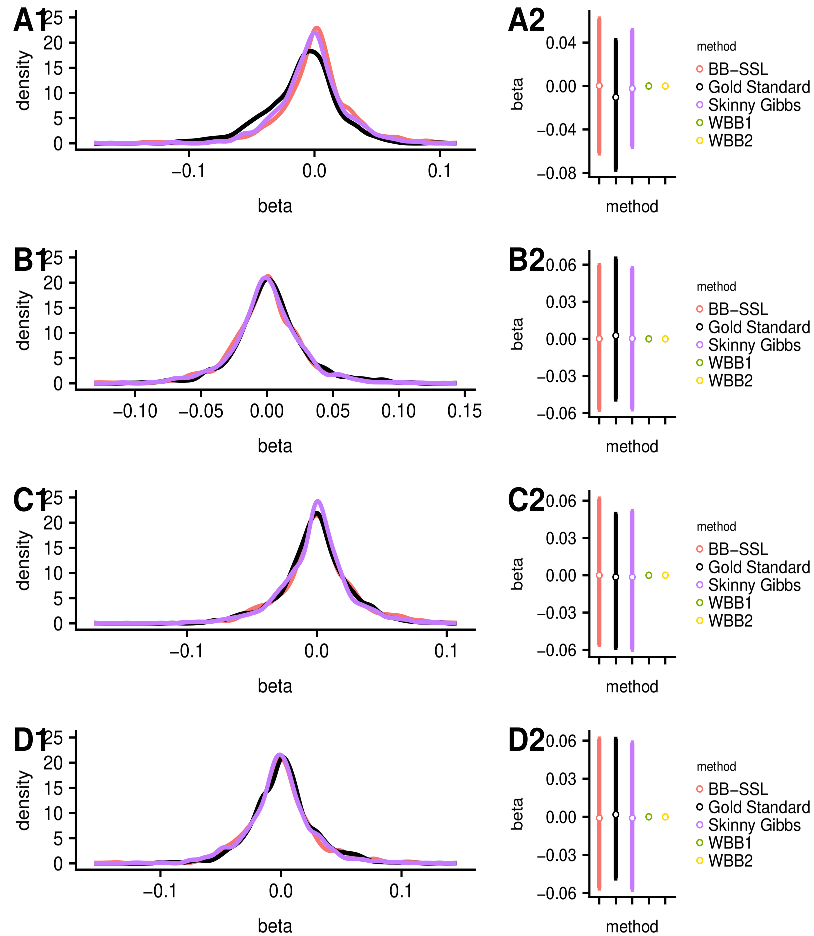
E.4 Influence of α on the Posterior Approximation

In this section we investigate the influence of α on the BB-SSL posterior under (a) high-dimensional, block-wise correlated ($\rho = 0.6$) setting, as shown in Figure 11, and (b) Durable Goods Marketing Data Set, as shown in Figure 12.

Figure 12 shows that the variability of BB-SSL draws decreases as α increases. In fact, when α is large (for instance as in Figure 11), the posterior approximation will not be materially different from setting w_i's equal to one. In practice, however, one would not choose α as large since our recommended choice is $\alpha \asymp \sigma^2 \log[\frac{(1-\theta)\lambda_0}{\theta\lambda_1}]$ (according to Corollary 4.1). With this choice, as n and p become larger, w_i's are more and more concentrated towards 1 and μ_i's are more concentrated towards 0 (assuming λ_0 as in Theorem 4.1 or Theorem 4.3). However, the variance of μ_i's decreases at a faster rate than that of w_i's. Consequently, as n and p grow larger the randomness from μ_i becomes more and more negligible relative to μ_i's.

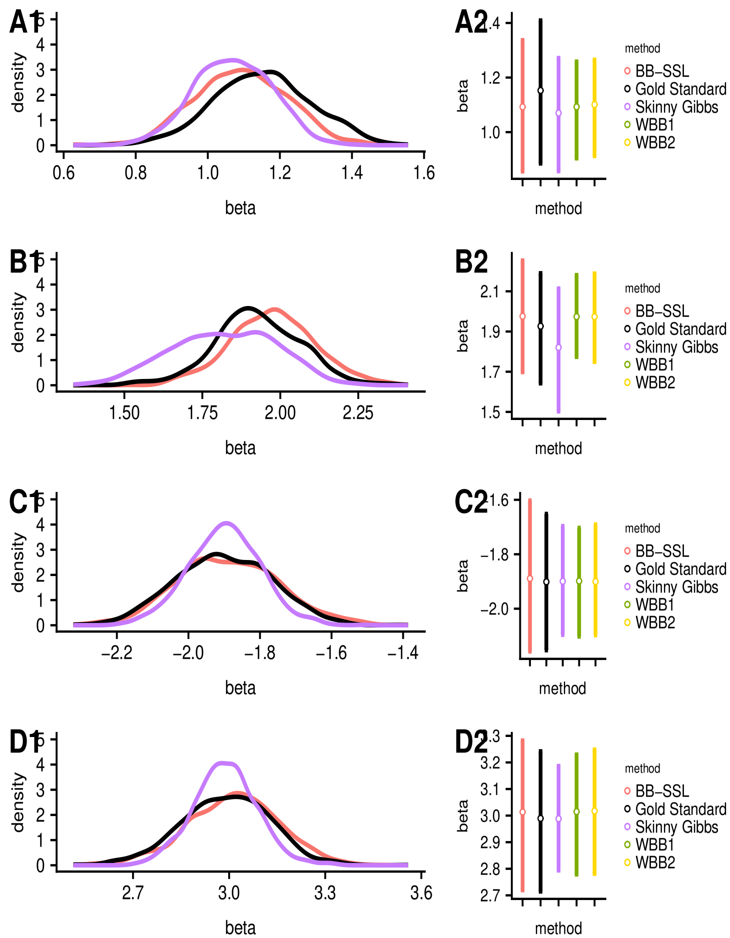


(a) Active predictors, from top to bottom: $\beta_1, \beta_{11}, \beta_{21}, \beta_{31}$

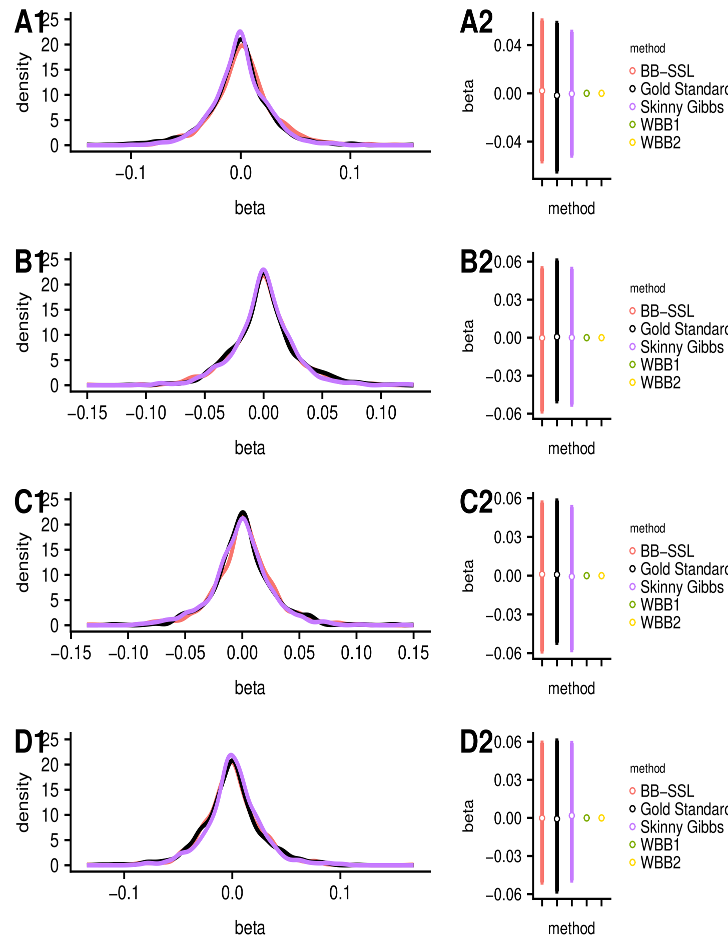


(b) Inactive predictors, from top to bottom: $\beta_2, \beta_{12}, \beta_{22}, \beta_{32}$

Figure 4: Estimated posterior density (left panel) and credible intervals (right panel) of β_i's in the high-dimensional independent case ($\rho = 0$). We have $n = 100, p = 1000, \beta_{active} = (1, 2, -2, 3)', \lambda_0 = 50, \lambda_1 = 0.05$. Each method has 5 000 sample points (after thinning for SSVS and Skinny Gibbs). BB-SSL is fitted using a single λ_0 and initialized at `SSLASSO` solution on the original $\boldsymbol{X}, \boldsymbol{y}$. Since WBB1 and WBB2 produce a point mass at zero, we exclude them from density comparisons.

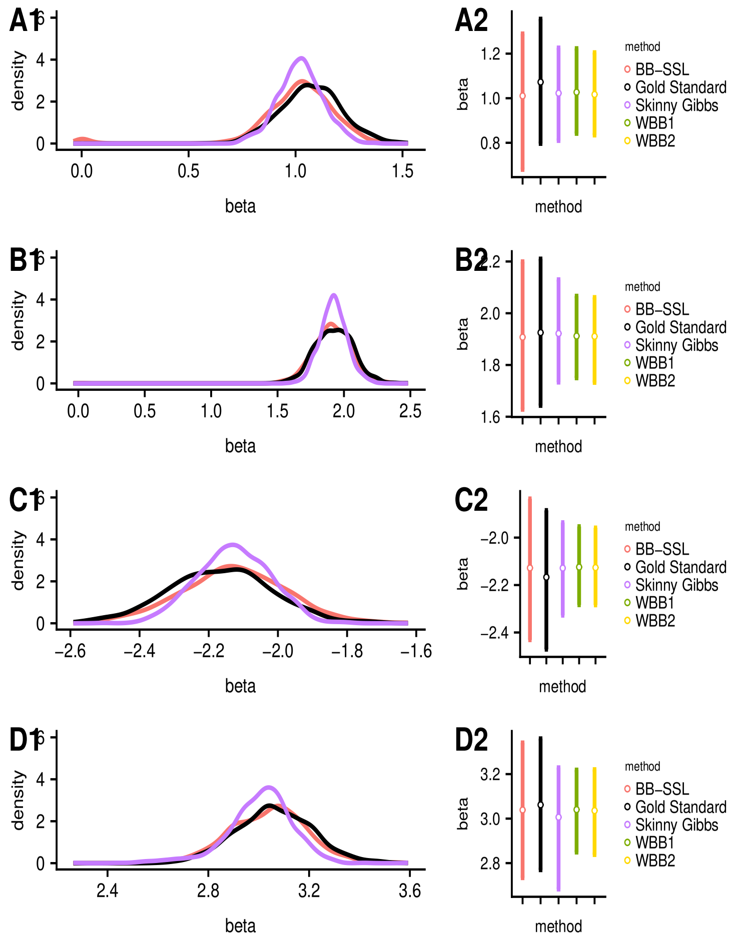


(a) Active predictors, from top to bottom: $\beta_1, \beta_{11}, \beta_{21}, \beta_{31}$

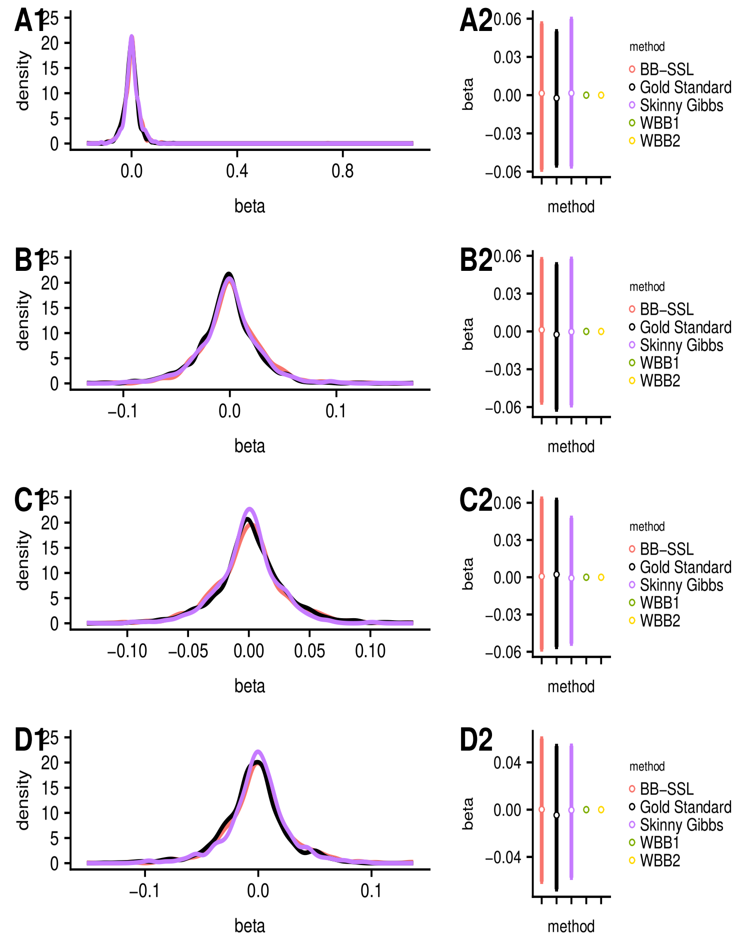


(b) Inactive predictors, from top to bottom: $\beta_2, \beta_{12}, \beta_{22}, \beta_{32}$

Figure 5: Estimated posterior density (left panel) and credible intervals (right panel) of β_i's in the high-dimensional block-wise correlated case ($\rho = 0.6$). We have $n = 100, p = 1000, \beta_{active} = (1, 2, -2, 3)', \lambda_0 = 50, \lambda_1 = 0.05$. Each method has 5 000 sample points (after thinning for SSVS and Skinny Gibbs). BB-SSL is fitted using a single λ_0 and initialized at `SSLASSO` solution on the original $\boldsymbol{X}, \boldsymbol{y}$. Since WBB1 and WBB2 produce a point mass at zero, we exclude them from density comparisons.

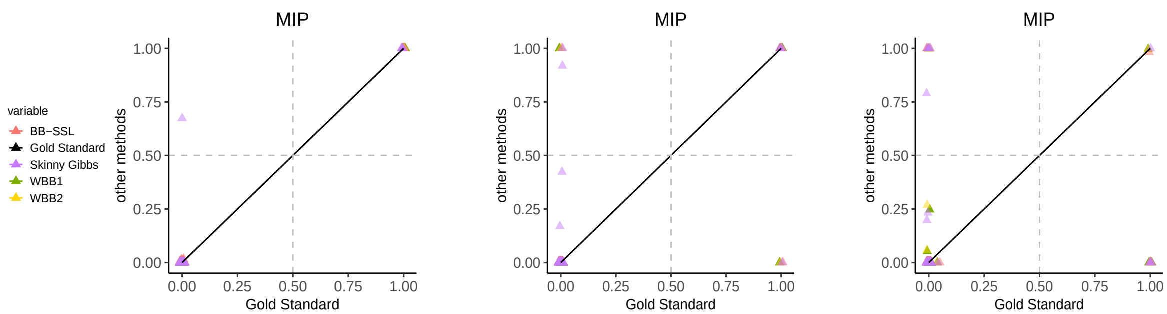


(a) Active predictors, from top to bottom: $\beta_1, \beta_{11}, \beta_{21}, \beta_{31}$



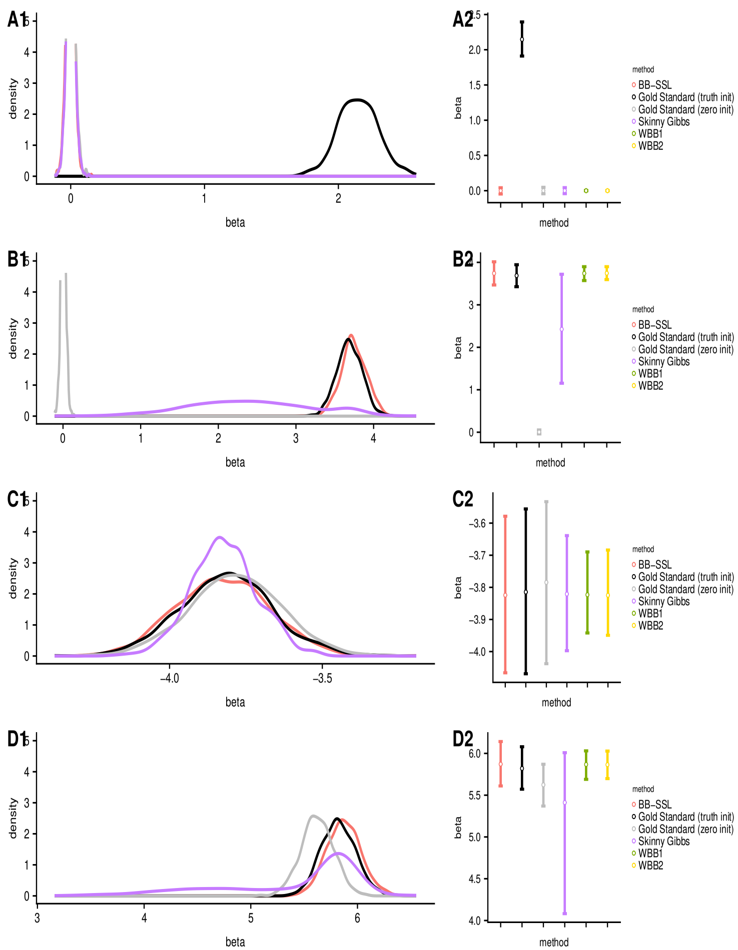
(b) Inactive predictors, from top to bottom: $\beta_2, \beta_{12}, \beta_{22}, \beta_{32}$

Figure 6: Estimated posterior density (left panel) and credible intervals (right panel) of β_i's in the high-dimensional, block-wise correlated case ($\rho = 0.9$). We have $n = 100, p = 1000, \beta_{active} = (1, 2, -2, 3)', \lambda_0 = 50, \lambda_1 = 0.05$. Each method has 5 000 sample points (after thinning for SSVS and Skinny Gibbs). BB-SSL is fitted using a single λ_0 and initialized at `SSLASSO` solution on the original $\boldsymbol{X}, \boldsymbol{y}$. Since WBB1 and WBB2 produce a point mass at zero, we exclude them from density comparisons.

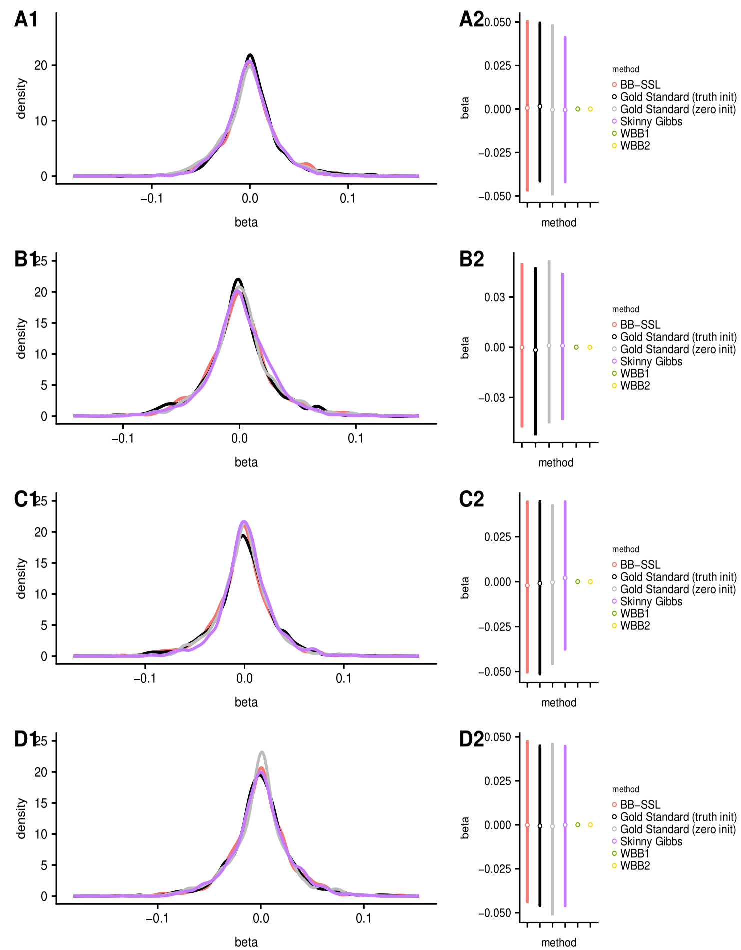


(a) Block-wise $\rho = 0.6$ (b) Block-wise $\rho = 0.99$ (c) Equi-correlation $\rho = 0.9$

Figure 7: Posterior means of γ_i's (i.e. a marginal inclusion probabilities) in high-dimensional settings with $n = 100, p = 1000$. We set $\lambda_0 = 50, \lambda_1 = 0.05$.

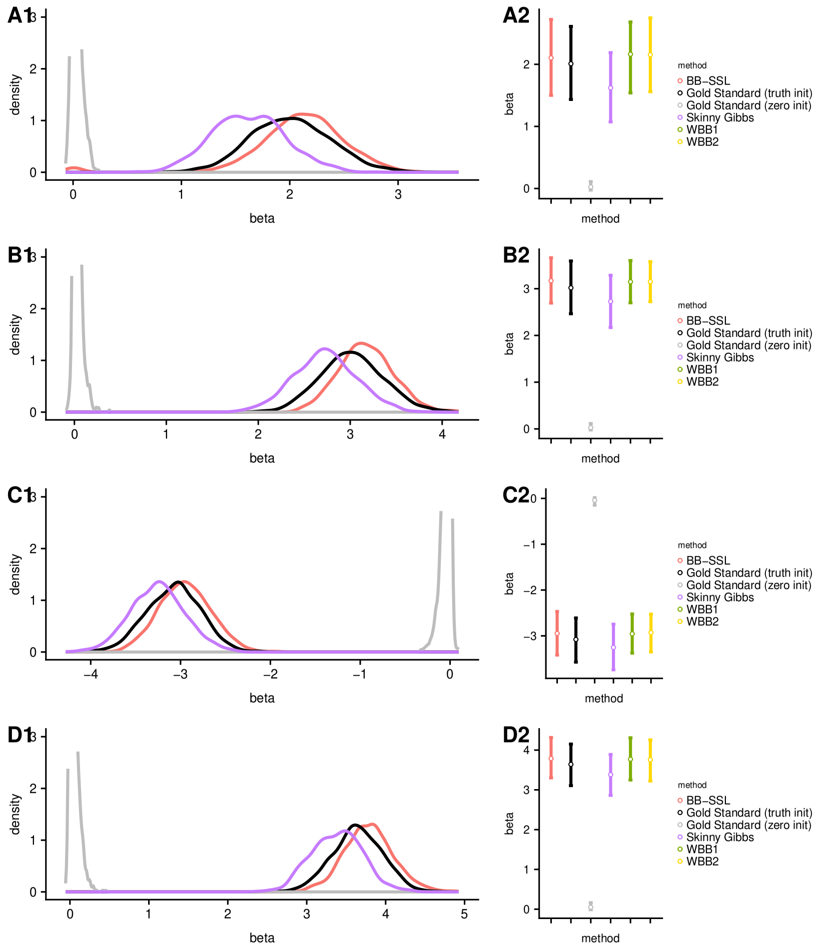


(a) Active predictors, from top to bottom: $\beta_1, \beta_{11}, \beta_{21}, \beta_{31}$

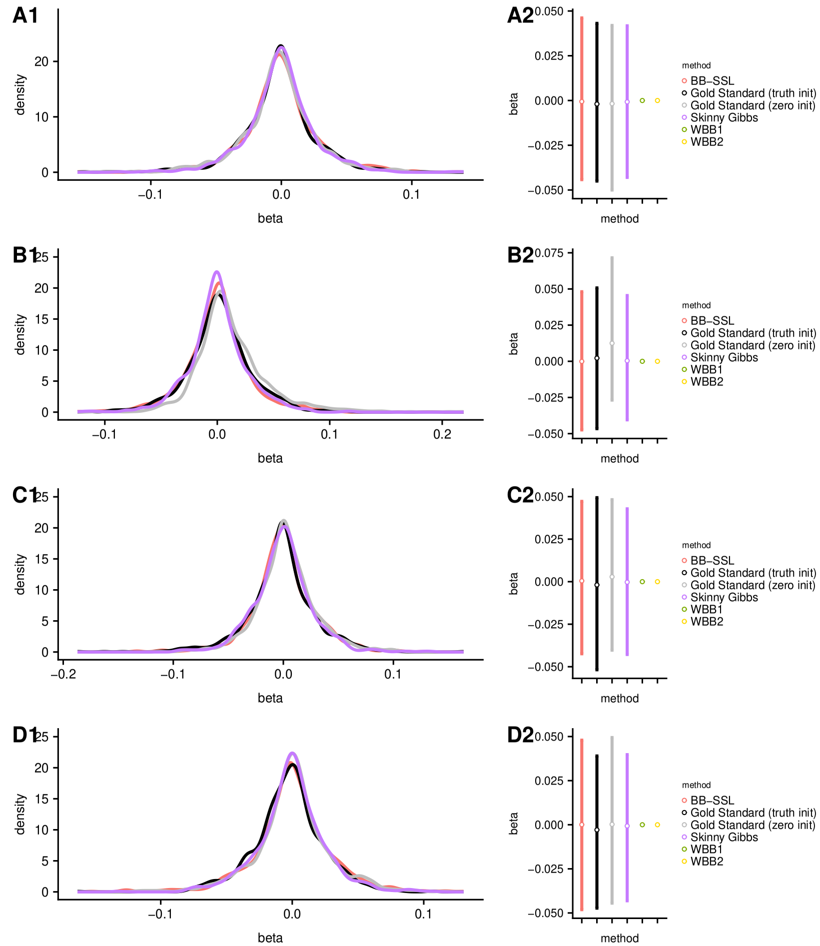


(b) Inactive predictors, from top to bottom: $\beta_2, \beta_{12}, \beta_{22}, \beta_{32}$

Figure 8: Estimated posterior density (left panel) and credible intervals (right panel) of β_i's in the high-dimensional block-wise correlated case ($\rho = 0.99$). We have $n = 100, p = 1000, \beta_{active} = (2, 4, -4, 6)', \lambda_0 = 50, \lambda_1 = 0.05$. Each method has 5 000 sample points (after thinning for SSVS and Skinny Gibbs). BB-SSL is fitted using a single λ_0 and initialized at `SSLASSO` solution on the original $\boldsymbol{X}, \boldsymbol{y}$. Since WBB1 and WBB2 produce a point mass at zero, we exclude them from density comparisons.

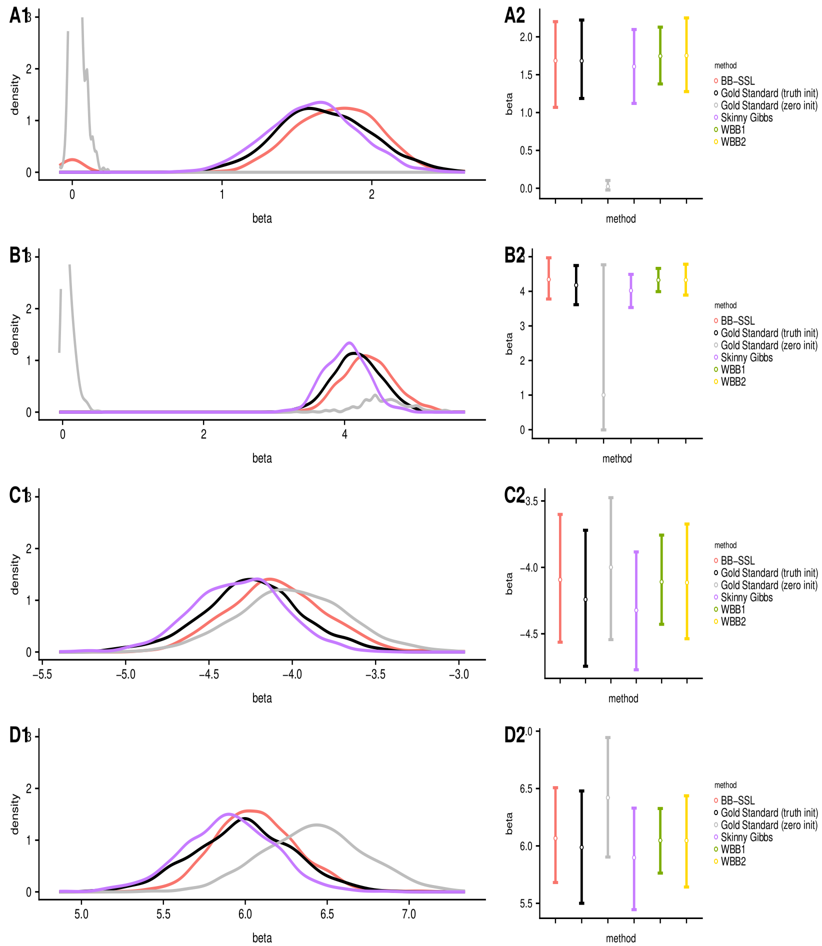


(a) Active predictors, from top to bottom: $\beta_1, \beta_2, \beta_3, \beta_4$

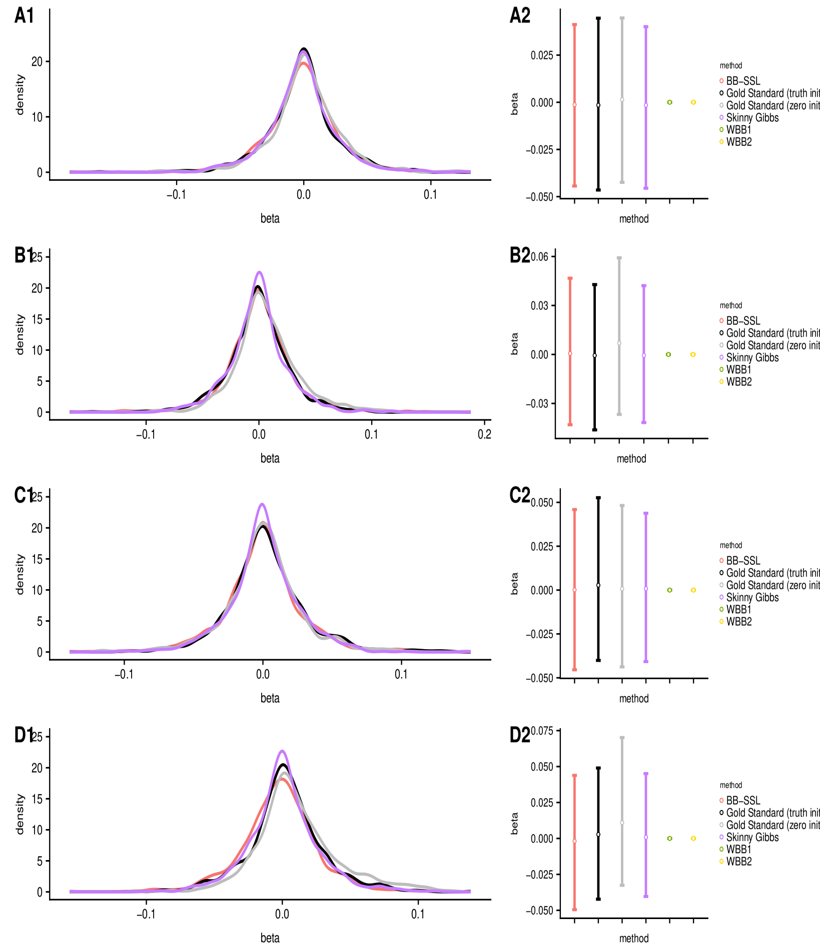


(b) Inactive predictors, from top to bottom: $\beta_5, \beta_6, \beta_7, \beta_8$

Figure 9: Estimated posterior density (left panel) and 90% credible intervals (right panel) of β_i's when all covariatess are correlated with $\rho = 0.9$.

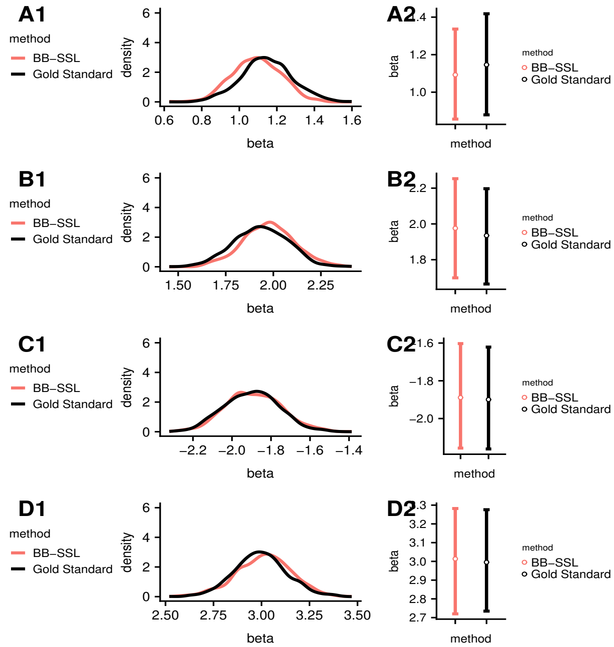


(a) Active predictors, from top to bottom: $\beta_1, \beta_2, \beta_3, \beta_4$

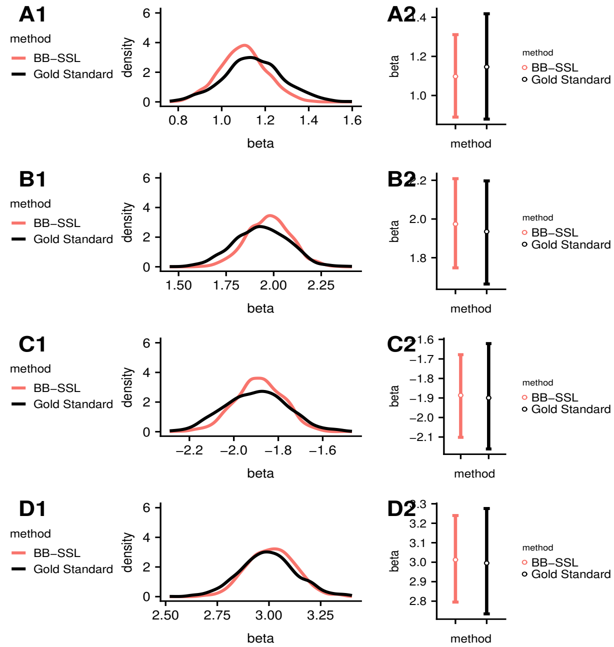


(b) Inactive predictors, from top to bottom: $\beta_5, \beta_6, \beta_7, \beta_8$

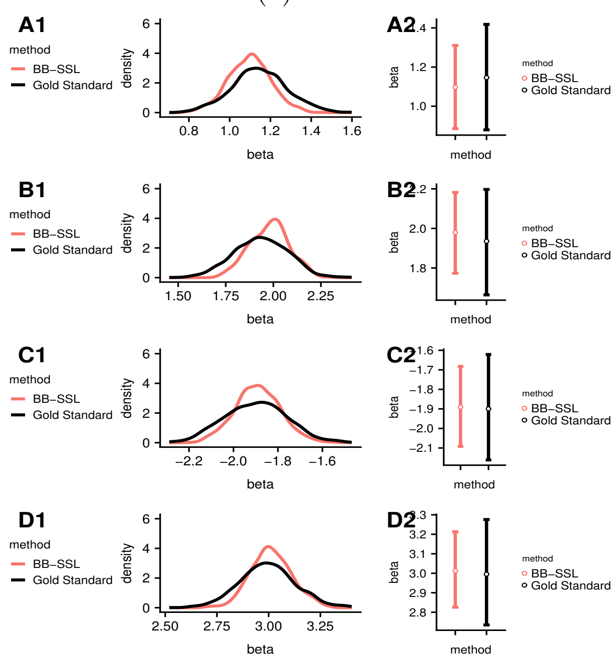
Figure 10: Estimated posterior density (left panel) and 90% credible intervals (right panel) of β_i's when all covariatess are correlated with $\rho = 0.9$.



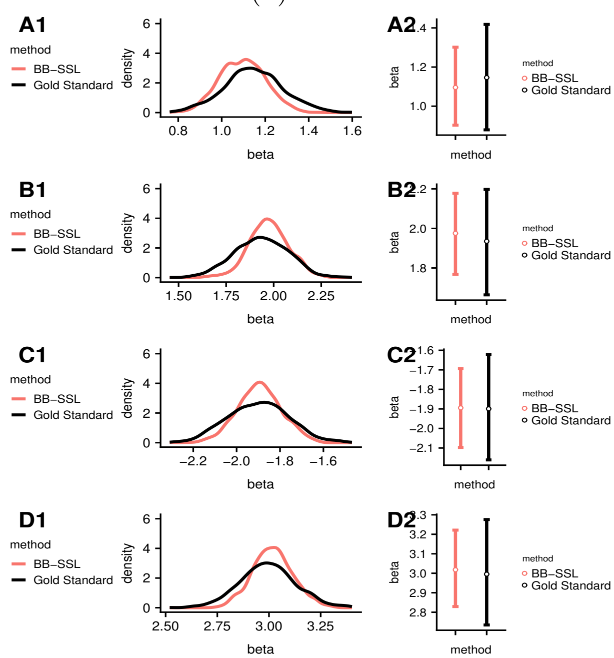
(a) $\alpha = 2$.



(b) $\alpha = 10$.



(c) $\alpha = 100$.



(d) $\alpha = 1000$.

Figure 11: Comparison of posterior density for active β_i's when choosing different α in high-dimensional, block-wise correlated setting ($\rho = 0.6$). `SSLASSO` is fitted with λ_0 being an equal difference sequence of length 10 starting at 0.05 and ending at 50. We set $\lambda_1 = 0.05$. Since WBB1 and WBB2 produce a point mass and do not fit into the y-axis, we exclude it in the density plot.

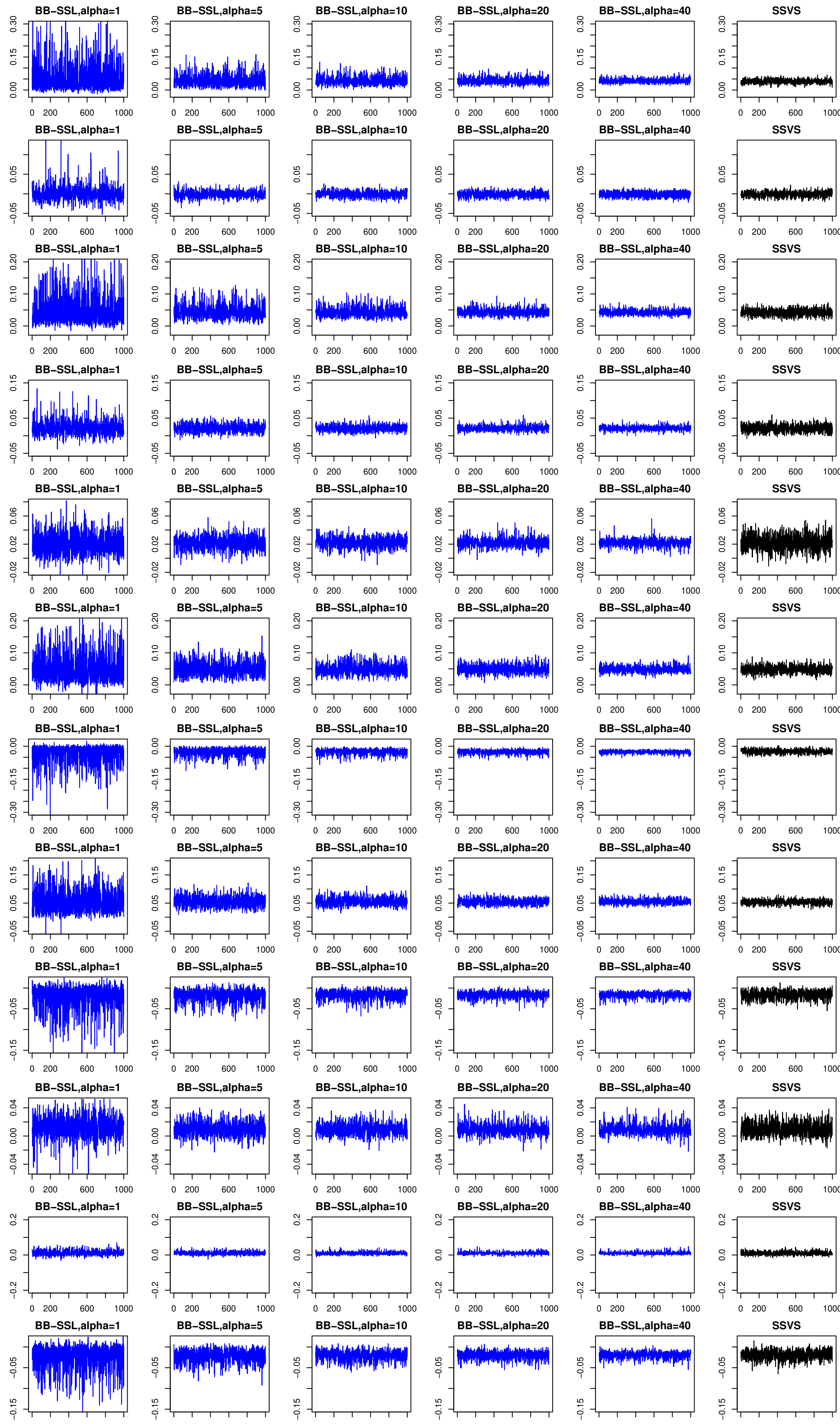


Figure 12: Trace plots of $\beta_i, i, 1, 2, \cdots, 12$ under varying α's in model (19) in the Durable Goods Marketing Data Set.

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